



Top 121 CAT Arrangement Questions With Video Solutions

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Questions

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five women decided to go shopping to M.G. Road, Bangalore. They arrived at the designated meeting place in the following order:

1. Archana,
2. Chellamma,
3. Dhenuka,
4. Helen, and
5. Shahnaz.

Each woman spent at least Rs. 1000. Below are some additional facts about how much they spent during their shopping spree.

- i. The woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193.
- ii. One woman spent Rs. 1340 and she was not Dhenuka.
- iii. One woman spent Rs. 1378 more than Chellamma.
- iv. One woman spent Rs. 2517 and she was not Archana.
- v. Helen spent more than Dhenuka.
- vi. Shahnaz spent the largest amount and Chellamma the smallest.

Question 1

Which of the following amounts was spent by one of them?

- ☒ A Rs. 1139
- ☐ B Rs. 1378
- ☐ C Rs. 2571
- ☐ D Rs. 2718

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Question 2

What was the amount spent by Helen?

- ☐ A Rs. 1193
- ☒ B Rs. 1340
- ☐ C Rs. 2234
- ☐ D Rs. 2517

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Question 3

The woman who spent Rs. 1193 is

- A Archana
- B Chellamma
- C Dhenuka**
- D Helen

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Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Seven varsity basketball players (A, B, C, D, E, F, and G) are to be honoured at a special luncheon. The players will be seated on the dais in a row. A and G have to leave the luncheon early and so must be seated at the extreme right. B will receive the most valuable player's trophy and so must be in the centre to facilitate presentation. C and D are bitter rivals and therefore must be seated as far apart as possible.

Question 4

Which of the following pairs cannot occupy the seats on either side of B?

- A F & D
- B D & E
- C E & G**
- D C & F

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Question 5

Which of the following cannot be seated at either end?

- A C
- B D
- C F**
- D G

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Question 6

Which of the following pairs cannot be seated together?

- A B & D
- B C & F**

C D & G

D E & A

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Instructions

DIRECTIONS for the following two questions: Answer the questions on the basis of the information given below.

The Head of a newly formed government desires to appoint five of the six elected members A, B, C, D, E and F to portfolios of Home, Power, Defence, Telecom and Finance. F does not want any portfolio if D gets one of the five. C wants either Home or Finance or no portfolio. B says that if D gets either Power or Telecom then she must get the other one. E insists on a portfolio if A gets one.

Question 7

If A gets Home and C gets Finance, then which is NOT a valid assignment of Defense and Telecom?

- A D-Defence, B-Telecom.
- B F-Defence, B-Telecom.
- C B-Defence, E-Telecom.
- D** B-Defence, D-Telecom.

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Question 8

Which is a valid assignment?

- A A-Home, B-Power, C-Defence, D-Telecom, E-Finance.
- B** C-Home, D-Power, A-Defence, B-Telecom, E-Finance.
- C A-Home, B-Power, E-Defence, D-Telecom, F-Finance.
- D B-Home, F-Power, E-Defence, C-Telecom, A-Finance.

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Instructions

Answer the questions on the basis of the information given below.

A, B, C, D, E, and F are a group of friends. There are two housewives, one professor, one engineer, one accountant and one lawyer in the group. There are only two married couples in the group. The lawyer is married to D, who is a housewife. No woman in the group is either an engineer or an accountant. C, the accountant, is married to F, who is a professor. A is married to a housewife. E is not a housewife.

Question 9

Which of the following is one of the married couples?

- A** A & B

- B B & E
- C D & E
- D** A & D

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Question 10

How many members of the group are males?

- A 2
- B 3
- C 4
- D** Cannot be determined

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Question 11

What is E's profession?

- A** Engineer
- B Lawyer
- C Professor
- D Accountant

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Instructions

The pages of a book are numbered 0, 1, 2 . upto M, $M > 0$. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are :

1. OPEN : Position the book at page No. 1
2. CLOSE : Position the book at page No. 0
3. FORWARD, n : From the current page move forward by n pages; if, in this process, page number M is reached, stop at M.
4. BACKWARD, n : From the current page, move backward by n pages; if in this process, page number 0 is reached, stop at page number 0.

In each of the following questions, you will find a sequence of instructions formed from the above categories. In each case, let n_1 be the page number before the instructions are executed and n_2 be the page number at which the book is positioned after the instructions are executed.

Question 12

FORWARD, 5; BACKWARD, 4. Which of the following statements about the above instructions is true?

- A $n_2 = n_1 + 4$ provided $1 < n_1 < 7$
- B $n_2 = n_1$ provided $M < 6$
- C $n_2 = n_1 + 1$ provided $M - n_1 > 5$
- D $n_2 - n_1 < 0$ provided $M > 0$



The pages of a book are numbered 0, 1, 2, ..., upto M, M+1. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are:

1. OPEN : Position the book at page No. 1
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3. FORWARD, n : From the current page move forward by n pages; if, in this process, page number M is reached, stop at M.
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In each of the following questions, you will find a sequence of instructions formed from the above categories. In each case, let n_1 be the page number before the instructions are executed and n_2 be the page number at which the book is positioned after the instructions are executed.

BACKWARD, 5; FORWARD, 5. Which of the following statements is true about the above set of instructions?

A) $n_1 = n_2$ provided $n_1 >= 5$ B) $n_1 = n_2$ provided $n_1 > 0$
 C) $n_2 = 5$ provided $M > 0$ D) $n_2 = 5$ provided $M > 0$

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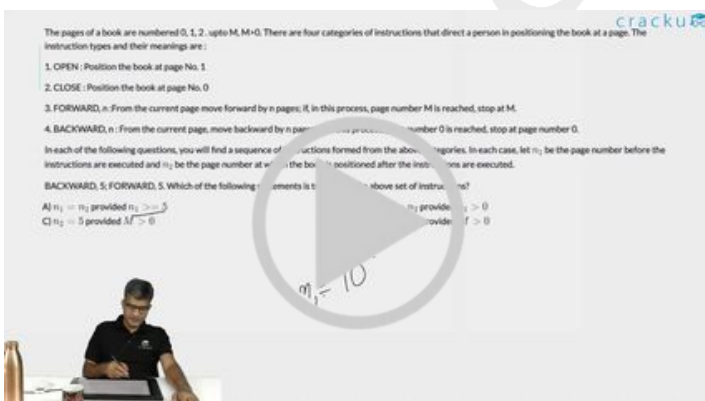
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Question 13

BACKWARD, 5; FORWARD, 5. Which of the following statements is true about the above set of instructions?

- A $n_1 = n_2$ provided $n_1 >= 5$
- B $n_1 = n_2$ provided $n_1 > 0$
- C $n_2 = 5$ provided $M > 0$
- D $n_1 > n_2$ provided $M > 0$



The pages of a book are numbered 0, 1, 2, ..., upto M, M+1. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are:

1. OPEN : Position the book at page No. 1
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A) $n_1 = n_2$ provided $n_1 >= 5$ B) $n_1 = n_2$ provided $n_1 > 0$
 C) $n_2 = 5$ provided $M > 0$ D) $n_2 = 5$ provided $M > 0$

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Question 14

FORWARD, 25 ; BACKWARD, 10. which of the following statements is true?

- A** $n_1 = n_2$ if $M=10$ and $n_1 = 0$
- B** $M = 20$ provided $n_1 > 0$
- C** $n_1 > 30$ provided $M = 900$
- D** $n_1 = 37$ provided $M = 25$



The pages of a book are numbered 0, 1, 2, ... upto M, $M > 0$. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are:

1. OPEN : Position the book at page No. 1
2. CLOSE : Position the book at page No. 0
3. FORWARD, n : From the current page move forward by n pages; if, in this process, page number M is reached, stop at M .
4. BACKWARD, n : From the current page, move backward by n pages; if, in this process, page number 0 is reached, stop at page number 0.

In each of the following questions, you will find a sequence of instructions formed from the above categories. In each case, let n_1 be the page number before the instructions are executed and n_2 be the page number at which the book is positioned after the instructions are executed.

BACKWARD, 5; FORWARD, 5. Which of the following statements is true about the above set of instructions?

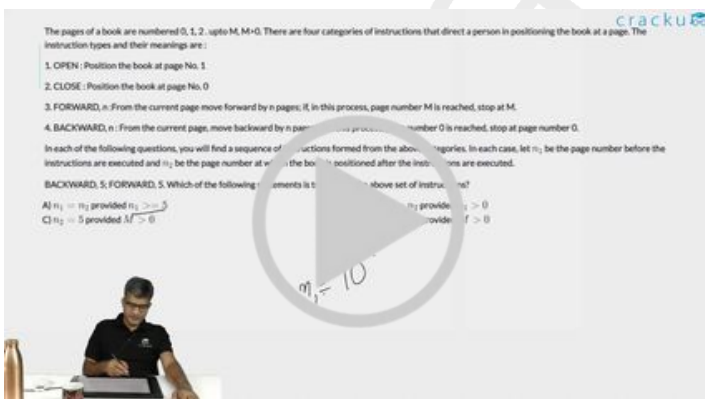
A) $n_1 = n_2$ provided $n_1 > 5$ B) $n_1 = n_2$ provided $n_1 > 0$
C) $n_2 = 5$ provided $M > 0$ D) $n_2 = 5$ provided $M > 5$

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Question 15

FORWARD, 10; FORWARD, 10. Which of the following statements about the above instructions is true?

- A** $n_2 - n_1 = 20$ only if $n_1 = 0$
- B** $n_2 - n_1 = 20$ if $M > 20$ and $n_1 = 1$
- C** $n_2 - n_1 = 10$ if $M = 21$ and $n_1 = 0$
- D** $n_2 > n_1$ if $M > 0$



The pages of a book are numbered 0, 1, 2, ... upto M, $M > 0$. There are four categories of instructions that direct a person in positioning the book at a page. The instruction types and their meanings are:

1. OPEN : Position the book at page No. 1
2. CLOSE : Position the book at page No. 0
3. FORWARD, n : From the current page move forward by n pages; if, in this process, page number M is reached, stop at M .
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In each of the following questions, you will find a sequence of instructions formed from the above categories. In each case, let n_1 be the page number before the instructions are executed and n_2 be the page number at which the book is positioned after the instructions are executed.

BACKWARD, 5; FORWARD, 5. Which of the following statements is true about the above set of instructions?

A) $n_1 = n_2$ provided $n_1 > 5$ B) $n_1 = n_2$ provided $n_1 > 0$
C) $n_2 = 5$ provided $M > 0$ D) $n_2 = 5$ provided $M > 5$

 VIDEO SOLUTION



Instructions

Directions for the following three questions: Answer the following questions based on the statements given below:

- (i) There are three houses on each side of the road.
- (ii) These six houses are labeled as P, Q, R, S, T and U.
- (iii) The houses are of different colours, namely, Red, Blue, Green, Orange, Yellow and White.
- (iv) The houses are of different heights.
- (v) T, the tallest house, is exactly opposite to the Red coloured house.
- (vi) The shortest house is exactly opposite to the Green coloured house.
- (vii) U, the Orange coloured house, is located between P and S.
- (viii) R, the Yellow coloured house, is exactly opposite to P.
- (ix) Q, the Green coloured house, is exactly opposite to U.
- (x) P, the White coloured house, is taller than R, but shorter than S and Q.

Question 16

What is the colour of the tallest house?

- A Red
- B Blue**
- C Green
- D Yellow
- E none of these

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Question 17

Which is the second tallest house?

- A P
- B S
- C Q
- D R
- E cannot be determined**

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Question 18

What is the colour of the house diagonally opposite to the Yellow coloured house?

- A White
- B Blue
- C Green
- D Red**
- E none of these

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Instructions

Directions for the following four questions: Answer the questions based on the following information.

A and B are two sets (e.g. A = Mothers, B = Women).

The elements that could belong to both the sets (e.g. women who are mothers) is given by the set $C = A \cap B$.

The elements which could belong to either A or B, or both, is indicated by the set $D = A \cup B$.

A set that does not contain any elements is known as a null set represented by ϕ (e.g. if none of the women in the set B is a mother, then $C = A \cap B$ is a null set, or $C = \phi$).

Let 'V' signify the set of all vertebrates, 'M' the set of all mammals, 'D' dogs, 'F' fish, 'A' alsatian and 'P', a dog named Pluto.

Question 19

If $P \cap A = \phi$ and $P \cup A = D$, then which of the following is true?

- A Pluto and alsatians are dogs
- B Pluto is an alsatian
- C Pluto is not an alsatian
- D Both A and C**

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Question 20

If $Y = F \cap (D \cap V)$ is not a null set, it implies that

- A all fish are vertebrates
- B all dogs are vertebrates
- C some fish are dogs**
- D None of these

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Question 21

Given that $X = M \cap D$ is such that $X = D$. Which of the following is true?

- ☒ A All dogs are mammals
- ☐ B Some dogs are mammals
- ☐ C $X = \phi$
- ☐ D All mammals are dogs

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Question 22

If $Z = (P \cap D) \cup M$, then

- ☒ A the elements of Z consist of Pluto, the dog, or any other mammal
- ☐ B Z implies any dog or mammal
- ☐ C Z implies Pluto or any dog that is a mammal
- ☐ D Z is a null set

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Instructions

Directions for the following two questions: Answer the following questions based on the information given below.

Elle is three times older than Yogesh. Zaheer is half the age of Wahida. Yogesh is older than Zaheer.

Question 23

Which of the following information will be sufficient to estimate Elle's age?

- ☐ A Zaheer is 10-year-old
- ☐ B Both Yogesh and Wahida are older than Zaheer by the same number of years
- ☒ C Both the above statements are needed
- ☐ D None of these

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Question 24

Which of the following can be inferred?

- A Yogesh is older than Wahida
- B** Elle is older than Wahida
- C Elle may be younger than Wahida
- D None of these

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Instructions

DIRECTIONS for the following two questions: These questions are based on the situation given below: Seven university cricket players are to be honored at a special luncheon. The players will be seated on the dais along one side of a single rectangular table. A and G have to leave the luncheon early and must be seated at the extreme right end of the table, which is closest to the exit. B will receive the Man of the Match award and must be in the center chair. C and D who are bitter rivals for the position of wicket keeper, dislike one another and should be seated as far apart as possible. E and F are best friends and want to sit together.

Question 25

Which of the following pairs may not be seated together?

- A** E & A
- B B & D
- C C & F
- D G & D

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Question 26

Which of the following may not be seated at either end of the table?

- A C
- B D
- C G
- D** F

[VIEW SOLUTION](#)

Instructions

Answer the questions on the basis of the information given below.

These questions are based on the situation given below: A, B, C, D, E and F are a group of friends from a club. There are two housewives, one lecturer, one architect, one accountant and one lawyer in the group. There are two married couples in the group. The lawyer is married to D who is a housewife. No lady in the group is either

an architect or an accountant. C, the accountant, is married to F who is a lecturer. A is married to D and E is not a housewife.

Question 27

What is E?

- A Lawyer
- B Architect
- C Lecturer
- D Accountant

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Question 28

How many members of the group are male?

- A 2
- B 3
- C 4
- D None of these

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Instructions

There were seven elective courses - E1 to E7 - running in a specific term in a college. Each of the 300 students enrolled had chosen just one elective from among these seven. However, before the start of the term, E7 was withdrawn as the instructor concerned had left the college. The students who had opted for E7 were allowed to join any of the remaining electives. Also, the students who had chosen other electives were given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased; however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
From Elective	E1	9	5	10	1	4	2
	E2		34	8		2	2
	E3	2	6	25			2
	E4		3	2	14		4
	E5		5			30	
	E6		7	3		2	9
	E7	4	16	30	5	5	41

Further, the following are known:

- Before the change process there were 6 more students in E1 than in E4, but after the reshuffle, the number of students in E4 was 3 more than that in E1.

- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

Question 29

How many elective courses among E1 to E6 had a decrease in their enrollments after the change process?

- A** 4
- B** 1
- C** 2
- D** 3

Each student was given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased, however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
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	E2	8	8	8	2	2	2
	E3	2	6	25			
	E4	3	2	14		4	
	E5	5			30		
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- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

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Question 30

After the change process, which of the following is the correct sequence of number of students in the six electives E 1 to E6?

- A** 19, 76, 79, 21, 45, 60
- B** 19, 76, 78, 22, 45, 60
- C** 18, 76, 79, 23, 43, 61
- D** 18, 76, 79, 21, 45, 61

Each student was given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased, however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
From Elective	E1	9	5	10	3	4	2
	E2	8	8	8	2	2	2
	E3	2	6	25			
	E4	3	2	14		4	
	E5	5			30		
	E6	7	3	2	9		
	E7	4	14	30	5	5	41

Further, the following are known:

- Before the change process there were 6 more students in E1 than in E4, but after the reshuffle, the number of students in E1 was 3 more than that in E1.
- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

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Question 31

Later, the college imposed a condition that if after the change of electives, the enrollment in any elective (other than E7) dropped to less than 20 students, all the students who had left that course will be required to re-enroll for that elective.

Which of the following is a correct sequence of electives in decreasing order of their final enrollments?

- A** E2, E3, E6, E5, E1, E4
- B** E3, E2, E6, E5, E4, E1
- C** E2, E5, E3, E1, E4, E6
- D** E2, E3, E5, E6, E1, E3

Students were given one chance to change their choice. The table below captures the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased; however, it is known that these were either 0 or 1.

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	E2	8	8	8	2	2	2
	E3	2	6	25			
	E4	3	2	14			4
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Further, the following are known:

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- The number of students in E2 increased by 30 after the change process.
- Before the change process, E4 had 2 more students than E6, while E2 had 10 more students than E3.

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Question 32

After the change process, which course among E1 to E6 had the largest change in its enrollment as a percentage of its original enrollment?

- A** E1
- B** E2
- C** E3
- D** E6

Each student is given one chance to change their choice. The table below tracks the movement of the students from one elective to another during this process. Movement from one elective to the same elective simply means no movement. Some numbers in the table got accidentally erased, however, it is known that these were either 0 or 1.

		To Elective					
		E1	E2	E3	E4	E5	E6
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VIDEO SOLUTION

Instructions

A supermarket has to place 12 items (coded A to L) in shelves numbered 1 to 16. Five of these items are types of biscuits, three are types of candies and the rest are types of savouries. Only one item can be kept in a shelf. Items are to be placed such that all items of same type are clustered together with no empty shelf between items of the same type and at least one empty shelf between two different types of items. At most two empty shelves can have consecutive numbers.

The following additional facts are known.

- A and B are to be placed in consecutively numbered shelves in increasing order.
- I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
- D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
- K is to be placed in shelf number 16.
- L and J are items of the same type, while H is an item of a different type.
- C is a candy and is to be placed in a shelf preceded by two empty shelves.
- L is to be placed in a shelf preceded by exactly one empty shelf.

Question 33

In how many different ways can the items be arranged on the shelves?

- A 8
- B 4
- C 2
- D 1

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The following additional facts are known.

1. A and B are to be placed in consecutively numbered shelves in increasing order.
2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
4. K is to be placed in shelf number 16.
5. L and J are items of the same type, while H is an item of a different type.
6. C is a candy and is to be placed in a shelf preceded by two empty shelves.
7. L is to be placed in a shelf preceded by exactly one empty shelf.

Handwritten notes:

- 12 items → A, B, ..., L
- 16 shelves
- 5 Biscuits
- 3 Candies
- 4 Savouries
- I, J



CAT Daily Targets

Step by step, heading towards 99+ percentile



Question 34

Which of the following statements is necessarily true?

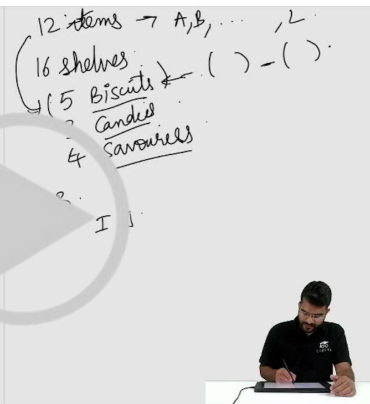
- A All biscuits are kept before candies.
- B There are two empty shelves between the biscuits and the candies.
- C All candies are kept before biscuits.
- D There are at least four shelves between items B and C.

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A supermarket has to place 12 items (coded A to L) in shelves numbered 1 to 16. Five of these items are types of biscuits, three are types of candies and the rest are types of savouries. Only one item can be kept in a shelf. Items are to be placed such that all items of same type are clustered together with no empty shelf between items of the same type and at least one empty shelf between two different types of items. At most two empty shelves can have consecutive numbers.

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- 1. A and B are to be placed in consecutively numbered shelves in increasing order.
- 2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
- 3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
- 4. K is to be placed in shelf number 16.
- 5. L and J are items of the same type, while H is an item of a different type.
- 6. C is a candy and is to be placed in a shelf preceded by two empty shelves.



Handwritten notes on the right side of the video player:

- 12 items → A, B, ..., L
- 16 shelves → () - ()
- 5 Biscuits
- 3 Candies
- 4 Savouries

Question 35

Which of the following items is not a type of biscuit?

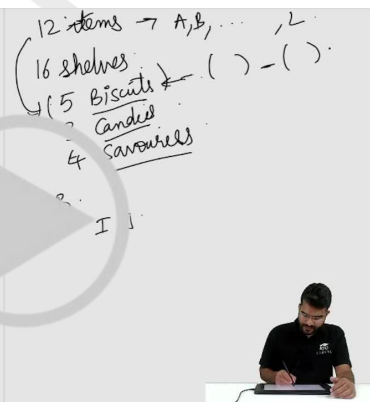
- A L
- B A
- C B
- D G

cracku

A supermarket has to place 12 items (coded A to L) in shelves numbered 1 to 16. Five of these items are types of biscuits, three are types of candies and the rest are types of savouries. Only one item can be kept in a shelf. Items are to be placed such that all items of same type are clustered together with no empty shelf between items of the same type and at least one empty shelf between two different types of items. At most two empty shelves can have consecutive numbers.

The following additional facts are known.

- 1. A and B are to be placed in consecutively numbered shelves in increasing order.
- 2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
- 3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
- 4. K is to be placed in shelf number 16.
- 5. L and J are items of the same type, while H is an item of a different type.
- 6. C is a candy and is to be placed in a shelf preceded by two empty shelves.



Handwritten notes on the right side of the video player:

- 12 items → A, B, ..., L
- 16 shelves → () - ()
- 5 Biscuits
- 3 Candies
- 4 Savouries

Question 36

Which of the following can represent the numbers of the empty shelves in a possible arrangement?

- A 1, 7, 11, 12
- B 1, 5, 6, 12
- C 1, 2, 6, 12
- D 1, 2, 8, 12

cracku

A supermarket has to place 12 items (coded A to L) in shelves numbered 1 to 16. Five of these items are types of biscuits, three are types of candies and the rest are types of savouries. Only one item can be kept in a shelf. Items are to be placed such that all items of same type are clustered together with no empty shelf between items of the same type and at least one empty shelf between two different types of items. At most, two empty shelves can have consecutive numbers.

The following additional facts are known:

- 1. A and B are to be placed in consecutively numbered shelves in increasing order.
- 2. I and J are to be placed in consecutively numbered shelves both higher numbered than the shelves in which A and B are kept.
- 3. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies.
- 4. K is to be placed in shelf number 16.
- 5. L and J are items of the same type, while H is an item of a different type.
- 6. C is a candy and is to be placed in a shelf preceded by two empty shelves.

Handwritten notes: 12 items → A, B, ..., L. 16 shelves. (5 Biscuits) - () - (). 3 Candies. 4 Savouries. I J.

FREE

CAT Sectional Tests

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Take Here

Instructions

Directions for the next 2 questions: There are three bottles of water, A, B, C, whose capacities are 5 litres, 3 litres, and 2 litres respectively. For transferring water from one bottle to another and to drain out the bottles, there exists a piping system. The flow through these pipes is computer controlled. The computer that controls the flow through these pipes can be fed with three types of instructions, as explained below:

Instruction type	Explanation for the instruction
FILL (X, Y)	Fill bottle labeled X from the water in bottle labeled Y, where the remaining capacity of X is less than or equal to the amount of water in Y.
EMPTY (X, Y)	Empty out water in bottle labeled X into bottle labeled Y, where the amount of water in X is less than or equal to remaining capacity of Y.
DRAIN (X)	Drain out all the water contained in bottle labeled X.

Initially, A is full with water, and B and C are empty.

Question 37

After executing a sequence of three instructions, bottle A contains one litre of water. The first and the third of these instructions are shown below:

First instruction: FILL (C, A)

Third instruction: FILL (C, A)

Then which of the following statements about the instruction is true?

- A The second instruction is FILL (B, A)
- B The second instruction is EMPTY (C, B)
- C The second instruction transfers water from B to C
- D The second instruction involves using the water in bottle A

[VIEW SOLUTION](#)

Question 38

Consider the same sequence of three instructions 'and the same initial state mentioned above. Three more instructions are added at the end of the above sequence to have A contain 4 litres of water. In this total sequence of six instructions, the fourth one is DRAIN (A). This is the only DRAIN instruction in the entire sequence. At the end of the execution of the above sequence, how much water (in litres) is contained in C?

- A One
- B Two
- C Zero
- D None of these

[VIEW SOLUTION](#)

Instructions

Seven children, Aarav, Bina, Chirag, Diya, Eshan, Farhan, and Gaurav, are sitting in a circle facing inside (not necessarily in the same order) and playing a game of 'Passing the Buck'.

The game is played over 10 rounds. In each round, the child holding the Buck must pass it directly to a child sitting in one of the following positions:

- Immediately to the left;
- Immediate to the right;
- Second to the left; or
- Second to the right.

The game starts with Bina passing the Buck and ends with Chirag receiving the Buck. The table below provides some information about the pass types and the child receiving the Buck. Some information is missing and labelled as '?'.
?

Round	Pass Type	Received by
1	Immediately to the left	Aarav
2	Second to the right	?
3	Immediately to the right	Diya
4	?	?
5	?	Aarav
6	Second to the left	?
7	Immediately to the left	Gaurav
8	Immediately to the left	?
9	?	Farhan
10	?	Chirag

Question 39

For which of the following children is it possible to determine how many times they received the Buck?

- A Farhan
- B Eshan
- C Bina
- D Gaurav

[VIEW SOLUTION](#)



Question 40

Who is sitting third to the left of Eshan?

- A Gaurav
- B Divya
- C Chirag
- D Aarav

[VIEW SOLUTION](#)

Question 41

For which of the following pass types can the total number of occurrences be uniquely determined?

- A Immediately to the left
- B Second to the right
- C Immediately to the right
- D Second to the left

[VIEW SOLUTION](#)

Question 42

Who is sitting immediately to the right of Bina?

- A Aarav
- B Eshan
- C Farhan
- D Chirag

[VIEW SOLUTION](#)

Instructions

For the following questions answer them individually

Question 43

Five boys went to a store to buy sweets. One boy had Rs. 40. Another boy had Rs. 30. Two other boys had Rs. 20 each. The remaining boy had Rs. 10. Below are some more facts about the initial and final cash positions.

- I. Alam started with more than Jugraj.
- II. Sandeep spent Rs. 1.50 more than Daljeet.
- III. Ganesh started with more money than just only one other person.
- IV. Daljeet started with $\frac{2}{3}$ of what Sandeep started with.
- V. Alam spent the most, but did not end with the least.
- VI. Jugraj spent the least and ended with more than Alam or Daljeet.
- VII. Ganesh spent Rs.3.50.
- VIII. Alam spent 10 times than what Ganesh did.

In the choices given below, all statements except one are false. Which one of the following statements can be true?

- A Alam started with Rs. 40 and ended with Rs. 9.50
- B Sandeep started with Rs. 30 and ended with Re. 1
- C Ganesh started with Rs. 20 and ended with Rs. 4
- D Jugraj started with Rs. 10 and ended with Rs. 7

[VIEW SOLUTION](#)

Question 44

At a village mela, the following six nautankis (plays) are scheduled as shown in the table below.

No.	Nautanki	Duration	Show Times
1	Sati Savitri	1 hr	9 am and 2 pm
2	Joru ka Ghulam	1 hr	10:30 am and 11:30 am
3	Sundar Kand	30 min	10 am and 11 am
4	Veer Abhimanyu	1 hr	10 am and 11 am
5	Reshma aur Shera	1 hr	9:30 am, 12 noon and 2 pm
6	Jhansi ki Rani	30 min	11 am and 1:30 pm

You wish to see all the six nautankis. Further, you wish to ensure that you get a lunch break from 12.30 p.m. to 1.30 p.m. Which of the following ways can you do this?

- A Sati Savitri is viewed first; Sundar Kand is viewed third, and Jhansi ki Rani is viewed last
- B Sati Savitri is viewed last; Veer Abhimanyu is viewed third, and Reshma aur Shera is viewed first
- C Sati Savitri is viewed first; Sundar Kand is viewed third, and Joru ka Ghulam is viewed fourth
- D Veer Abhimanyu is viewed third; Reshma aur Shera is viewed fourth, and Jahansi ki Rani is viewed fifth

[VIEW SOLUTION](#)

Question 45

Three children won the prizes in the Bournvita Quiz contest. They are from the schools: Loyola, Convent and Little Flowers, which are located at different cities. Below are some of the facts about the schools, the children and the city they are from.

- I. One of the children is Bipin.
- II. Loyola School's contestant did not come first.
- III. Little Flower's contestant was named Riaz.
- IV. Convent School is not in Hyderabad.
- V. The contestant from Pune is not from Loyola School.
- VI. The contestant from Bangalore did not come first.
- VII. Convent School's contestant's name is not Balbir.
- VIII. The contestant from Pune came in third.

Which of the following statements is true?

- A 1st prize: Riaz (Little Flowers), 2nd prize: Bipin (Convent), 3rd prize: Balbir (Loyola)
- B 1st prize: Bipin (Convent), 2nd prize: Riaz (Little Flowers), 3rd prize: Balbir (Loyola)
- C 1st prize: Riaz (Little Flowers), 2nd prize: Balbir (Loyola), 3rd prize: Bipin (Convent)
- D 1st prize: Bipin (Convent), 2nd prize: Balbir (Loyola), 3rd prize: Riaz (Little Flowers)

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100+ CAT Quant Questions



Question 46

The Bannerjees, the Sharmas, and the Pattabhiramans each have a tradition of eating Sunday lunch as a family. Each family serves a special meal at a certain time of day. Each family has a particular set of chinaware used for this meal. Use the clues below to answer the following question.

- . The Sharma family eats at noon.
- . The family that serves fried brinjal uses blue chinaware.
- . The Bannerjee family eats at 2 o'clock.
- . The family that serves sambar does not use red chinaware.
- . The family that eats at 1 o'clock serves fried brinjal.
- . The Pattabhiraman family does not use white chinaware.
- . The family that eats last likes makkai-ki-roti.

Which one of the following statements is true?

- A The Bannerjees eat makkai-ki-roti at 2 o'clock, the Sharmas eat fried brinjal at 12 o'clock and the Pattabhiramans eat sambar from red chinaware
- B The Sharmas eat sambar served in white chinaware, the Pattabhiramans eat fried brinjal at 1 o'clock, and the Bannerjees eat makkai-ki-roti served in blue chinaware
- C The Sharmas eat sambar at noon, the Pattabhiramans eat fried brinjal served in blue chinaware, and the Bannerjees eat makkai-ki-roti served in red chinaware

- D** The Bannerjees eat makkai-ki-roti served in white chinaware, the Sharmas eat fried brinjal at 12 o'clock and the Pattabhiramans eat sambar from red chinaware

 [VIEW SOLUTION](#)

Question 47

Albert, David, Jerome and Tommy were plucking mangoes in a grove to earn some pocket money during the summer holidays. Their earnings were directly related to the number of mangoes plucked and had the following relationship: Jerome got less money than Tommy. Jerome and Tommy together got the same amount as Albert and David taken together. Albert and Tommy together got less than David and Jerome taken together. Who earned the most pocket money? Who plucked the least number of mangoes?

- A** David, Jerome
- B** David, Albert
- C** Jerome, Tommy
- D** Jerome, Albert

 [VIEW SOLUTION](#)

Question 48

Eight people carrying food baskets are going for a picnic on motorcycles.

Their names are A, B, C, D, E, F, G, and H. They have 4 motorcycles M1, M2, M3 and M4 among them. They also have 4 food baskets O, P, Q and R of different sizes and shapes and each can be carried only on motorcycles M1, M2, M3 and M4 respectively.

No more than 2 persons can travel on a motorcycle and no more than one basket can be carried on a motorcycle. There are 2 husband-wife pairs in this group of 8 people and each pair will ride on a motorcycle together.

C cannot travel with A or B. E cannot travel with B or F. G cannot travel with F, or H, or D.

The husband-wife pairs must carry baskets O and P. Q is with A and P is with D.

F travels on M1 and E travels on M2 motorcycles.

G is with Q, and B cannot go with R.

Who is travelling with H?

- A** A
- B** B
- C** C
- D** D

cracku

Eight people carrying food baskets are going for a picnic on motorcycles. Their names are A, B, C, D, E, F, G, and H. They have 4 motorcycles M1, M2, M3 and M4 among them. They also have 4 food baskets O, P, Q and R of different sizes and shapes and each can be carried only on motorcycles M1, M2, M3 and M4 respectively.

No more than 2 persons can travel on a motorcycle and no more than one basket can be carried on a motorcycle. There are 2 husband-wife pairs in this group of 8 people and each pair will ride on a motorcycle together.

C cannot travel with A or B. E cannot travel with B or F. G cannot travel with F, C, or D.

The husband-wife pairs must carry baskets O and P. A is with A and P is with D.

F travels on M1 and E travels on M2 motorcycle.

G is with Q, and B cannot go with R.

Who is travelling with H?

A) A
B) B
C) C
D) D



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100+ CAT DILR Questions



Question 49

Mrs Ranga has three children and has difficulty remembering their ages and months of their birth. The clue below may help her remember.

- . The boy, who was born in June, is 7 years old.
- . One of the children is 4 years old but it was not Anshuman.
- . Vaibhav is older than Suprita.
- . One of the children was born in September, but it was not Vaibhav.
- . Suprita's birthday is in April.
- . The youngest child is only 2-year-old.

Based on the above clues, which one of the following statements is true?

- A Vaibhav is the oldest, followed by Anshuman who was born in September, and the youngest is Suprita who was born in April
- B Anshuman is the oldest being born in June, followed by Suprita who is 4-year-old, and the youngest is Vaibhav who is 2-year-old
- C Vaibhav is the oldest being 7-year-old, followed by Suprita who was born in April, and the youngest is Anshuman who was born in September
- D Suprita is the oldest who was born in April, followed by Vaibhav who was born in June, and Anshuman who was born in September

VIEW SOLUTION

Question 50

Six persons are playing a card game sitting around a circular table. Suresh is facing Raghubir who is to the left of Ajay and to the right of Pramod. Ajay is to the left of Dhiraj. Yogendra is to the left of Pramod. If Dhiraj exchanges his seat with Yogendra and Pramod exchanges with Raghubir, who will be sitting to the left of Dhiraj?

- A Yogendra
- B Raghubir

C Suresh

D Ajay

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Question 51

Four students — Ashish, Dhanraj, Felix and Sameer sat for the Common Entrance Exam for Management (CEEM). Each of the students has one profession among the four: Doctor, Chartered Accountant, Engineer, and Economist. One student got admission offers from three NIMs (National Institutes of Management), another from two NIMs, the third from one NIM, while the fourth got none. Below are some of the facts about who got admission offers from how many NIMs and what is their educational background.

- I. The one who is an engineer didn't get as many admissions as Ashish.
- II. The one who got the offer for admissions in two NIMs isn't Dhanraj nor is he a chartered accountant.
- III. Sameer is an economist.
- IV. Dhanraj isn't an engineer and received more admission offers than Ashish.
- V. The doctor got the most number of admission offers.

Which one of the following statements is necessarily true?

- A Ashish is a chartered accountant and got offer for admission in three NIMs.
- B Dhanraj is a doctor and got admission offer in one NIM.
- C Sameer is an economist who got admission offers in two NIMs.
- D Felix who is not an engineer did not get any offer for admission.

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100+ CAT VARC Questions



Question 52

Abraham, Border, Charlie, Dennis and Elmer and their respective wives recently dined together and were seated at a circular table. The seats were so arranged that men and women alternated and each woman was three places distant from her husband. Mrs. Charlie sat to the immediate left of Mr. Abraham. Mrs. Elmer sat two places to the right of Mrs. Border. Who sat to the right of Mr. Abraham?

- A Mrs. Dennis
- B Mrs. Elmer
- C Mrs. Border
- D Mrs. Border or Mrs. Dennis

cracku

Abraham, Border, Charlie, Dennis and Elmer and their respective wives recently dined together and were seated at a circular table. The seats were so arranged that men and women alternated and each woman was three places distant from her husband. Mrs. Charlie sat to the left of Mr. Abraham. Mrs. Elmer sat two places to the right of Mrs. Border. Who sat to the right of Mr. Abraham?

A) Mrs. Dennis
C) Mrs. Border

B) Mrs. Elmer
Mrs. Border or Mrs. Dennis

VIDEO SOLUTION

Question 53

I happened to be the judge in the all India Essay Competition on Nylon Dying, organized some time back by a dyestuff firm. Mill technicians were eligible to enter the competition. My work was simplified in assessing the essays, which had to be done under five heads-Language, Coherence, Subject Matter, Machinery and Recent Developments. Integral marks were to be given out of a maximum of 20 under each head. There were only five entries. The winner got 90 marks. Only a single person can be made the winner. Akhila got 13 in Coherence and Divya 10 in Machinery. Bhanu.s total was less than Akhila.s. Charulata has sent an entry. Ela had got as many marks as Divya. None got 20 under any head. Who was the winner?

- A Divya
- B Charulata
- C Ela
- D Bhanu

VIEW SOLUTION

Question 54

In a 'keep-fit' gymnasium class there are 15 females enrolled in a weight-loss programme. They all have been grouped in any one of the five weight-groups W1, W2, W3, W4, or W5. One instructor is assigned to one weight-group only.

Sonali, Shalini, Shubhra and Shahira belong to the same weight-group.

Sonali and Rupa are in one weight-group, Rupali and Renuka are also in one weight-group.

Rupa, Radha, Renuka, Ruchika, and Ritu belong to different weight groups.

Somya cannot be with Ritu, and Tara cannot be with Radha.

Komal cannot be with Radha, Somya, or Ritu.

Shahira is in W1 and Somya is in W4 with Ruchika.

Sweta and Jyotika cannot be with Rupali, but are in a weight-group with total membership of four.

No weight-group can have more than five or less than one member.

Amita, Babita, Chandrika, Deepika and Elina are instructors of weight-groups with membership sizes 5, 4, 3, 2 and 1 respectively. Who is the instructor of Radha?

- A Babita
- B Elina
- C Chandrika

CAT Expected Questions PDF

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Question 55

There is a vertical stack of books marked 1, 2 and 3 on Table-A, with 1 at the bottom and 3 on top. These are to be placed vertically on Table-B with 1 at the bottom and 2 on the top, by making a series of moves from one table to the other. During a move, the topmost book, or the topmost two books, or all the three, can be moved from one of the tables to the other. If there are any books on the other table, the stack being transferred should be placed, on top of the existing books, without changing the order of books in the stack that is being moved in that move. If there are no books on the other table, the stack is simply placed on the other table without disturbing the order of books in it. What is the minimum number of moves in which the above task can be accomplished?

- A One
- B Two
- C Three
- D Four

Question 56

Five persons with names P, M, U, T and X live separately in any one of the following: a palace, a hut, a fort, a house or a hotel. Each one likes two different colours from among the following: blue, black, red, yellow and green. U likes red and blue. T likes black. The person living in a palace does not like black or blue. P likes blue and red. M likes yellow. X lives in a hotel. M lives in a:

- A hut
- B palace
- C fort
- D house

Question 57

Persons X, Y, Z and Q live in red, green, yellow or blue coloured houses placed in a sequence on a street. Z lives in a yellow house. The green house is adjacent to the blue house. X does not live adjacent to Z. The yellow house is the only house in between the green and red houses. The colour of the house X lives in is:

- A blue
- B green
- C red

D not possible to determine

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CAT Important Topics



Instructions

A, B, C and D are to be seated in a row. But C and D cannot be together. Also B cannot be at the third place.

Question 58

If A and B are together, then which of the following must be necessarily true?

- A C is not at the first place
- B A is at the third place
- C D is at the first place
- D C is at the first place

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Question 59

If A is not at the third place, then which of the following options does C have?

- A The first place only
- B The third place only
- C The first and third place only
- D Any of the places

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Question 60

Which of the following must be false?

- A A is at the first place
- B A is at the second place
- C A is at the third place
- D A is at the fourth place

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CAT WhatsApp Group



Instructions

Direction for the following four questions: Answer the questions based on the following information. Mr Bankatlal acted as a judge for the beauty contest. There were four participants, viz. Ms Andhra Pradesh, Ms Uttar Pradesh, Ms West Bengal and Ms Maharashtra. Mrs Bankatlal, who was very anxious about the result, asked him about it as soon as he was back home. Mr Bankatlal just told that the one who was wearing the yellow saree won the contest. When Mrs Bankatlal pressed for further details, he elaborated as follows: All of them were sitting in a row. All of them wore sarees of different colours, viz. green, yellow, white, red. There was only one runner-up and she was sitting beside Ms. Maharashtra. The runner-up was wearing the green saree. Ms West Bengal was not sitting at the ends and was not the runner up. The winner and the runner-up are not sitting adjacent to each other. Ms Maharashtra was wearing white saree. Ms Andhra Pradesh was not wearing the green saree. Participants wearing yellow saree and white saree were at the ends.

Question 61

Who was the runner-up?

- A Ms Andhra Pradesh
- B Ms West Bengal
- C Ms Uttar Pradesh
- D Ms Maharashtra

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Question 62

Who wore the red saree?

- A Ms Andhra Pradesh
- B Ms West Bengal
- C Ms Uttar Pradesh
- D Ms Maharashtra

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Question 63

Which saree was worn by Ms Andhra Pradesh?

- A Yellow
- B Red
- C Green
- D White

[VIEW SOLUTION](#)



CAT Mock Test

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Question 64

Ms. West Bengal was sitting adjacent to

- A Ms Andhra Pradesh and Ms Maharashtra
- B Ms Uttar Pradesh and Ms Maharashtra
- C Ms Andhra Pradesh and Ms Uttar Pradesh
- D Ms Uttar Pradesh

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Instructions

A round table has seven chairs around it. The chairs are numbered 1 through 7 in a clockwise direction. Four friends, Aslam, Bashir, Chhavi, and Davies, sit on four of the chairs. In the starting position, Aslam and Chhavi are sitting next to each other, while for Bashir as well as Davies, there are empty chairs on either side of the chairs that are sitting on.

The friends take turns moving either clockwise or counterclockwise from their chair. The friend who has to move in a turn occupies the first empty chair in whichever direction (s)he chooses to move. Aslam moves first (Turn 1), followed by Bashir, Chhavi, and Davies (Turns 2, 3, and 4, respectively). Then Aslam moves again followed by Bashir, and Chhavi (Turns 5, 6, and 7, respectively).

The following information is known

1. The four friends occupy adjacent chairs only at the end of Turn 2 and Turn 6.
2. Davies occupies Chair 2 after Turn 1 and Chair 4 after Turn 5, and Chhavi occupies Chair 7 after Turn 2.

Question 65

Which of the chairs are occupied at the end of Turn 6?

- A Chairs numbered 4, 5, 6, and 7
- B Chairs numbered 1, 2, 3, and 4
- C Chairs numbered 2, 3, 4, and 5
- D Chairs numbered 1, 2, 6, and 7

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Question 66

Who sits on the chair numbered 4 at the end of Turn 3?

- A Bashir
- B Chhavi
- C Davies
- D No one

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Question 67

Which of the following BEST describes the friends sitting on chairs adjacent to the one occupied by Bashir at the end of Turn 7?

- A Chhavi only
- B Davies only
- C Chhavi and Davies
- D Aslam and Chhavi

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Question 68

What is the number of the chair initially occupied by Bashir?

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Instructions

Princess, Queen, Rani and Samragini were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancers. The dancers performed their second items in the same sequence of their performance of their first items.

The following additional facts are known.

- i) No composer who assigned item to Princess, assigned any item to Queen.
- ii) No composer who assigned item to Rani, assigned any item to Samragini.
- iii) The first performance was by Princess; this item was assigned by Badal.
- iv) The last performance was by Rani; this item was assigned by Gagan.
- v) The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Question 69

The sixth performance was composed by:

- A Badal
- B Dyu
- C Ashman
- D Gagan

Princess, Queen, Rani and Samragni were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragni.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Finalist		Music Composers	
1	.		
2	.		
3	.		
4	.		
5	.		
6	.		
7	.		
8	.		



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Question 70

Which pair of performances were composed by the same composer?

- The first and the seventh
- The third and the seventh
- The second and the sixth
- The first and the sixth

Princess, Queen, Rani and Samragni were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragni.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Finalist		Music Composers	
1	.		
2	.		
3	.		
4	.		
5	.		
6	.		
7	.		
8	.		



VIDEO SOLUTION

Question 71

Which of the following is true?

- The second performance was composed by Dyu.
- The third performance was composed by Dyu.
- The third performance was composed by Ashman.
- The second performance was composed by Gagan.

cracku

Princess, Queen, Rani and Samragni were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragni.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Handwritten notes on a whiteboard:
 P, A, R, S. A, B, G, D.
 Finalist Music Composers.
 A table with 8 rows and 2 columns. The first column contains numbers 1 through 8. The second column contains dots. The first four rows are grouped under 'Finalist' and the last four rows are grouped under 'Music Composers'.



VIDEO SOLUTION

Question 72

Which of the following is FALSE?

- A Samragni did not perform in any item composed by Ashman.
- B Princess did not perform in any item composed by Dyu.
- C Rani did not perform in any item composed by Badal.
- D Queen did not perform in any item composed by Gagan.

cracku

Princess, Queen, Rani and Samragni were the four finalists in a dance competition. Ashman, Badal, Gagan and Dyu were the four music composers who individually assigned items to the dancers. Each dancer had to individually perform in two dance items assigned by the different composers. The first items performed by the four dancers were all assigned by different music composers. No dancer performed her second item before the performance of the first item by any other dancer. The dancers performed their second items in the same sequence of their performance of their first item.

The following additional facts are known.

- No composer who assigned item to Princess, assigned any item to Queen.
- No composer who assigned item to Rani, assigned any item to Samragni.
- The first performance was by Princess; this item was assigned by Badal.
- The last performance was by Rani; this item was assigned by Gagan.
- The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Handwritten notes on a whiteboard:
 P, A, R, S. A, B, G, D.
 Finalist Music Composers.
 A table with 8 rows and 2 columns. The first column contains numbers 1 through 8. The second column contains dots. The first four rows are grouped under 'Finalist' and the last four rows are grouped under 'Music Composers'.



VIDEO SOLUTION

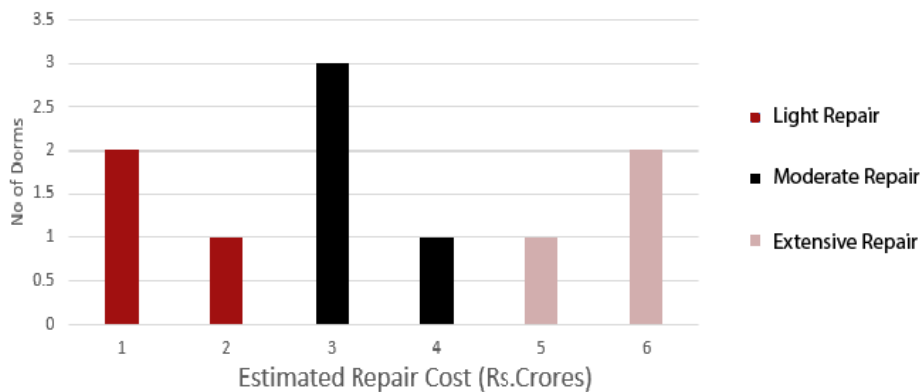


CAT Syllabus PDF



Instructions

At a management school, the oldest M dorms, numbered 1 to 10, need to be repaired urgently. The following diagram represents the estimated repair costs (in Rs. Crores for, the 10 dorms. For any dorm, the estimated repair cost (in Rs. Crores) is an integer. Repairs with estimated cost Rs. 1 or 2 Crores are considered light repairs, repairs with estimated cost Rs. 3 or 4 are considered moderate repairs and repairs with estimated cost Rs. 5 or 6 Crores are considered extensive repairs.



Further, the following information is known.

1. Odd-numbered dorms do not need light repair; even-numbered dorms do not need moderate repair and dorms, whose numbers are divisible by 3, do not need extensive repair.
2. Dorms 4 to 9 all need different repair costs, with Dorm 7 needing the maximum and Dorm 8 needing the minimum.

Question 73

Which of the following is NOT necessarily true?

- A Dorm 1 needs a moderate repair
- B Dorm 5 repair will cost no more than Rs. 4 Crores
- C Dorm 7 needs an extensive repair
- D Dorm 10 repair will cost no more than Rs. 4 Crores

At a management school, the oldest 10 dorms, numbered 1 to 10, need to be repaired urgently. This following diagram represents the estimated repair costs (in Rs. Crores) for the 10 dorms. For any dorm, the estimated repair cost (in Rs. Crores) is an integer. Repairs with estimated cost Rs. 1 or 2 Crores are considered light repairs, repairs with estimated cost Rs. 3 or 4 Crores are considered moderate repairs, repairs with estimated cost Rs. 5 or 6 Crores are considered extensive repairs.

Further, the following information is known.

1. Odd-numbered dorms do not need light repair; even-numbered dorms do not need moderate repair and dorms, whose numbers are divisible by 3, do not need extensive repair.
2. Dorms 4 to 9 all need different repair costs, with Dorm 7 needing the maximum and Dorm 8 needing the minimum.

VIDEO SOLUTION

Question 74

What is the total cost of repairing the odd-numbered dorms (in Rs. Crores)?



VIDEO SOLUTION

Question 75

Suppose further that:

1. 4 of the 10 dorms needing repair are women's dorms and need a total of Rs. 20 Crores for repair.
2. Only one of Dorms 1 to 5 is a women's dorm.

What is the cost for repairing Dorm 9 (in Rs. Crores)?



VIDEO SOLUTION



CAT Formula PDF



Question 76

Suppose further that:

1. 4 of the 10 dorms needing repair are women's dorms and need a total of Rs. 20 Crores for repair.
2. Only one of Dorms 1 to 5 is a women's dorm.

Which of the following is a women's dorm?

- A Dorm 2
- B Dorm 5
- C Dorm 8
- D Dorm 10

At a management school, the oldest M dorms, numbered 1 to 50, need to be repaired urgently. This following diagram represents the estimated repair costs (in Rs. Crores) for the 50 dorms. For any dorm, the estimated repair cost (in Rs. Crores) is an integer. Repairs with estimated cost Rs. 1 or 2 Crores are considered light repairs, repairs with estimated cost Rs. 3 or 4 Crores are considered moderate repairs and repairs with estimated cost Rs. 5 or 6 Crores are considered extensive repairs.

Further, the following information is known:

1. Odd-numbered dorms do not need light repair; even-numbered dorms do not need moderate repair and dorms, whose numbers are divisible by 3, do not need extensive repair.
2. Dorms 4 to 9 all need different repair costs, with Dorm 7 needing the maximum and Dorm 8 needing the minimum.

Handwritten notes on the right side of the screen show calculations for repair costs:

light: $2 - 1$
moderate: $3 - 3$
extensive: $5 - 1$
 $6 - 2$
10

VIDEO SOLUTION

Instructions

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- Tea from Ooty got the highest rating, but it was not in Cup 3.
- The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- Only two cups got ratings in even numbers.
- Cup 2 got the minimum rating and this rating was an even number.
- Tea in Cup 3 got a higher rating than that in Cup 1.
- The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Question 77

What was the second highest rating given?

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- Tea from Ooty got the highest rating, but it was not in Cup 3.
- The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- Only two cups got ratings in even numbers.
- Cup 2 got the minimum rating and this rating was an even number.
- Tea in Cup 3 got a higher rating than that in Cup 1.
- The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-10)
1		
2		
3		
4		
5		
6	Himachal	

VIDEO SOLUTION

Question 78

What was the number of the cup that contained tea from Ooty?

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 50. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- 2. Tea from Ooty got the highest rating, but it was not in Cup 3.
- 3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- 4. Only two cups got ratings in even numbers.
- 5. Cup 2 got the minimum rating and this rating was an even number.
- 6. Tea in Cup 3 got a higher rating than that in Cup 1.
- 7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-50)
1		
2		
3		
4		
5		
6	Himachal	

VIDEO SOLUTION

IIM Call Predictor
Accurate analysis of your profile



Question 79

If the tea from Munnar did not get the minimum rating, what was the rating of the tea from Wayanad?

- A 3
- B 5
- C 1
- D 6

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 50. He gave a unique integer rating to each tea. Some other information is given below:

- Cup 6 contained tea from Himachal.
- 2. Tea from Ooty got the highest rating, but it was not in Cup 3.
- 3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- 4. Only two cups got ratings in even numbers.
- 5. Cup 2 got the minimum rating and this rating was an even number.
- 6. Tea in Cup 3 got a higher rating than that in Cup 1.
- 7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-50)
1		
2		
3		
4		
5		
6	Himachal	

VIDEO SOLUTION

Question 80

If cups containing teas from Wayanad and Ooty had consecutive numbers, which of the following statements may be true?

- A Cup 5 contains tea from Assam
- B Cup 1 contains tea from Darjeeling
- C Tea from Wayanad has got a rating of 6

D Tea from Darjeeling got the minimum rating

A tea taster was assigned to rate teas from six different locations — Munnar, Wayanad, Ooty, Darjeeling, Assam and Himachal. These teas were placed in six cups, numbered 1 to 6, not necessarily in the same order. The tea taster was asked to rate these teas on the strength of their flavour on a scale of 1 to 10. He gave a unique integer rating to each tea. Some other information is given below:

- 1. Cup 6 contained tea from Himachal.
- 2. Tea from Ooty got the highest rating, but it was not in Cup 3.
- 3. The rating of tea in Cup 3 was double the rating of the tea in Cup 5.
- 4. Only two cups got ratings in even numbers.
- 5. Cup 2 got the minimum rating and this rating was an even number.
- 6. Tea in Cup 3 got a higher rating than that in Cup 1.
- 7. The rating of tea from Wayanad was more than the rating of tea from Munnar, but less than that from Assam.

Cup	City/State	Strength (1-10)
1		
2		
3		
4		
5		
6	Himachal	



VIDEO SOLUTION

Instructions

Eight friends: Ajit, Byomkesh, Gargi, Jayanta, Kikira, Manik, Prodosh and Tapesha are going to Delhi from Kolkata by a flight operated by Cheap Air. In the flight, sitting is arranged in 30 rows, numbered 1 to 30, each consisting of 6 seats, marked by letters A to F from left to right, respectively. Seats A to C are to the left of the aisle (the passage running from the front of the aircraft to the back), and seats D to F are to the right of the aisle. Seats A and F are by the windows and referred to as Window seats, C and D are by the aisle and are referred to as Aisle seats while B and E are referred to as Middle seats. Seats marked by consecutive letters are called consecutive seats (or seats next to each other). A seat number is a combination of the row number, followed by the letter indicating the position in the row; e.g., 1A is the left window seat in the first row, while 12E is the right middle seat in the 12th row.

Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as those have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

1. The eight friends were seated in six different rows.
2. They occupied 3 Window seats, 4 Aisle seats and 1 Middle seat.
3. Seven of them had to pay extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
4. Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
5. Gargi was sitting next to Kikira, and Manik was sitting next to Jayanta.
6. Prodosh and Tapesha were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but they paid different amounts for their choices of seat. One of these amounts may be zero.

Question 81

In which row was Manik sitting?

- A 10
- B 11
- C 12
- D 13

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.
 Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

- The eight friends were seated in six different rows.
- Seventeen paid 3 Window seats, 4 Aisle seats and 1 Middle seat.
- Seven of them had to pay extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Ajit and Gargi were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kira, and Manik was sitting next to Jayanta.
- Sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different seat. One of these amounts may be zero.

→ 1 - Rs. 1000
 2 - 300 for window
 500 for aisle

VIDEO SOLUTION

CAT Percentile Predictor



Question 82

How much extra did Gargi pay for her choice of seat?

- A 0
- B Rs. 300
- C Rs. 400
- D Rs. 1000

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.
 Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

- The eight friends were seated in six different rows.
- Seventeen paid 3 Window seats, 4 Aisle seats and 1 Middle seat.
- Seven of them had to pay extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Ajit and Gargi were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kira, and Manik was sitting next to Jayanta.
- Sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different seat. One of these amounts may be zero.

→ 1 - Rs. 1000
 2 - 300 for window
 500 for aisle

VIDEO SOLUTION

Question 83

How much extra did Jayanta pay for his choice of seat?

- A Rs. 300
- B Rs. 400
- C Rs. 500
- D Rs. 1000

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.

Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

- The eight friends were seated in six different rows.
- Seven of them paid extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Ali and Tapesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kikira, and Manik was sitting next to Jayanta.
- Two of them were sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different seat. One of these amounts may be zero.

→ 1 - 300 for window
2 - 500 for aisle

VIDEO SOLUTION

Question 84

Who among the following did not pay any extra amount for his his/her choice of seat?

- A Kikira
- B Manik
- C Gargi
- D Tapesh

	Row	Seat
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		

while 12E is the right middle seat in the 12th row.

Cheap Air charges Rs. 1000 extra for any seats in Rows 1, 12 and 13 as these have extra legroom. For Rows 2-10, it charges Rs. 300 extra for Window seats and Rs. 500 extra for Aisle seats. For Rows 11 and 14 to 20, it charges Rs. 200 extra for Window seats and Rs. 400 extra for Aisle seats. All other seats are available at no extra charge.

The following are known:

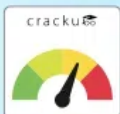
- The eight friends were seated in six different rows.
- Seven of them paid extra amounts, totaling to Rs. 4600, for their choices of seat. One of them did not pay any additional amount for his/her choice of seat.
- Jayanta, Ali and Tapesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat. One of these amounts may be zero.
- Kikira, and Manik was sitting next to Jayanta.
- Two of them were sitting in seats marked by the same letter, in increasing order of row numbers; but they paid different seat. One of these amounts may be zero.

→ 1 - 300 for window
2 - 500 for aisle

VIDEO SOLUTION



CAT Score Calculator



Instructions

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

- The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
- The number of educationalists in the teaching committee is less than the number of educationalists in the

research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.

3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

Question 85

Based on the given information, which of the following statements MUST be FALSE?

- A In the teaching committee the number of educationalists is equal to the number of politicians
- B In the administration committee the number of bureaucrats is equal to the number of educationalists
- C The size of the research committee is less than the size of the teaching committee
- D The size of the research committee is less than the size of the administration committee

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
P				
Total				24

[VIDEO SOLUTION](#)

Question 86

What is the number of bureaucrats in the administration committee?

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
P				
Total				24

[VIDEO SOLUTION](#)

Question 87

What is the number of educationalists in the research committee?

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
Total				24

VIDEO SOLUTION

CAT Crash Course
Live Classes by IIMA alumni

Question 88

Which of the following CANNOT be determined uniquely based on the given information?

- A The size of the teaching committee
- B The size of the research committee
- C The total number of bureaucrats in the three committees
- D The total number of educationalists in the three committees

Twenty four people are part of three committees which are to look at research, teaching, and administration respectively. No two committees have any member in common. No two committees are of the same size. Each committee has three types of people: bureaucrats, educationalists, and politicians, with at least one from each of the three types in each committee. The following facts are also known about the committees:

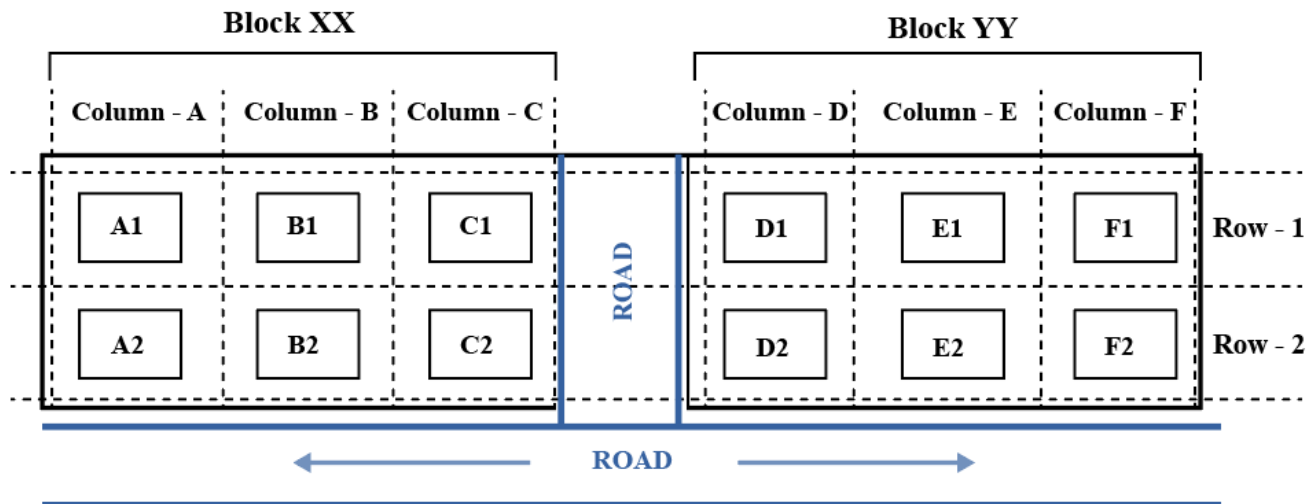
1. The numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee.
2. The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees.
3. 60% of the politicians are in the administration committee, and 20% are in the teaching committee.

	R	T	A	Total
B				
E				
Total				24

VIDEO SOLUTION

Instructions

The schematic diagram below shows 12 rectangular houses in a housing complex. House numbers are mentioned in the rectangles representing the houses. The houses are located in six columns - Column-A through Column-F, and two rows - Row-1 and Row-2. The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.



Some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The road adjacency value of a house is the number of its sides adjacent to a road. For example, the road adjacency values of C2, F2, and B1 are 2, 1, and 0, respectively. The neighbour count of a house is the number of sides of that house adjacent to occupied houses in the same block. For example, E1 and C1 can have the maximum possible neighbour counts of 3 and 2, respectively.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbour count). The following information is also known.

1. The maximum quoted price of a house in Block XX is Rs. 24 lakhs. The minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column-E.
2. Row-1 has two occupied houses, one in each block.
3. Both houses in Column-E are vacant. Each of Column-D and Column-F has at least one occupied house.
4. There is only one house with parking space in Block YY.

Question 89

Which of the following houses is definitely occupied?

- A A1
- B D2
- C B1
- D F2

Handwritten notes and a video solution for Question 89. The notes show calculations for road adjacency and neighbour count for various houses. The video solution shows a person explaining the problem and the calculations.

VIDEO SOLUTION

Question 90

How many houses are vacant in Block XX?

VIDEO SOLUTION

Question 91

Which of the following options best describes the number of vacant houses in Row-2?

- A Exactly 3
- B Either 3 or 4
- C Exactly 2
- D Either 2 or 3

VIDEO SOLUTION

Question 92

What is the maximum possible quoted price (in lakhs of Rs.) for a vacant house in Column-E?

10 12 ←

0 5 10 ← PAV

← No

Row-2. The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.

Block XX			Block YY		
Column-A	Column-B	Column-C	Column-D	Column-E	Column-F
A1	B1	C1	D1	E1	F1
A2	B2	C2	D2	E2	F2

Row-1

Row-2

Block XX

Block YY

Column-A

Column-B

Column-C

Column-D

Column-E

Column-F

Row-1

Row-2

The houses are occupied. The remaining ones are vacant and are the only ones for sale.

Value of a house is the number of its sides adjacent to a road.

Values of C2, F2, and B1 are 2, 1, and 0, respectively.

Count of a house is the number of sides of that house adjacent to a road.

For example, E1 and C1 can have the maximum possible neighbour count of 3 and 2, respectively.

The base price of a house is Rs. 10 lakhs if the house does not have parking space, 12 lakhs if it does. The quoted price (in lakhs of Rupees) is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbour count). The following information is for reference:

- The maximum quoted price of a house in Block XX is Rs. 24 lakhs.
- The minimum quoted price of a house in Block YY is Rs. 15 lakhs, and Column-E.
- Row-1 has two occupied houses, one in each block.
- Both houses in Column-E are vacant.

VIDEO SOLUTION

Question 93

Which house in Block YY has parking space?

- A E1
- B F2
- C E2
- D F1

10 12 ←

0 5 10 ← PAV

← No

Row-2. The houses are divided into two blocks - Block XX and Block YY. The diagram also shows two roads, one passing in front of the houses in Row-2 and another between the two blocks.

Block XX			Block YY		
Column-A	Column-B	Column-C	Column-D	Column-E	Column-F
A1	B1	C1	D1	E1	F1
A2	B2	C2	D2	E2	F2

Row-1

Row-2

Block XX

Block YY

Column-A

Column-B

Column-C

Column-D

Column-E

Column-F

Row-1

Row-2

The houses are occupied. The remaining ones are vacant and are the only ones for sale.

Value of a house is the number of its sides adjacent to a road.

Values of C2, F2, and B1 are 2, 1, and 0, respectively.

Count of a house is the number of sides of that house adjacent to a road.

For example, E1 and C1 can have the maximum possible neighbour count of 3 and 2, respectively.

The base price of a house is Rs. 10 lakhs if the house does not have parking space, 12 lakhs if it does. The quoted price (in lakhs of Rupees) is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbour count). The following information is for reference:

- The maximum quoted price of a house in Block XX is Rs. 24 lakhs.
- The minimum quoted price of a house in Block YY is Rs. 15 lakhs, and Column-E.
- Row-1 has two occupied houses, one in each block.
- Both houses in Column-E are vacant.

VIDEO SOLUTION

CAT Daily Targets

Step by step, heading towards 99+ percentile

cracku

Instructions

Fuel contamination levels at each of 20 petrol pumps P1, P2, ..., P20 were recorded as either high, medium, or low.

- Contamination levels at three pumps among P1 - P5 were recorded as high.
- P6 was the only pump among P1 - P10 where the contamination level was recorded as low.
- P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

4. High contamination levels were not recorded at any of the pumps P16 - P20.
5. The number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded.

Question 94

What best can be said about the number of pumps at which the contamination levels were recorded as medium?

- A At least 8
- B More than 4
- C Exactly 8
- D At most 9

[VIDEO SOLUTION](#)

Question 95

If contamination level at P15 was recorded as medium, then which of the following MUST be FALSE?

- A Contamination levels at P13 and P17 were recorded as the same.
- B Contamination levels at P11 and P16 were recorded as the same.
- C Contamination level at P14 was recorded to be higher than that at P15.
- D Contamination levels at P10 and P14 were recorded as the same.

[VIDEO SOLUTION](#)

Question 96

Which of the following **MUST** be true?

- A The contamination level at P20 was recorded as medium.
- B The contamination level at P13 was recorded as low.
- C The contamination level at P12 was recorded as high.
- D The contamination level at P10 was recorded as high.

Fuel contamination levels at each of 20 petrol pumps P1, P2, ..., P20 were recorded as either high, medium, or low.

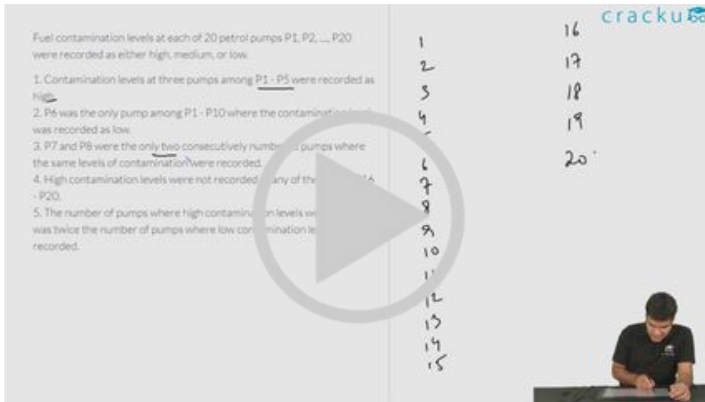
1. Contamination levels at three pumps among P1 - P5 were recorded as high.

2. P6 was the only pump among P1 - P10 where the contamination level was recorded as low.

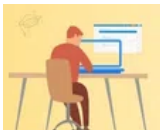
3. P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

4. High contamination levels were not recorded for any of the pumps P14 - P20.

5. The number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded.

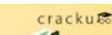


 VIDEO SOLUTION





CAT Sectional Tests



Take Here

Question 97

If the contamination level at P11 was recorded as low, then which of the following **MUST** be true?

- A The contamination level at P12 was recorded as high.
- B The contamination level at P15 was recorded as medium.
- C The contamination level at P18 was recorded as low.
- D The contamination level at P14 was recorded as medium.

Fuel contamination levels at each of 20 petrol pumps P1, P2, ..., P20 were recorded as either high, medium, or low.

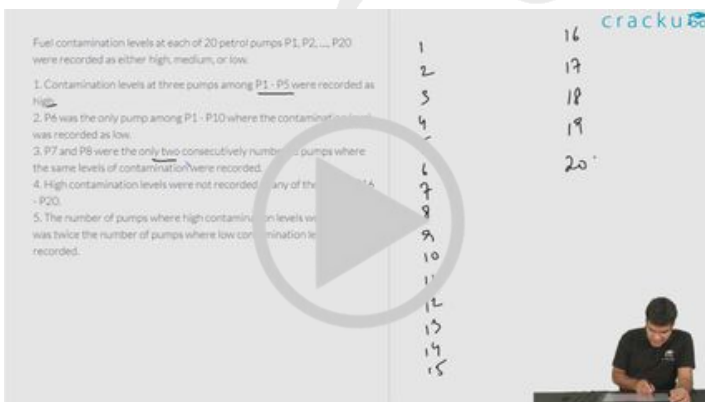
1. Contamination levels at three pumps among P1 - P5 were recorded as high.

2. P6 was the only pump among P1 - P10 where the contamination level was recorded as low.

3. P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

4. High contamination levels were not recorded for any of the pumps P14 - P20.

5. The number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded.



 VIDEO SOLUTION

Instructions

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared consecutively in MT as well as ET.
- Chetan prepared the third question in both MT and ET; and Esha prepared the eighth question in both.
- Fakir prepared the first question of MT and the last one in ET. Dave prepared the last question of MT and the first one in ET.

Question 98

The second question in ET was prepared by:

- A Chetan
- B Beti
- C Esha
- D Dave

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared consecutively in MT as well as ET.
- Chetan prepared the third question in both MT and ET; and Esha prepared the eighth question in both.
- Fakir prepared the first question of MT and the last one in ET. Dave prepared the last question of MT and the first one in ET.

VIDEO SOLUTION

Question 99

How many 5-mark questions were there in MT and ET combined?

- A 13
- B 12
- C 10
- D Cannot be determined

cracku

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared

Handwritten notes:

BTMT
 A B C D E F
 MT
 ET
 5 10 15
 3 3 2



VIDEO SOLUTION

cracku

CAT Study Material

All CAT Free Resources



Question 100

Who prepared 15-mark questions for MT and ET?

- A Only Beti, Dave, Esha and Fakir
- B Only Dave and Fakir
- C Only Esha and Fakir
- D Only Dave, Esha and Fakir

cracku

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared

Handwritten notes:

BTMT
 A B C D E F
 MT
 ET
 5 10 15
 3 3 2



VIDEO SOLUTION

Question 101

Which of the following questions did Beti prepare in ET?

- A Seventh question
- B Fourth question
- C Ninth question
- D Tenth question

cracku

The first year students in a business school are split into six sections. In 2019 the Business Statistics course was taught in these six sections by Annie, Beti, Chetan, Dave, Esha, and Fakir. All six sections had a common midterm (MT) and a common endterm (ET) worth 100 marks each. ET contained more questions than MT. Questions for MT and ET were prepared collectively by the six faculty members. Considering MT and ET together, each faculty member prepared the same number of questions.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks. In both MT and ET, all the 5-mark questions preceded the 10-mark questions, and all the 15-mark questions followed the 10-mark questions.

The following additional facts are known.

- Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks.
- Annie prepared one question for MT. Every other faculty member prepared more than one questions for MT.
- All questions prepared by a faculty member appeared

Handwritten notes on a whiteboard:

Sections: A, B, C, D, E, F

MT: 5, 10, 15

ET: 5, 10, 15

34, 33, 32

VIDEO SOLUTION

Instructions

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

- The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.
- Three persons, including the one who eats four vadas eat without chutney.
- Sandeep does not take any chutney.
- The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.
- Daljit eats idli with chutney and also eats vada.
- Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.
- Bimal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Bimal.

Question 102

Which of the following statements is true?

- Mukesh eats 8 idlis and 4 vadas but no chutney.
- The person who eats 5 idlis and 1 vada does not take chutney.
- The person who eats equal number of vadas and idlis also takes chutney.
- The person who eats 4 idlis and 2 vadas also takes chutney.

cracku

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

- The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.
- Three persons, including the one who eats four vadas eat without chutney.
- Sandeep does not take any chutney.
- The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.
- Daljit eats idli with chutney and also eats vada.
- Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.
- Bimal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Bimal.

Which one of the following statements is true?

A) Daljit eats 5 idlis
C) Bimal eats 1 idli.

B) Ignesh eats 8 idlis
D) Bimal eats 6 idlis.

	Idli	Vadas	Chutney
Ignesh			

Handwritten notes: 4, a, 3a, a=0

VIDEO SOLUTION



Question 103

Which one of the following statements is true?

- A Daljit eats 5 idlis
- B Ignesh eats 8 idlis
- C Bimal eats 1 idli.
- D Bimal eats 6 idlis.

cracku

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

- The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.
- Three persons, including the one who eats four vadas eat without chutney.
- Sandeep does not take any chutney.
- The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.
- Daljit eats idli with chutney and also eats vada.
- Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.
- Bimal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Bimal.

Which one of the following statements is true?

A) Daljit eats 5 idlis
C) Bimal eats 1 idli.

B) Ignesh eats 8 idlis
D) Bimal eats 6 idlis.

	Idli	Vadas	Chutney
Ignesh			

Handwritten notes: 4, a, 3a, a=0

VIDEO SOLUTION

Question 104

Which of the following statements is true?

- A Sandeep eats 2 vadas.
- B Mukesh eats 4 vadas.
- C Ignesh eats 6 vadas.
- D Bimal eats 2 vadas.

cracku

DIRECTIONS for the following three questions: Answer the questions on the basis of the information given below.

Five friends meet every morning at Sree Sagar restaurant for an idli-vada breakfast. Each consumes a different number of idlis and vadas. The number of idlis consumed are 1, 4, 5, 6, and 8, while the number of vadas consumed are 0, 1, 2, 4, and 6. Below are some more facts about who eats what and how much.

i. The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.

ii. Three persons, including the one who eats four vadas eat without chutney.

iii. Sandeep does not take any chutney.

iv. The one who eats one idli a day does not eat any vadas or chutney. Further, he is not Mukesh.

v. Daljit eats idli with chutney and also eats vada.

vi. Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.

vii. Binal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Binal.

Which one of the following statements is true?

A) Daljit eats 5 idlis
B) Ignesh eats 8 idlis
C) Binal eats 1 idli
D) Binal eats 6 idlis

	Idli	Vadas	Chutney
Ignesh			

VIDEO SOLUTION

Instructions

DIRECTIONS for the following two questions: Answer the questions on the basis of the information given below.

Some children were taking free throws at the basketball court in school during lunch break. Below are some facts about how many baskets these children shot.

- Ganesh shot 8 baskets less than Ashish.
- Dhanraj and Ramesh together shot 37 baskets.
- Jugraj shot 8 baskets more than Dhanraj.
- Ashish shot 5 baskets more than Dhanraj.
- Ashish and Ganesh together shot 40 baskets.

Question 105

Which of the following statements is true?

- Ramesh shot 18 baskets and Dhanraj shot 19 baskets
- Ganesh shot 24 baskets and Ashish shot 16 baskets
- Jugraj shot 19 baskets and Dhanraj shot 27 baskets
- Dhanraj shot 11 baskets and Ashish shot 16 baskets

VIEW SOLUTION

100+ CAT Quant Questions



Question 106

Which of the following statements is true?

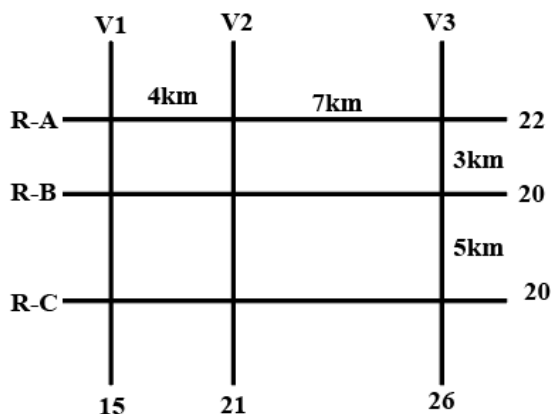
- Dhanraj and Jugraj together shot 46 baskets.
- Ganesh shot 18 baskets and Ramesh shot 21 baskets.
- Dhanraj shot 3 more baskets than Ramesh.
- Ramesh and Jugraj together shot 29 baskets.

Instructions

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km.



Question 107

How many ATMs have cash requirements of Rs. 10 Lakhs or more?

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km.

Question 108

Which of the following statements is correct?

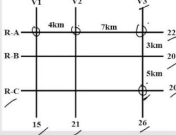
- A There is no ATM placed at the (R-C, V2) intersection.
- B The ATM placed at the (R-C, V2) intersection has a cash requirement of Rs. 9 Lakhs.
- C The ATM placed at the (R-C, V2) intersection has a cash requirement of Rs. 8 Lakhs.

D The cash requirement of the ATM placed at the (R-C, V2) intersection cannot be uniquely determined.

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R and V3 is 12 km.



VIDEO SOLUTION

100+ CAT DILR Questions



Question 109

Which of the following two statements is/are DEFINITELY true?

Statement A: Each of R-A, R-B, and R-C has two ATMs.

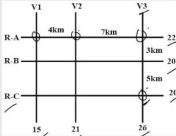
Statement B: Each of V1, V2, and V3 has two ATMs.

- A Only Statement A
- B Neither Statement A nor Statement B
- C Only Statement B
- D Both Statement A and Statement B

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R and V3 is 12 km.



VIDEO SOLUTION

Question 110

What best can be said about the road distance (in km) between the ATMs having the second highest and the second lowest cash requirements?

- A 4 km

- B 7 km
- C 5 km
- D Either 4 km or 7 km

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-B and V3 is 12 km.

VIDEO SOLUTION

Question 111

What is the number of ATMs whose locations and cash requirements can both be uniquely determined?

The figure below shows a network with three parallel roads represented by horizontal lines R-A, R-B, and R-C and another three parallel roads represented by vertical lines V1, V2, and V3. The figure also shows the distance (in km) between two adjacent intersections. Six ATMs are placed at six of the nine road intersections. Each ATM has a distinct integer cash requirement (in Rs. Lakhs), and the numbers at the end of each line in the figure indicate the total cash requirements of all ATMs placed on the corresponding road. For example, the total cash requirement of the ATM(s) placed on road R-A is Rs. 22 Lakhs.

The following additional information is known.

1. The ATMs with the minimum and maximum cash requirements of Rs. 7 Lakhs and Rs. 15 Lakhs are placed on the same road.
2. The road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-B and V3 is 12 km.

VIDEO SOLUTION

100+ CAT VARC Questions



Instructions

A shopping mall has a large basement parking lot with parking slots painted in it along a single row. These slots are quite narrow; a compact car can fit in a single slot but an SUV requires two slots. When a car arrives, the parking attendant guides the car to the first available slot from the beginning of the row into which the car can fit.

For our purpose, cars are numbered according to the order in which they arrive at the lot. For example, the first car to arrive is given a number 1, the second a number 2, and so on. This numbering does not indicate whether a car is a compact or an SUV. The configuration of a parking lot is a sequence of the car numbers in each slot. Each single vacant slot is represented by letter V.

For instance, suppose cars numbered 1 through 5 arrive and park, where cars 1, 3 and 5 are compact cars and 2 and 4 are SUVs. At this point, the parking lot would be described by the sequence 1, 2, 3, 4, 5. If cars 2 and 5 now vacate their slots, the parking lot would now be described as 1, V, V, 3, 4. If a compact car (numbered 6) arrives subsequently followed by an SUV (numbered 7), the parking lot would be described by the sequence 1, 6, V, 3, 4, 7.

Answer the following questions INDEPENDENTLY of each other.

Question 112

Suppose eight cars have arrived, of which two have left. Also suppose that car 4 is a compact and car 7 is an SUV. Which of the following is a POSSIBLE current configuration of the parking lot?

- A 8, 2, 3, V, 6, 5, 7
- B V, 2, 3, 7, 5, 6, 8
- C 8, 2, 3, V, 5, 7, 6
- D 8, 2, 3, V, 5, 6, 7

cracku

A shopping mall has a large basement parking lot with parking slots painted in it along a single row. These slots are quite narrow; a compact car can fit in a single slot but an SUV requires two slots. When a car arrives, the parking attendant guides the car to the first available slot from the beginning of the row into which the car can fit.

For our purpose, cars are numbered according to the order in which they arrive at the lot. For example, the first car to arrive is given number 1, the second a number 2, and so on. This numbering does not indicate whether a car is a compact or an SUV. The configuration of a parking lot is a sequence of the car numbers in each slot. Each single vacant slot is represented by letter V.

For instance, suppose cars numbered 1 through 5 arrive and park, where cars 1, 3 and 5 are compact cars and 2 and 4 are SUVs. At this point, the parking lot would be described by the sequence 1, 2, 3, 4, 5. If cars 2 and 5 now vacate their slots, the parking lot would now be described as 1, V, V, 3, 4. If a compact car (numbered 6) arrives subsequently followed by an SUV (numbered 7), the parking lot would be described by the sequence 1, 6, V, 3, 4, 7.

Answer the following questions INDEPENDENTLY of each other.

Question (1)

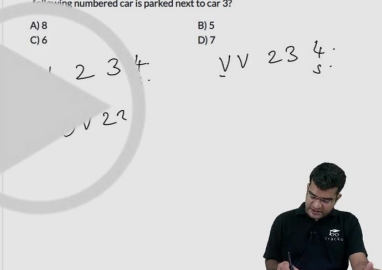
Initially cars numbered 1, 2, 3, and 4 arrive among which 1 and 4 are SUVs while 2 and 3 are compact cars. Car 1 then leaves, followed by the arrivals of car 5 (a compact car) and car 6 (an SUV). Car 4 then leaves. Then car 7 (an SUV) and car 8 (a compact car) arrive. At this moment, which among the following numbered car is parked next to car 3?

A) 8

C) 6

B) 5

D) 7



VIDEO SOLUTION

Question 113

Suppose the sequence at some point of time is 4, 5, 6, V, 3. Which of the following is NOT necessarily true?

- A Car 4 is a compact.
- B Car 1 is an SUV.
- C Car 3 is an SUV
- D Car 5 is a compact.

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A shopping mall has a large basement parking lot with parking slots painted in it along a single row. These slots are quite narrow; a compact car can fit in a single slot but an SUV requires two slots. When a car arrives, the parking attendant guides the car to the first available slot from the beginning of the row into which the car can fit.

For our purpose, cars are numbered according to the order in which they arrive at the lot. For example, the first car to arrive is given number 1, the second a number 2, and so on. This numbering does not indicate whether a car is a compact or an SUV. The configuration of a parking lot is a sequence of the car numbers in each slot. Each single vacant slot is represented by letter 'V'.

For instance, suppose cars numbered 1 through 5 arrive and park, where cars 1, 3 and 5 are compact cars and 2 and 4 are SUVs. At this point, the parking lot would be described by the sequence 1, 2, 3, 4, 5. If cars 2 and 5 now vacate their slots, the parking lot would now be described as 1, V, V, 3, 4. If a compact car (numbered 6) arrives subsequently followed by an SUV (numbered 7), the parking lot would be described by the sequence 1, 6, V, 3, 4, 7.

Answer the following questions INDEPENDENTLY of each other.

Question (1)

Initially cars numbered 1, 2, 3, and 4 arrive among which 1 and 4 are SUVs while 2 and 3 are compact cars. Car 1 then leaves, followed by the arrivals of car 5 (a compact car) and car 6 (an SUV). Car 4 then leaves. Then car 7 (an SUV) and car 8 (a compact car) arrive. At this moment, which among the following numbered car is parked next to car 3?

A) 8
B) 5
C) 6
D) 7

VIDEO SOLUTION

Question 114

Suppose that car 4 is not the first car to leave and that the sequence at a time between the arrival of the car 7 and car 8 is V, 7, 3, 6, 5. Then which of the following statements MUST be false?

- A Car 2 is a compact.
- B Car 7 is a compact.
- C Car 4 is an SUV.
- D Car 6 is a compact.

cracku

A shopping mall has a large basement parking lot with parking slots painted in it along a single row. These slots are quite narrow; a compact car can fit in a single slot but an SUV requires two slots. When a car arrives, the parking attendant guides the car to the first available slot from the beginning of the row into which the car can fit.

For our purpose, cars are numbered according to the order in which they arrive at the lot. For example, the first car to arrive is given number 1, the second a number 2, and so on. This numbering does not indicate whether a car is a compact or an SUV. The configuration of a parking lot is a sequence of the car numbers in each slot. Each single vacant slot is represented by letter 'V'.

For instance, suppose cars numbered 1 through 5 arrive and park, where cars 1, 3 and 5 are compact cars and 2 and 4 are SUVs. At this point, the parking lot would be described by the sequence 1, 2, 3, 4, 5. If cars 2 and 5 now vacate their slots, the parking lot would now be described as 1, V, V, 3, 4. If a compact car (numbered 6) arrives subsequently followed by an SUV (numbered 7), the parking lot would be described by the sequence 1, 6, V, 3, 4, 7.

Answer the following questions INDEPENDENTLY of each other.

Question (1)

Initially cars numbered 1, 2, 3, and 4 arrive among which 1 and 4 are SUVs while 2 and 3 are compact cars. Car 1 then leaves, followed by the arrivals of car 5 (a compact car) and car 6 (an SUV). Car 4 then leaves. Then car 7 (an SUV) and car 8 (a compact car) arrive. At this moment, which among the following numbered car is parked next to car 3?

A) 8
B) 5
C) 6
D) 7

VIDEO SOLUTION

CAT Expected Questions PDF



Question 115

Initially cars numbered 1, 2, 3, and 4 arrive among which 1 and 4 are SUVs while 2 and 3 are compact cars. Car 1 then leaves, followed by the arrivals of car 5 (a compact car) and car 6 (an SUV). Car 4 then leaves. Then car 7 (an SUV) and car 8 (a compact car) arrive. At this moment, which among the following numbered car is parked next to car 3?

- A 8

B 5

C 6

D 7

cracku

A shopping mall has a large basement parking lot with parking slots painted in it along a single row. These slots are quite narrow; a compact car can fit in a single slot but an SUV requires two slots. When a car arrives, the parking attendant guides the car to the first available slot from the beginning of the row into which the car can fit.

For our purpose, cars are numbered according to the order in which they arrive at the lot. For example, the first car to arrive is given number 1, the second a number 2, and so on. This numbering does not indicate whether a car is a compact or an SUV. The configuration of a parking lot is a sequence of the car numbers in each slot. Each single vacant slot is represented by letter 'V'.

For instance, suppose cars numbered 1 through 5 arrive and park, where cars 1, 3 and 5 are compact cars and 2 and 4 are SUVs. At this point, the parking lot would be described by the sequence 1, 2, 3, 4, 5. If cars 2 and 5 now vacate their slots, the parking lot would now be described as 1, V, V, 3, 4. If a compact car (numbered 6) arrives subsequently followed by an SUV (numbered 7), the parking lot would be described by the sequence 1, 6, V, 3, 4, 7.

Answer the following questions INDEPENDENTLY of each other.

Question (1)

Initially cars numbered 1, 2, 3, and 4 arrive among which 1 and 4 are SUVs while 2 and 3 are compact cars. Car 1 then leaves, followed by the arrivals of car 5 (a compact car) and car 6 (an SUV). Car 4 then leaves. Then car 7 (an SUV) and car 8 (a compact car) arrive. At this moment, which among the following numbered car is parked next to car 3?

A) 8
B) 5
C) 6
D) 7

Handwritten notes: 1 2 3 4, V V 2 3 4, V V 2 3 4.



VIDEO SOLUTION

Instructions

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

Question 116

What is the minimum number of Blue beads in any configuration?

cracku

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

G	B			
B				
	B	G	B	
B	G	B	B	



VIDEO SOLUTION

Question 117

Two Red beads have been placed in 'second row, third column' and 'third row, second column'. How many more Red beads can be placed so as to maximise the number of Red beads used in the configuration?

cracku

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

G ₁	B ₁			
B ₁				
	B	G	B	
B	G	B	B	

VIDEO SOLUTION

CAT Important Topics

Question 118

What is the maximum possible number of Red beads that can appear in any configuration?

cracku

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

G ₁	B ₁			
B ₁				
	B	G	B	
B	G	B	B	

VIDEO SOLUTION

Question 119

The total number of possible configurations using beads of only two colours is:

cracku

Twenty five coloured beads are to be arranged in a grid comprising of five rows and five columns. Each cell in the grid must contain exactly one bead. Each bead is coloured either Red, Blue or Green. While arranging the beads along any of the five rows or along any of the five columns, the rules given below are to be followed:

1. Two adjacent beads along the same row or column are always of different colours.
2. There is at least one Green bead between any two Blue beads along the same row or column.
3. There is at least one Blue and at least one Green bead between any two Red beads along the same row or column.

Every unique, complete arrangement of twenty five beads is called a configuration.

G ₁	B ₁			
B ₁				
	B	G	B	
B	G	B	B	

 VIDEO SOLUTION

Instructions

Direction for the following two questions: Answer the questions based on the following information. Amar, Akbar and Anthony are three friends. Only three colours are available for their shirts, viz. red, green and blue. Amar does not wear red shirt. Akbar does not wear green shirt. Anthony does not wear blue shirt.

Question 120

If two of them wear the same colour, then how many of the following must be false?

- I. Amar wears blue and Akbar does not wear green
- II. Amar does not wear blue and Akbar wears blue
- III. Amar does not wear blue and Akbar does not wear blue
- IV. Amar wears green, Akbar does not wear red, Anthony does not wear green

- A None
- B One
- C Two
- D Three

 VIEW SOLUTION



CAT WhatsApp Group

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Question 121

If Akbar and Anthony wear the same colour of shirts, then which of the following is not true?

- A Amar wears blue and Akbar wears green
- B Amar wears green and Akbar wears red
- C Amar wears blue and Akbar does not wear blue
- D Anthony wears red

 VIEW SOLUTION

Answers

1.A	2.B	3.C	4.C	5.C	6.D	7.D	8.B
9.D	10.B	11.A	12.C	13.A	14.A	15.B	16.B
17.E	18.D	19.D	20.C	21.A	22.A	23.C	24.B
25.A	26.D	27.B	28.B	29.C	30.D	31.A	32.D
33.A	34.D	35.D	36.C	37.B	38.C	39.D	40.C
41.C	42.B	43.D	44.C	45.C	46.C	47.B	48.C
49.C	50.C	51.C	52.D	53.B	54.B	55.D	56.B
57.A	58.B	59.C	60.A	61.C	62.B	63.A	64.C
65.A	66.D	67.B	68.4	69.A	70.D	71.A	72.D
73.D	74.19	75.3	76.D	77.7	78.4	79.B	80.B
81.A	82.D	83.C	84.D	85.C	86.4	87.3	88.A
89.C	90.3	91.D	92.21	93.A	94.C	95.B	96.D
97.D	98.D	99.A	100.D	101.D	102.C	103.A	104.C
105.A	106.A	107.3	108.B	109.A	110.D	111.3	112.D
113.C	114.D	115.D	116.6	117.6	118.9	119.2	120.B
121.A							

Explanations

1.A

From the given statements, we can infer that the amounts spent by the women are Rs.2234, Rs.1193, Rs.1340, and Rs.2517. Also, we know that one of the 5 women spent Rs.1378 more than Chellamma. Also, we know that each woman spent at least Rs.1000.

The person who spent Rs.2234 cannot be the person who spent Rs.1378 more than Chellamma. Also, we know that Chellamma spent the least among the five women.

Case 1:

Chellamma spent Rs.1193.

=> One of the 5 women spent $1193 + 1378 = \text{Rs.}2571$.

We know that Shahnaz spent the most. Therefore, Shahnaz should have spent Rs.2571.

Archana didn't spend Rs.2517. Also, Helen spent more than Dhenuka.

Therefore, Helen should have spent Rs.2517.

Dhenuka didn't spend Rs.1340.

Therefore, Dhenuka should have spent Rs. 2234 and Archana should have spent Rs.1340.

However, it has been given that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. According to the given order, Archana arrived before Dhenuka. Therefore, we can eliminate this case.

Case 2:

Rs.2517 is the highest amount spent. Shahnaz spent Rs.2517.

=> Amount spent by Chellamma = $2517 - 1378 = \text{Rs.}1139$

Dhenuka didn't spend Rs.1340. Helen spent more than Dhenuka. Therefore, Dhenuka should have spent

Rs.1193.

Now, we know that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. Helen did not arrive before Dhenuka but Archana did. Therefore, Archana should have spent Rs.2234 and Helen should have spent Rs.1340.

According to given conditions amount spent by everyone is,

ARCHANA	CHELLAMMA	DHENUKA	HELEN	SHAHNAZ
2234	1139	1193	1340	2517
	=2517-1378			

Hence, option A is the right answer.

[VIEW SOLUTION](#)

2. B

From the given statements, we can infer that the amounts spent by the women are Rs.2234, Rs.1193, Rs.1340, and Rs.2517. Also, we know that one of the 5 women spent Rs.1378 more than Chellamma. Also, we know that each woman spent at least Rs.1000.

The person who spent Rs.2234 cannot be the person who spent Rs.1378 more than Chellamma. Also, we know that Chellamma spent the least among the five women.

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=> Amount spent by Chellamma = $2517 - 1378 = \text{Rs.}1139$

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Now, we know that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. Helen did not arrive before Dhenuka but Archana did. Therefore, Archana should have spent Rs.2234 and Helen should have spent Rs.1340.

According to given conditions amount spent by everyone is,

ARCHANA	CHELLAMMA	DHENUKA	HELEN	SHAHNAZ
2234	1139	1193	1340	2517
	=2517-1378			

Hence, option B is the right answer.

[VIEW SOLUTION](#)

3. C

From the given statements, we can infer that the amounts spent by the women are Rs.2234, Rs.1193, Rs.1340, and Rs.2517. Also, we know that one of the 5 women spent Rs.1378 more than Chellamma. Also, we know that each woman spent at least Rs.1000.

The person who spent Rs.2234 cannot be the person who spent Rs.1378 more than Chellamma. Also, we know that Chellamma spent the least among the five women.

Case 1:

Chellamma spent Rs.1193.

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Archana didn't spend Rs.2517. Also, Helen spent more than Dhenuka. Therefore, Helen should have spent Rs.2517.

Dhenuka didn't spend Rs.1340.

Therefore, Dhenuka should have spent Rs. 2234 and Archana should have spent Rs.1340.

However, it has been given that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. According to the given order, Archana arrived before Dhenuka. Therefore, we can eliminate this case.

Case 2:

Rs.2517 is the highest amount spent. Shahnaz spent Rs.2517.

=> Amount spent by Chellamma = $2517 - 1378 = \text{Rs.}1139$

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Now, we know that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. Helen did not arrive before Dhenuka but Archana did. Therefore, Archana should have spent Rs.2234 and Helen should have spent Rs.1340.

According to given conditions amount spent by everyone is,

ARCHANA	CHELLAMMA	DHENUKA	HELEN	SHAHNAZ
2234	1139	1193	1340	2517
	=2517-1378			

Hence, option C is the right answer.

[VIEW SOLUTION](#)

4. C

We are given that A and G sit to the extreme right on the dias, but the order is not mentioned. So, seats 6 and 7 must be A and G in some order.

B sits in the middle of the presentation, occupying seat number 4.

C and D have to be seated as far as possible, and seats 4, 6 and 7 are already occupied. So, C and D must be placed in seats 1 and 5 in some order.

The seats left are 2 and 3, and E and F have to be placed in those seats in some order.

Putting all the information in the table, we get,

LEFT						RIGHT
SEAT-1	SEAT-2	SEAT-3	SEAT-4	SEAT-5	SEAT-6	SEAT-7
C/D	E/F	F/E	B	D/C	G/A	A/G

All the combinations mentioned in options A, B and D are possible from the above configuration. However, the combination mentioned in option C is not possible as E and G cannot be seated on either side of B.

Hence, the correct answer is option C.

[VIEW SOLUTION](#)

5. C

We are given that A and G sit to the extreme right on the dias, but the order is not mentioned. So, seats 6 and 7 must be A and G in some order.

B sits in the middle of the presentation, occupying seat number 4.

C and D have to be seated as far as possible, and seats 4, 6 and 7 are already occupied. So, C and D must be placed in seats 1 and 5 in some order.

The seats left are 2 and 3, and E and F have to be placed in those seats in some order.

Putting all the information in the table, we get,

LEFT						RIGHT
SEAT-1	SEAT-2	SEAT-3	SEAT-4	SEAT-5	SEAT-6	SEAT-7
C/D	E/F	F/E	B	D/C	G/A	A/G

We can see that A, C, D or G can be placed in the corner. But F cannot be seated at either end.

Hence, the correct answer is option C.

[VIEW SOLUTION](#)

A banner for 'CAT Mock Test' featuring two people on the left, the text 'CAT Mock Test' in large blue letters in the center, and a yellow box below it saying 'Mocks designed exactly like the actual CAT'. On the right, there is a small icon of a laptop with a clock and the word 'MOCK' above it, with the 'cracku' logo in the top right corner.

6. D

We are given that A and G sit to the extreme right on the dias, but the order is not mentioned. So, seats 6 and 7 must be A and G in some order.

B sits in the middle of the presentation, occupying seat number 4.

C and D have to be seated as far as possible, and seats 4, 6 and 7 are already occupied. So, C and D must be placed in seats 1 and 5 in some order.

The seats left are 2 and 3, and E and F have to be placed in those seats in some order.

Putting all the information in the table, we get,

LEFT						RIGHT
SEAT-1	SEAT-2	SEAT-3	SEAT-4	SEAT-5	SEAT-6	SEAT-7
C/D	E/F	F/E	B	D/C	G/A	A/G

We can clearly see that E and A cannot be seated together.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)

7. D

B says that if D gets power or telecom then he must get the other one. Option D clearly violates that.

[VIEW SOLUTION](#)

8. **B**

Since C wants either home or finance or none so options A and D are eliminated.

Since F does not want any portfolio if D gets one, Option C is eliminated.

[VIEW SOLUTION](#)

9. **D**

We are given that there are two housewives, one professor, one engineer, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an engineer or an accountant.

C is an accountant, has to be male, and is married to F, who is a professor, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Professor	Housewife	Engineer
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to the housewife. We know that there are only two married couples in the group, and C and F are already one couple in that group. We also know that D is a housewife and married to a lawyer, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and engineer professions. Given that E is not a housewife, E has to be the Engineer and B has to be the housewife. Since the engineer cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Professor	Housewife	Engineer
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

From the table, we can see that A and D are a married couple.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)

10. **B**

We are given that there are two housewives, one professor, one engineer, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an engineer or an accountant.

C is an accountant, has to be male, and is married to F, who is a professor, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Professor	Housewife	Engineer
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to the housewife. We know that there are only two married couples in the group, and C and F are already one couple in that group. We also know that D is a housewife and married to a lawyer, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and engineer professions. Given that E is not a housewife, E has to be the Engineer and B has to be the housewife. Since the engineer cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Professor	Housewife	Engineer
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

We can see that A, C and E are the three males in the group.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)



11. A

We are given that there are two housewives, one professor, one engineer, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an engineer or an accountant.

C is an accountant, has to be male, and is married to F, who is a professor, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Professor	Housewife	Engineer
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to the housewife. We know that there are only two married couples in the group, and C and F are already one couple in that group. We also know that D is a housewife and married to a lawyer, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and engineer professions. Given that E is not a housewife, E has to be the Engineer and B has to be the housewife. Since the engineer cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Professor	Housewife	Engineer
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

We can see that E is an engineer.

Hence, the correct answer is option A.

[VIEW SOLUTION](#)

12. C

For option A, put $n_1 = 1$, then n_2 will be 2, hence equation won't hold true.

Now for option B, if we say $M = 5$, then n_1 will be 0 and n_2 will be 1, hence equation won't hold true.

And equation in option D will not hold true too as put $M = 5$, then n_1 will be 0 and n_2 will be 1.

Now in option C, equation is valid as put $n_1 = 1$, then n_2 will be 2. Which is holding the equation valid too.

Hence our answer will be C.

[VIDEO SOLUTION](#)

13. A

In option A, $n_1 \geq 5$ which is before instruction value so after going 5 backward and 5 forward value of n_2 will remain equal to n_1

So our answer will be A as other options have insufficient information

[VIDEO SOLUTION](#)

14. A

Option B and C are irrelevant and in option D, n_1 should be 0

Hence option A will be our answer as n_1 is the value before instruction as 0 and n_2 is value after instructions which is also 0.

[VIDEO SOLUTION](#)

15. B

Let's consider each statement one by one

A) Suppose $M=50$, $n_1=5$

In this case, $n_2 = 5 + 10 + 10 = 25$. Hence, $n_2 - n_1 = 20$ even though $n_1 \neq 0$. Hence, statement A is false.

B) Suppose $M=21$ and $n_1=1$

In this case, $n_2 = 1 + 10 + 10 = 21$. Hence, $n_2 - n_1 = 20$. Thus, if $M > 20$ then this statement is true.

If $M = 19$

In this case, the process stops when M is reached. Hence, $n_2=19$ and $n_2 - n_1 = 18$. Hence, option B is true.

C) As we have seen earlier, $n_2 - n_1 = 20$ in this case. Hence, option C is false.

D) Suppose $n_1 = M$. In this case, $n_2 = n_1$. Hence, option D is false.

[VIDEO SOLUTION](#)

16. B

Before directly trying to answer the question, it is important to gather all the information given by the question.
There are three houses on each side of the road => Draw 6 lines, 3 in each row, to accommodate P, Q, R, S, T and U.

The houses are of different colours and different heights.

T is tallest and is opposite to red house => Let's number T as 1.

Shortest house is opposite to green house.

U is orange and is between P and S => Two cases arise here. P-U-S is one possibility and the other possibility is S-U-P.

P	U	S
R	Q	T

R is yellow and is opposite to P.

Q is green and is opposite to U. We know that green house is opposite to the shortest house. This implies that U is the shortest house => Number of U is 6.

P is white and is taller than R but shorter than S and Q => Apart from T, S and Q are also taller than P => S and Q can be 2 and 3 in any order => Number of P is 4 and number of R is 5.

Name	Colour	Height-Rank
P	White	4
Q	Green	2 or 3
R	Yellow	5
S	Red	3 or 2
T	Blue	1
U	Orange	6

We know that P is opposite to R and Q is opposite to U => S is opposite to T

It is given that T is opposite to red house => S is the red house and hence T is the blue house.

T is the tallest house and hence the colour of the tallest house is blue.

[VIEW SOLUTION](#)

17. E

Before directly trying to answer the question, it is important to gather all the information given by the question.

There are three houses on each side of the road => Draw 6 lines, 3 in each row, to accommodate P, Q, R, S, T and U.

The houses are of different colours and different heights.

T is tallest and is opposite to red house => Let's number T as 1.

Shortest house is opposite to green house.

U is orange and is between P and S => Two cases arise here. P-U-S is one possibility and the other possibility is S-U-P.

P	U	S
R	Q	T

R is yellow and is opposite to P.

Q is green and is opposite to U. We know that green house is opposite to the shortest house. This implies that U is the shortest house => Number of U is 6.

P is white and is taller than R but shorter than S and Q => Apart from T, S and Q are also taller than P => S and Q can be 2 and 3 in any order => Number of P is 4 and number of R is 5.

Name	Colour	Height-Rank
P	White	4
Q	Green	2 or 3
R	Yellow	5
S	Red	3 or 2
T	Blue	1
U	Orange	6

We only know that the second tallest house is either Q or S. Hence the answer is cannot be determined.

[VIEW SOLUTION](#)

18. D

Before directly trying to answer the question, it is important to gather all the information given by the question.

There are three houses on each side of the road => Draw 6 lines, 3 in each row, to accommodate P, Q, R, S, T and U.

The houses are of different colours and different heights.

T is tallest and is opposite to red house => Let's number T as 1.

Shortest house is opposite to green house.

U is orange and is between P and S => Two cases arise here. P-U-S is one possibility and the other possibility is S-U-P.

P	U	S
R	Q	T

R is yellow and is opposite to P.

Q is green and is opposite to U. We know that green house is opposite to the shortest house. This implies that U is the shortest house => Number of U is 6.

P is white and is taller than R but shorter than S and Q => Apart from T, S and Q are also taller than P => S and Q can be 2 and 3 in any order => Number of P is 4 and number of R is 5.

Name	Colour	Height-Rank
P	White	4
Q	Green	2 or 3
R	Yellow	5
S	Red	3 or 2
T	Blue	1
U	Orange	6

We know that P is opposite to R and Q is opposite to U => S is opposite to T

It is given that T is opposite to red house => S is the red house and hence T is the blue house.

So, we know the colours of all houses and heights of P, R, T and U.

In this question, we are asked to find the house that is opposite to yellow house. R is the yellow house, P is opposite to R and S is on the other corner in P's row. Hence S is the house that is diagonally opposite to yellow house and the colour of S is Red.

[VIEW SOLUTION](#)

19. **D**

We know that $P \cap A = \phi$ means that the pluto is not an alsatian and $P \cup A = D$ means that the pluto along with alsatian makes the set D . hence option D - Both A and C.

[VIEW SOLUTION](#)

20. **C**

We know that all dogs are vertebrates. Hence $D \cap V = D$. Ahead it is given that $Y = F \cap D$ is not a null set which implies that some fish are dogs. Hence , option C.

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21. **A**

It is given that $X = M \cap D$ is such that $X = D$, which means D is a subset of M . Which means all dogs are mammals. Hence , option A.

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22. **A**

From $P \cap D$ is a set containing just pluto ,as pluto is subset of dog D, hence $P \cap D = P$. Also $P \cup M$ would be set of pluto and all mammals together.

[VIEW SOLUTION](#)

23. **C**

$$E = 3Y$$

$$Z = W/2$$

$$Y > Z$$

Using statement A alone, Z and W can be determined but not E

Using statement B, $Y = W$. So, using this statement alone, we can get a relation between all the four ages but cannot determine the absolute values.

Using both the statements, we can determine the value of E.

$$10 = W/2$$

$$W = 20 = Y$$

$$E = 3 \times 20 = 60$$

So, c) is the correct answer.

[VIEW SOLUTION](#)

24. **B**

$$E = 3Y$$

$$Z = W/2$$

$$Y > Z$$

$$\text{So, } Y > W/2 \Rightarrow 2Y > W \Rightarrow 2E/3 > W \Rightarrow E > 3W/2 \Rightarrow E > W$$

So, Elle is older than W

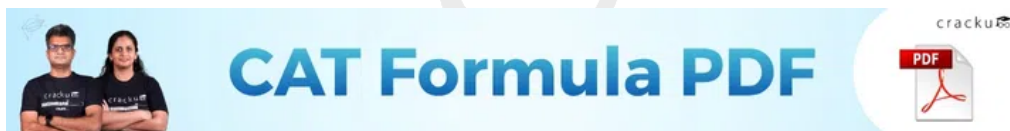
[VIEW SOLUTION](#)

25. **A**

The arrangement of people is as follows: C/D F/E E/F B D/C A/G G/A

So, E and A can never sit together.

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26. **D**

--- B ---

A and G will sit in 6th and 7th position from left in any order.

C and D must be as far apart as possible. Hence, they'll sit in 1st and 5th positions from left in any order.

Hence, only A, C, D and G can sit in the corners.

Among, the given options F cannot sit in the extreme end.

[VIEW SOLUTION](#)

27. **B**

We are given that there are two housewives, one lecturer, one architect, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an architect or an accountant.

C is an accountant, has to be male, and is married to F, who is a lecturer, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Lecturer	Housewife	Architect
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to D, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and architect professions. Given that E is not a housewife, E has to be the Architect and B has to be the housewife. Since the Architect cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Lecturer	Housewife	Architect
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

From the table, we can see that E is an Architect.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

28. B

We are given that there are two housewives, one lecturer, one architect, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an architect or an accountant.

C is an accountant, has to be male, and is married to F, who is a lecturer, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Lecturer	Housewife	Architect
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to D, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and architect professions. Given that E is not a housewife, E has to be the Architect and B has to be the housewife. Since the Architect cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountant	Lecturer	Housewife	Architect
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

From the table, we can see 3 males in the group.

Hence, the correct answer is option B.

29. C

From the table we can say that number of students who opted for E2 after reshuffle = $5 + 34 + 6 + 3 + 5 + 7 + 16 = 76$.

It is given us that the number of students in E2 increased by 30 after the change process. Hence, we can say that the number of students who were enrolled in E2 before reshuffle = $76 - 30 = 46$.

It is given that before the change process there were 10 more students in E2 than in E3. Therefore, the number of students who were enrolled in E3 before reshuffle = $46 - 10 = 36$.

Number of students who moved from E1 to all other electives are known. Therefore, the number of students who were enrolled in E1 before reshuffle = $9 + 5 + 10 + 1 + 4 + 2 = 31$.

It is given that before the change process there were 6 more students in E1 than in E4. Therefore, the number of students who were enrolled in E4 before reshuffle = $31 - 6 = 25$.

Also, it is given that E4 had 2 more students than E6 before reshuffle. Therefore, the number of students who were enrolled in E6 before reshuffle = $25 - 2 = 23$.

All the students from E7 moved to one of electives among E1 to E6.

Therefore, the number of students who were enrolled in E7 before reshuffle = $4 + 16 + 30 + 5 + 5 + 41 = 101$.

Except E5 we know the number of students who were enrolled in all electives. We also know that there were total 300 students who opted for exactly 1 elective.

Hence, the the number of students who were enrolled in E7 before reshuffle = $300 - (46+36+31+25+23+101) = 38$.

For each elective, the number of students who were enrolled before reshuffle will be same as sum of the number of students who moved from that elective to another elective including no movement cases.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2		34	8		2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E2,

Number of students who moved to E1 + 34 + 8 + Number of students who moved to E4 + 2 + 2 = 46

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 0

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25

i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

It is given that after the reshuffle, the number of students in E4 was 3 more than that in E1.

As of now the number of students enrolled in E4 after reshuffle = 1 + 0 + E3 to E4 + 14 + E5 to E4 + 1 + 5 = 21 + {E3 to E4} + {E5 to E4}

Also, the number of students enrolled in E1 after reshuffle = 9 + 0 + 2 + 1 + E5 to E1 + 1 + 4 = 17 + E5 to E1.

Hence, it is possible only when E5 to E1 = 1 and E3 to E4 = E5 to E4 = 0.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0		2	36
	E4	1	3	2	14	1	4	25
	E5	1	5		0	30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

Remaining blank places can be filled easily as we know the total sum of each row.

Therefore, the number of students who moved from E3 to E5 = the number of students who moved from E5 to E3 = the number of students who moved from E5 to E6 = 1.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
Total		18	76	79	21	45	61	300

From the table we can see that the number of students who enrolled for E1 and E4 decreased from 31 and 25 to 18 and 21 respectively.

Therefore, option C is the correct answer.

[VIDEO SOLUTION](#)

30. D

From the table we can say that number of students who opted for E2 after reshuffle = $5 + 34 + 6 + 3 + 5 + 7 + 16 = 76$.

It is given us that the number of students in E2 increased by 30 after the change process. Hence, we can say that the number of students who were enrolled in E2 before reshuffle = $76 - 30 = 46$.

It is given that before the change process there were 10 more students in E2 than in E3. Therefore, the number of students who were enrolled in E3 before reshuffle = $46 - 10 = 36$.

Number of students who moved from E1 to all other electives are known. Therefore, the number of students who were enrolled in E1 before reshuffle = $9 + 5 + 10 + 1 + 4 + 2 = 31$.

It is given that before the change process there were 6 more students in E1 than in E4. Therefore, the number of students who were enrolled in E4 before reshuffle = $31 - 6 = 25$.

Also, it is given that E4 had 2 more students than E6 before reshuffle. Therefore, the number of students who were enrolled in E6 before reshuffle = $25 - 2 = 23$.

All the students from E7 moved to one of electives among E1 to E6.

Therefore, the number of students who were enrolled in E7 before reshuffle = $4 + 16 + 30 + 5 + 5 + 41 = 101$.

Except E5 we know the number of students who were enrolled in all electives. We also know that there were total 300 students who opted for exactly 1 elective.

Hence, the the number of students who were enrolled in E5 before reshuffle = $300 - (46+36+31+25+23+101) = 38$.

For each elective, the number of students who were enrolled before reshuffle will be same as sum of the number of students who moved from that elective to another elective including no movement cases.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2		34	8		2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E2,

Number of students who moved to E1 + 34 + 8 + Number of students who moved to E4 + 2 + 2 = 46

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 0

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25

i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

It is given that after the reshuffle, the number of students in E4 was 3 more than that in E1.

As of now the number of students enrolled in E4 after reshuffle = 1 + 0 + E3 to E4 + 14 + E5 to E4 + 1 + 5 = 21 + {E3 to E4} + {E5 to E4}

Also, the number of students enrolled in E1 after reshuffle = 9 + 0 + 2 + 1 + E5 to E1 + 1 + 4 = 17 + E5 to E1.

Hence, it is possible only when E5 to E1 = 1 and E3 to E4 = E5 to E4 = 0.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0		2	36
	E4	1	3	2	14	1	4	25
	E5	1	5		0	30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

Remaining blank places can be filled easily as we know the total sum of each row.

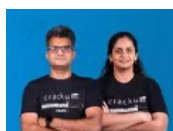
Therefore, the number of students who moved from E3 to E5 = the number of students who moved from E5 to E3 = the number of students who moved from E5 to E6 = 1.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
Total		18	76	79	21	45	61	300

Form the table, we can see that after the reshuffle the number of students in electives E1 to E6 are 18, 76, 79, 21, 45 and 61 in that order.

Therefore, option D is the correct answer.

 VIDEO SOLUTION



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31. A

From the table we can say that number of students who opted for E2 after reshuffle = $5 + 34 + 6 + 3 + 5 + 7 + 16 = 76$.

It is given us that the number of students in E2 increased by 30 after the change process. Hence, we can say that the number of students who were enrolled in E2 before reshuffle = $76 - 30 = 46$.

It is given that before the change process there were 10 more students in E2 than in E3. Therefore, the number of students who were enrolled in E3 before reshuffle = $46 - 10 = 36$.

Number of students who moved from E1 to all other electives are known. Therefore, the number of students who were enrolled in E1 before reshuffle = $9 + 5 + 10 + 1 + 4 + 2 = 31$.

It is given that before the change process there were 6 more students in E1 than in E4. Therefore, the number of students who were enrolled in E4 before reshuffle = $31 - 6 = 25$.

Also, it is given that E4 had 2 more students than E6 before reshuffle. Therefore, the number of students who were enrolled in E6 before reshuffle = $25 - 2 = 23$.

All the students from E7 moved to one of electives among E1 to E6.

Therefore, the number of students who were enrolled in E7 before reshuffle = $4 + 16 + 30 + 5 + 5 + 41 = 101$.

Except E5 we know the number of students who were enrolled in all electives. We also know that there were total 300 students who opted for exactly 1 elective.

Hence, the the number of students who were enrolled in E7 before reshuffle = $300 - (46+36+31+25+23+101) = 38$.

For each elective, the number of students who were enrolled before reshuffle will be same as sum of the number of students who moved from that elective to another elective including no movement cases.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2		34	8		2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E2,

Number of students who moved to E1 + 34 + 8 + Number of students who moved to E4 + 2 + 2 = 46

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 0

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25

i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25

i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

It is given that after the reshuffle, the number of students in E4 was 3 more than that in E1.

As of now the number of students enrolled in E4 after reshuffle = 1 + 0 + E3 to E4 + 14 + E5 to E4 + 1 + 5 = 21 + {E3 to E4} + {E5 to E4}

Also, the number of students enrolled in E1 after reshuffle = 9 + 0 + 2 + 1 + E5 to E1 + 1 + 4 = 17 + E5 to E1.

Hence, it is possible only when E5 to E1 = 1 and E3 to E4 = E5 to E4 = 0.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0		2	36
	E4	1	3	2	14	1	4	25
	E5	1	5		0	30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

Remaining blank places can be filled easily as we know the total sum of each row.

Therefore, the number of students who moved from E3 to E5 = the number of students who moved from E5 to E3 = the number of students who moved from E5 to E6 = 1.

		To Elective						Total
		E1	E2	E3	E4	E5	E6	
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
Total		18	76	79	21	45	61	300

We are asked the largest change in its enrollment as a percentage of its original enrollment for all 6 electives but as we can see there are only 4 electives. Hence, we will check only for E1, E2, E3 and E6.

The percentage change in the number of students for E1 = $\frac{18 - 31}{31} \times 100 \approx 42\%$

The percentage change in the number of students for E2 = $\frac{76 - 46}{46} \times 100 \approx 65\%$

The percentage change in the number of students for E3 = $\frac{79 - 36}{36} \times 100 \approx 119\%$

The percentage change in the number of students for E6 = $\frac{61 - 23}{23} \times 100 \approx 165\%$

We can see that the percent change in the number of student for E6 is the largest. Therefore, option D is the correct answer.

VIDEO SOLUTION

33. A

The total number of biscuits = 5, the total number of candies = 3 and the total number of savouries = 12 - (3+5) = 4

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D, E, F and K are savouries and placed at 13, 14, 15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C,H are candies and I,J,L are biscuits. It is given that A, B are placed consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I,J,L,A,B are biscuits and C,H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	L	A	B	I/J	J/I	---	---	C	H/G	G/H	---	D	E	F	K
Item Type	---	B	B	B	B	B	---	---	C	C	C	---	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	---	C	H/G	G/H	---	L	A	B	I/J	J/I	---	D	E	F	K
Item Type	---	---	C	C	C	---	B	B	B	B	B	---	S	S	S	S

The number of arrangements for the first case = $2 \times 2 = 4$

The number of arrangements for the second case = $2 \times 2 = 4$

The total number of arrangements = $4 + 4 = 8$

 VIDEO SOLUTION

34. D

The total number of biscuits = 5, the total number of candies = 3 and the total number of savouries = $12 - (3 + 5) = 4$

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D,E,F and K are savouries and placed at 13,14,15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C,H are candies and I,J,L are biscuits. It is given that A, B are placed consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I,J,L,A,B are biscuits and C,H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	L	A	B	I/J	J/I	---	---	C	H/G	G/H	---	D	E	F	K
Item Type	---	B	B	B	B	B	---	---	C	C	C	---	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	---	C	H/G	G/H	---	L	A	B	I/J	J/I	---	D	E	F	K
Item Type	---	---	C	C	C	---	B	B	B	B	B	---	S	S	S	S

Option A and C are wrong as candies can come before biscuits and vice versa. B is not necessarily true as there can be one empty shelf too as shown in the table. Option D is true as there are at least 4 shelves between B and C. Hence D is the answer.

[VIDEO SOLUTION](#)

35. D

The total number of biscuits = 5, the total number of candies = 3 and the total number of savouries = $12 - (3 + 5) = 4$

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D, E, F and K are savouries and placed at 13, 14, 15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C, H are candies and I, J, L are biscuits. It is given that A, B are placed consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I, J, L, A, B are biscuits and C, H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	L	A	B	I/J	J/I	---	---	C	H/G	G/H	---	D	E	F	K
Item Type	---	B	B	B	B	B	---	---	C	C	C	---	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	---	C	H/G	G/H	---	L	A	B	I/J	J/I	---	D	E	F	K
Item Type	---	---	C	C	C	---	B	B	B	B	B	---	S	S	S	S

G is a candy. Hence D is the answer.

[VIDEO SOLUTION](#)

CAT Percentile Predictor



36. C

The total number of biscuits = 5, the total number of candies = 3 and the total number of savouries = $12 - (3 + 5) = 4$

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D, E, F and K are savouries and placed at 13, 14, 15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C, H are candies and I, J, L are biscuits. It is given that A, B are placed consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I, J, L, A, B are biscuits and C, H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	L	A	B	I/J	J/I	---	---	C	H/G	G/H	---	D	E	F	K
Item Type	---	B	B	B	B	B	---	---	C	C	C	---	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	---	---	C	H/G	G/H	---	L	A	B	I/J	J/I	---	D	E	F	K
Item Type	---	---	C	C	C	---	B	B	B	B	B	---	S	S	S	S

From the table(case 2), only 1, 2, 6 and 12 are empty in the same arrangement. Hence, C is the answer.

[VIDEO SOLUTION](#)

37. B

After first instruction: $A = 5 - 2 = 3L$. $C = 2L$. In the third instruction, A yields 2L to C. It means the capacity of A before third operation was 3L. It means that C's water was either drained or transferred to B. Option B is correct.

[VIEW SOLUTION](#)

38. C

We know that after 3 operations A contain 1 ltr and 4 th operation is Drain(a) , so 1 ltr is drained and only 4 ltrs remain in all there cylinders. Also after 6 operations A should have all the 4 ltrs in it. hence the other 2 vessels must contain 0 ltrs because there is only one drain operation in the entire sequence.

[VIEW SOLUTION](#)

39. D

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

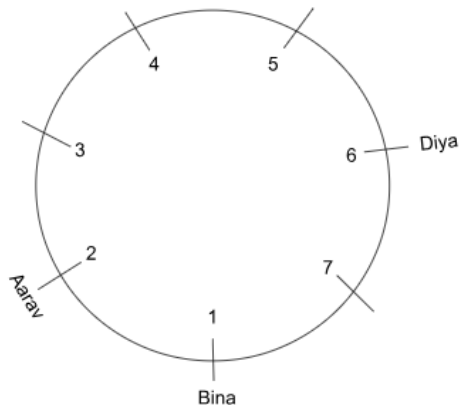
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

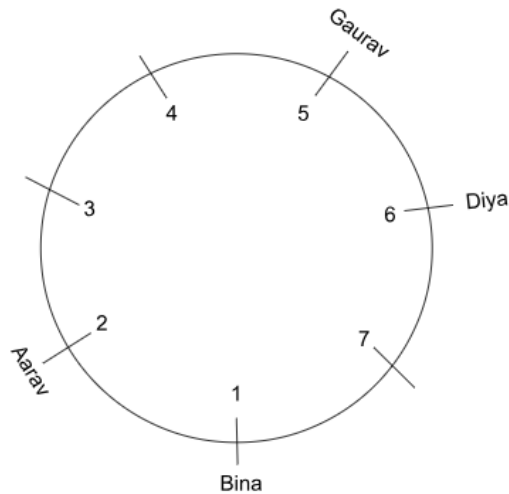
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



Round 8:

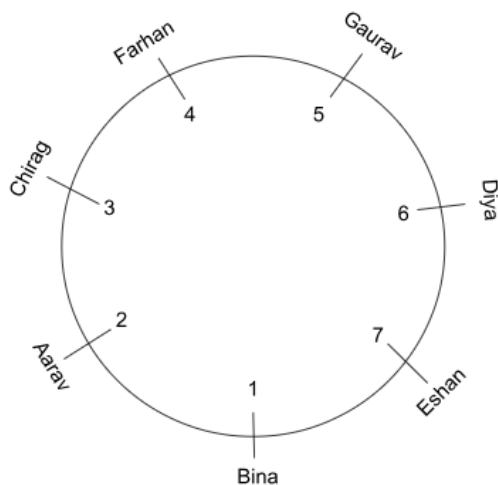
The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.

If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



In rounds 4 and 5, the possibilities were

- 1) Passing the buck second to the right in round 4 and second to the right in round 5. Here, the buck would have gone to Farhan in round 4.
- 2) Passing the buck second to the left in round 4 and to the left in round 5. Here, the buck would have gone to Bina in round 4.
- 3) Passing the buck to the left in round 4 and second to the left in round 5. Here, the buck would have gone to Eshan in round 4.

In round 4, the buck might have gone to either Farhan, Bina or Eshan, which is not known to us. Therefore, the count of the number of times the buck was received by them cannot be determined uniquely, whereas it can be uniquely determined in the case of Gaurav, which is 1 in round 7.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)

40. C

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

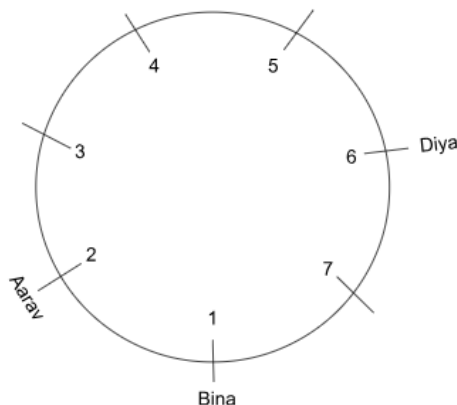
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

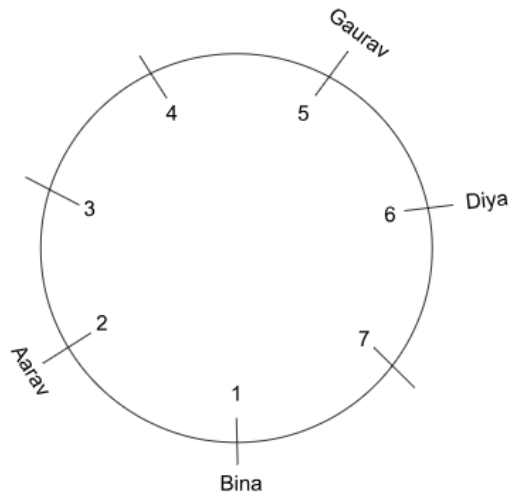
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



Round 8:

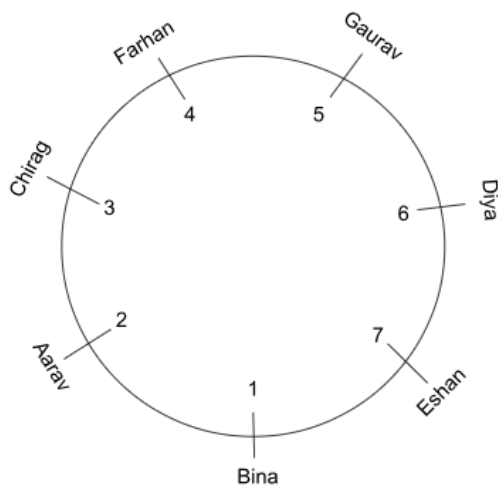
The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.

If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.

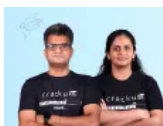


This is the final arrangement.

The person third to the left of Eshan is Chirag.

Hence, the correct answer is option C.

[VIEW SOLUTION](#)



CAT Score Calculator



41.C

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

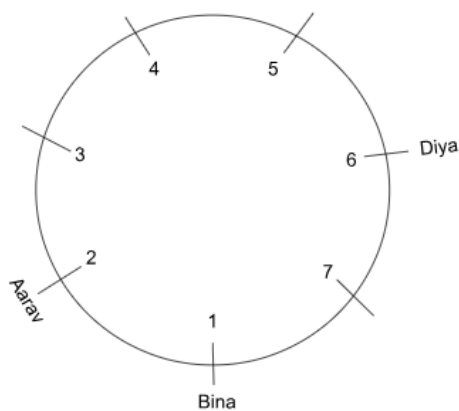
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

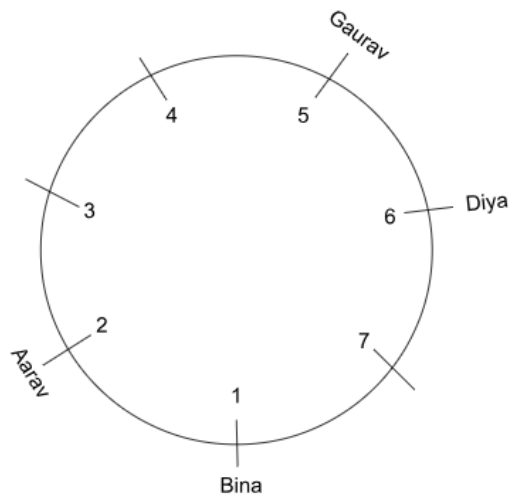
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



Round 8:

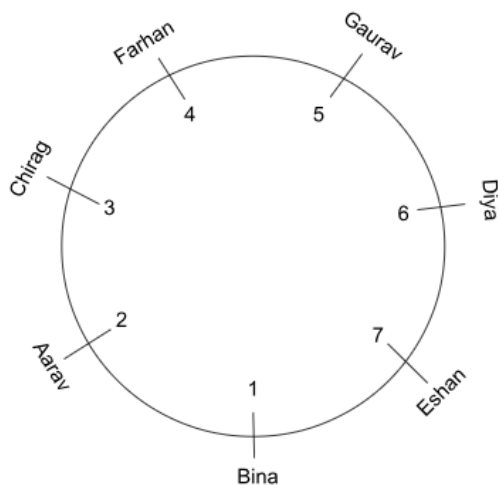
The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.

If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



In rounds 4 and 5, the possibilities were

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

Here, Buck could have been passed second to the right, second to the left or immediately to the left in rounds 4 and 5, which is unknown to us. The only pass type that we are certain to have happened is immediately to the right.

Hence, the correct answer is option C.

42. B

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

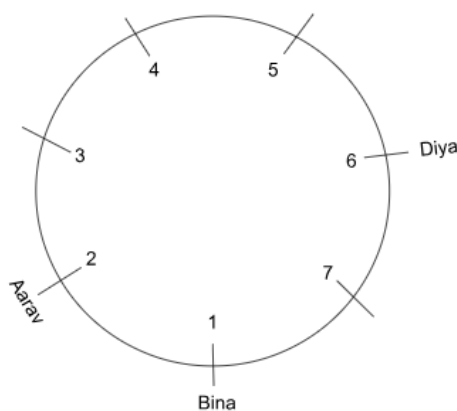
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

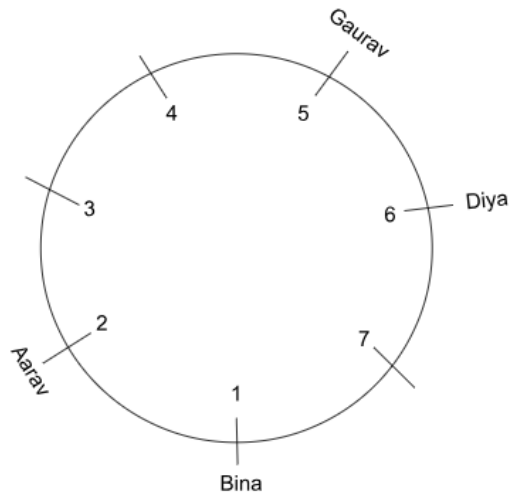
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



Round 8:

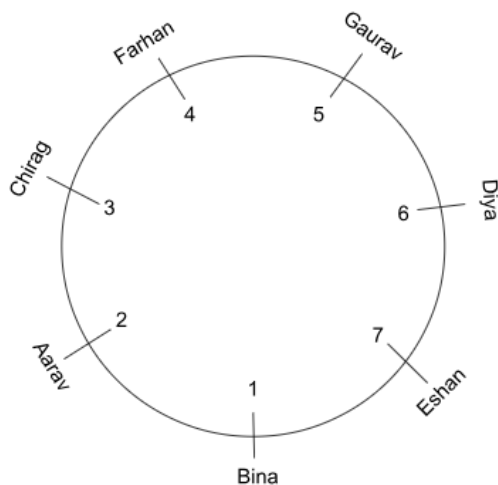
The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.

If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7 and the only person left to be assigned is Eshan making the person at position 7 to be Eshan.



This is the final arrangement.

The person immediately to the right of Bina is Eshan.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

43. D

According to given condition we know that Alam started with 40 spent 35 and saved 5, Daljeet started with 20, ganesh started with 20 and saved 16.5, Jugraj started with 10 and spent less than 3 rs, sandeep started with 30 and spent more than 1.5 rs. So among the options only option D satisfies given requirements.

[VIEW SOLUTION](#)

44. C

According to given conditions , following time table is possible:

SS	VA	SK	JKG	JKR	RAS
9-10	10-11	11-11:30	11:30-12:30	1:30-2:00	2:00

hence option C.

[VIEW SOLUTION](#)

45. C

We can get the following from the information provided.

	Student	City	School
1	Riaz	Hyderabad	Little Flower
2	Balbir	Bangalore	Loyola
3	Bipin	Pune	Convent

Hence, Option C is right.

[VIEW SOLUTION](#)

46. C

According to given conditions we have -

Sharmas	Banerjees	Pattabhiraman
12 noon	2:00 PM	1:00 PM
Sambar	makkey ki roti	fried brinjal
White chinaware	red chinaware	blue chinaware

[VIEW SOLUTION](#)

47. B

According to given question $\text{Jerome} + \text{Tommy} = \text{Albert} + \text{David}$ eq.(1) ; $\text{Jerome} < \text{Tommy}$ eq. (3)
 $\text{Albert} + \text{Tommy} < \text{David} + \text{jerome}$ eq. (2)

Now adding above two equations we will get

$2\text{David} > 2\text{Tommy}$

or $\text{David} > \text{Tommy}$

or $\text{Tommy} > \text{Jerome} > \text{Albert}$ (from eq.(3))

Now David should earn most pocket money as to hold equation (2) to be true.

Hence $\text{David} > \text{Tommy} > \text{Jerome} > \text{Albert}$.

[VIEW SOLUTION](#)

48. C

It is given in the statements, that C cannot travel with A or B. E cannot travel with B or F. G cannot travel with F, or H, or D. By formulating the table we get

	A	B	C	D	E	F	G	H
A	×		×					
B		×	×		×			
C	×	×	×					
D				×			×	
E		×			×	×		
F					×	×	×	
G				×		×	×	×
H							×	×

Q is with A and G is with Q => G and Q are travelling together on motorcycle M3

F travels on M1 and E travels on M2 motorcycles.

D is travelling with P on M2 => D and E are traveling together on M2

B cannot go with R => F and B go together on M1

Therefore, C and H go together on M4

So, the table can be formed as below :

	A	B	C	D	E	F	G	H
A	×	×	×	×	×	×	✓	×
B	×	×	×	×	×	✓	×	×
C	×	×	×	×	×	×	×	✓
D	×	×	×	×	✓	×	×	×
E	×	×	×	✓	×	×	×	×
F	×	✓	×	×	×	×	×	×
G	✓	×	×	×	×	×	×	×
H	×	×	✓	×	×	×	×	×

Basket	Motorcycle		
O	M1	F	B
P	M2	E	D
Q	M3	A	G
R	M4	C	H

Person	A	B	C	D	E	F	G	H
Basket	Q	O	R	P	P	O	Q	R
Motorcycle	M3	M1	M4	M2	M2	M1	M3	M4

Hence, C would be travelling with H.

[VIDEO SOLUTION](#)

49. C

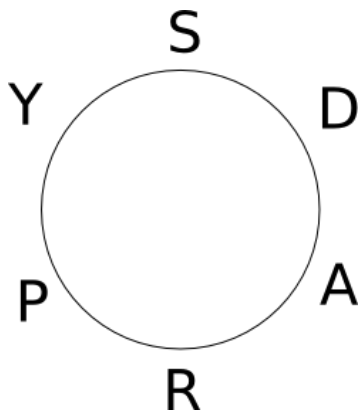
According to given conditions ,we get

Vaibhav	Suprita	Anshuman
June	April	September
7 yrs	4 yrs	2 yrs

[VIEW SOLUTION](#)

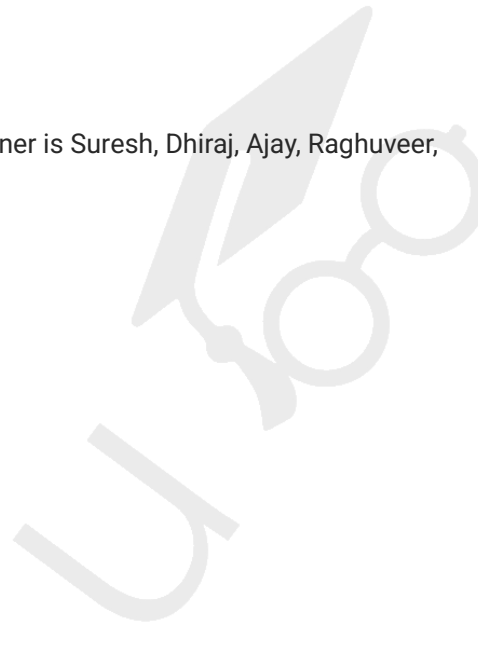
50. C

The correct original circular arrangement sequence in clockwise manner is Suresh, Dhiraj, Ajay, Raghuveer, Pramod, Yogendra.



So after the changes, Suresh is to the left of Dhiraj.

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51. C

From Statement I, we know that Ashish is not an engineer, and from Statement IV Dhanraj is not an engineer either. Since Sameer is an Economist, Felix must be the engineer.

Name	Ashish	Dhanraj	Felix	Sameer
Profession			Engineer	Economist

From Statement IV, Dhanraj got more offers than Ashish. Since Doctor got the most number of offers, Ashish is not the doctor:

Name	Ashish	Dhanraj	Felix	Sameer
Profession	CA	Doctor	Engineer	Economist

We know that the doctor got offers from 3 NIMs and Ashish got more offers than the engineer. Since Ashish is a CA and did not get 2 offers, he can only get 1 offer and the engineer gets none. The final table will look like this:

Name	Ashish	Dhanraj	Felix	Sameer
Profession	CA	Doctor	Engineer	Economist
No. of offers	1	3	0	2

Here, we can see that Sameer is an economist with offers from 2 NIMs. Hence only Option C is correct.

[VIEW SOLUTION](#)

52. D

Mrs. Abraham can't be sitting next to him as per the seating arrangement. Wives sit three places away from their husbands.

Mrs. Charlie is sitting to the left of Mr. Abraham. So, she can't be sitting to his right.

Mrs. Elenor is sitting two places to the right of Mrs. Border (and not Mrs. Charlie). So, she can't be sitting right next to Mr. Abraham.

Mrs. Border and Mrs. Dennis are the remaining two wives and each is equally likely to be sitting to the right of Mr. Abraham.

[VIDEO SOLUTION](#)

53. B

Names	Language	Coherence	Subject-Matter	Machinary	Recent-develop.
Akhila		13			
Bhanu					
Charulata					
Divya				10	
Ela					

Given informations are as follows:

Akhila got 13 marks in coherence and divya got 10 marks in Machinary

Divya's marks = Ela's Marks (Hence they both are not winners as winner is only one)

Bhanu's marks < Akhila's marks (Hence bhanu is also not a winner)

So winner will be either Akhila or Charulata

As Akhila got 13 marks in coherence so she must have 20 marks in some head as highest marks are 90

That's why winner will be Charulata.

[VIEW SOLUTION](#)

54. B

According to given conditions, the groups are as follows:

Case 1:

W1 : Sonali, Shalini, Shubhra ,Shahira and Rupa

W4: Somya , Ruchika, Jyotika, sweta

: Renuka, Rupali, Komal

: Ritu and Tara

: Radha

Case 2:

W1 : Sonali, Shalini, Shubhra ,Shahira and Rupa

W4: Somya , Ruchika

: Renuka, Rupali, Komal

: Ritu, Tara, Jyotika, sweta

: Radha

Hence, Radha must be alone in a group . Therefore, her trainer is Elina.

[VIEW SOLUTION](#)

55. D

For min steps 1st move would be to place book 3 from A to B
2nd move would be to place book 2 from A to B,
3rd move would be to place book 2 and 3 from B to A.
Last move would be to place books 2,3,1 from table A to Table B.
So 4 moves are atleast needed.

[VIEW SOLUTION](#)

A banner for 'CAT Daily Targets' featuring two people on the left, the text 'CAT Daily Targets' in large bold letters in the center, and a 'cracku' logo with a calendar icon on the right. Below the main text is a tagline: 'Step by step, heading towards 99+ percentile'.

56. **B**

According to given condition we know that P likes Blue and red, M likes yellow, U likes Red and blue, T likes Black and X lives in a hotel. Since the person in palace doesn't like blue or black. Only 1 such person is possible i.e. M

[VIEW SOLUTION](#)

57. **A**

According to given condition the correct sequence of houses is 1st is blue, 2nd green, 3rd is yellow and last red. Now in the yellow house i.e. 3rd Z lives and X doesn't live as a neighbour i.e. Green and Red house. So, X lives in blue house.

[VIEW SOLUTION](#)

58. **B**

If A and B are together, then it can't be at first and second place as C and D will be together which is not possible. Now if A can be at third place and B is at second place then C and D can be placed accordingly. Answer will be B).

[VIEW SOLUTION](#)

59. **C**

If A is not at the third place, then it has only two choices either 2nd or 4th position. So if A is at 2nd position, then B has only fourth position as choice and C can have either first or third position. And if A is at 4th position, then B has only 2nd position and again C can have either first or third position. Hence answer will be C.

[VIEW SOLUTION](#)

60. **A**

Among all positions for A, if it has first position then either B will be at the 3rd position or C and D will be together which is not possible. Hence answer will be A).

[VIEW SOLUTION](#)

A banner for 'CAT Sectional Tests' with a 'FREE' tag above the text. On the left is an illustration of a person at a desk, and on the right is the 'cracku' logo with a 'Take Here' button.

61. C

Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

[VIEW SOLUTION](#)

62. B

Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

[VIEW SOLUTION](#)

63. A

Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

[VIEW SOLUTION](#)

64. C

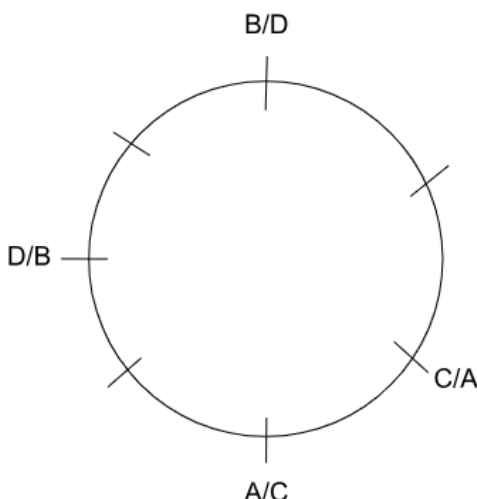
Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

[VIEW SOLUTION](#)

65. A

Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,

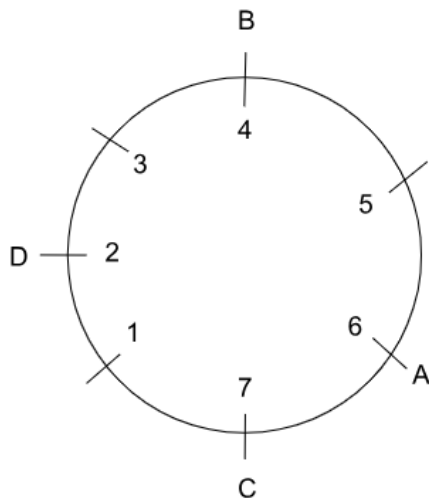


This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies'

initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6.

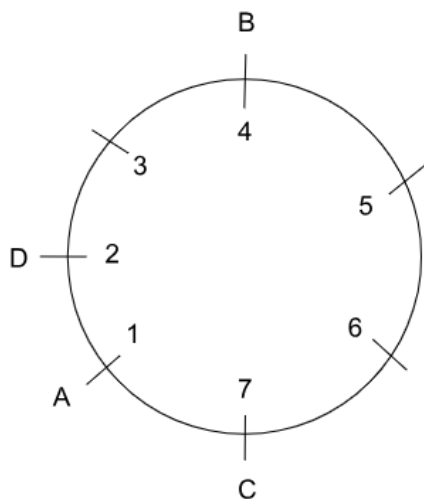
We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

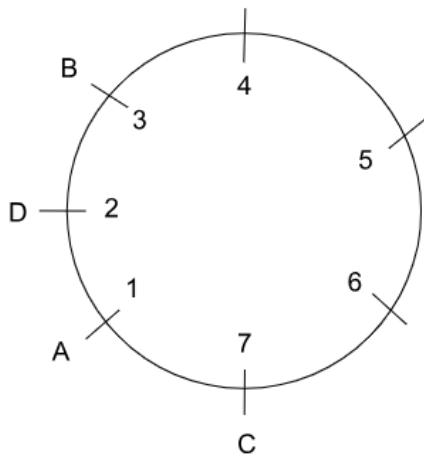
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



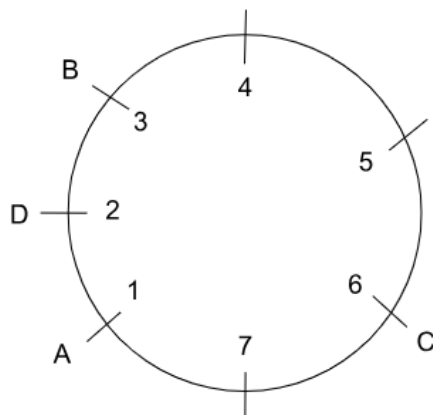
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



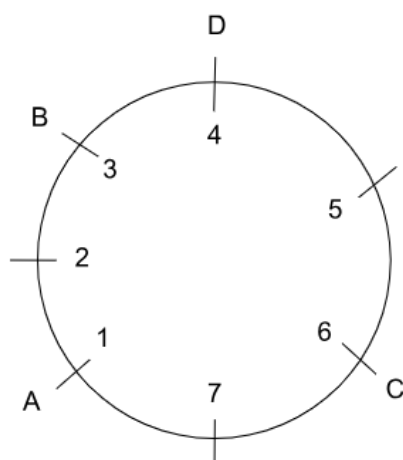
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

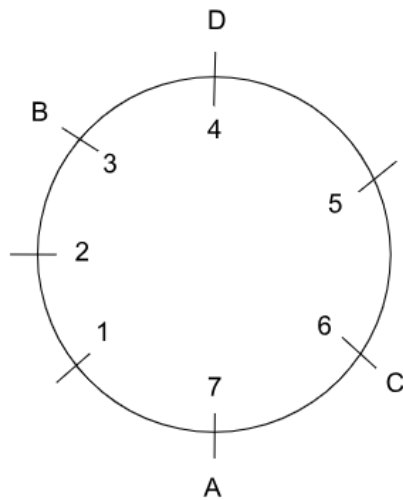
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

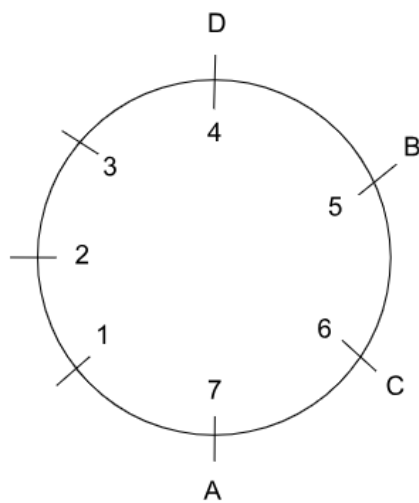
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



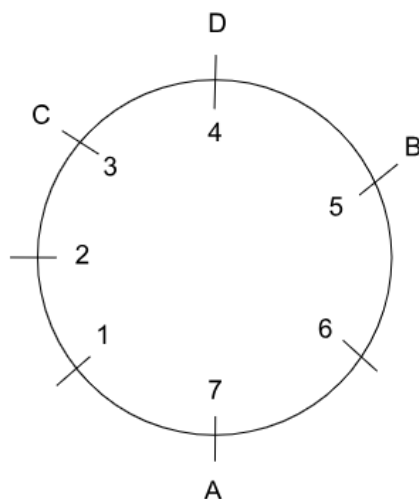
Turn 6: Bashir moves

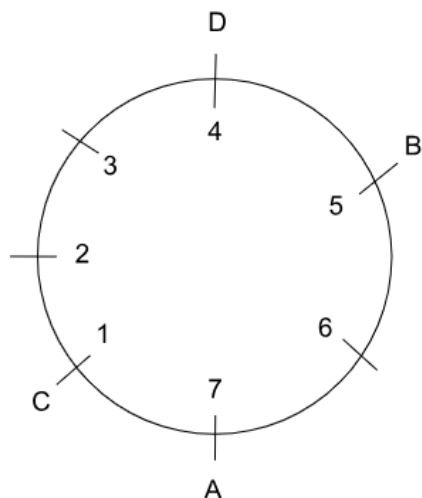
Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.





The chairs occupied after turn 6 are 4, 5, 6 and 7.

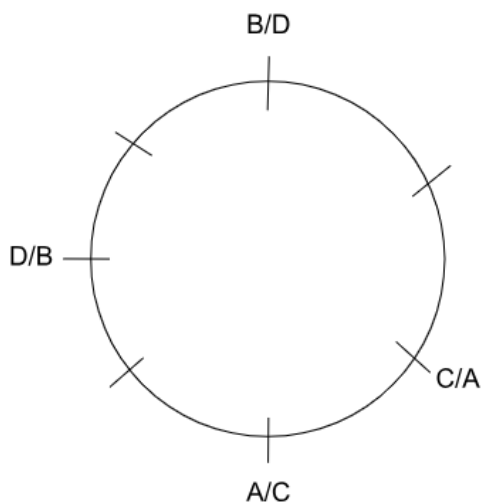
Hence, the correct answer is option A.

[VIEW SOLUTION](#)



66. D

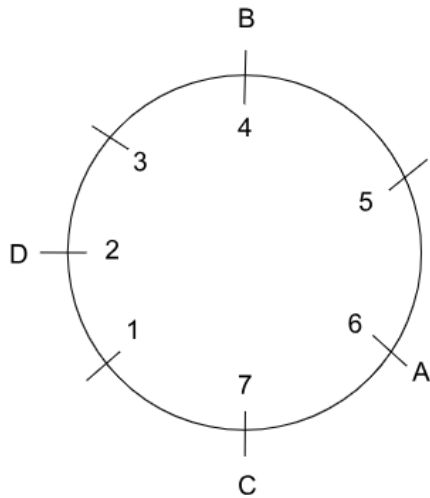
Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,



This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



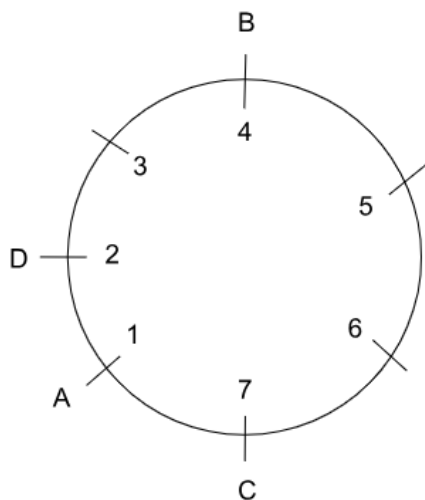
We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6. We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

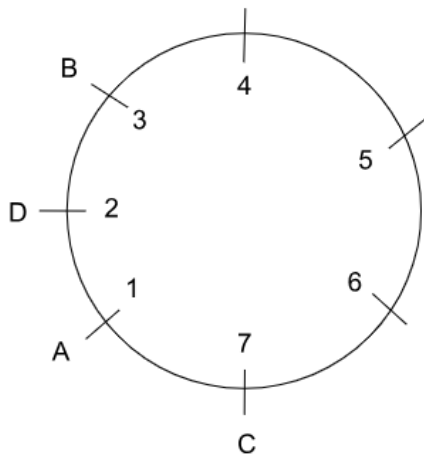
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



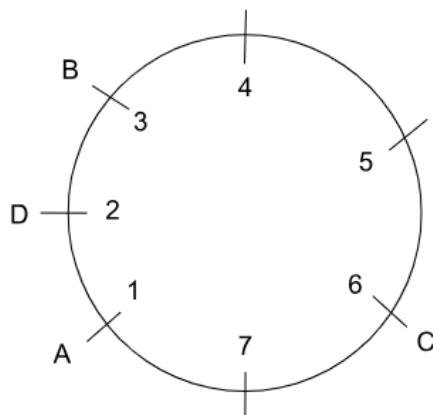
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



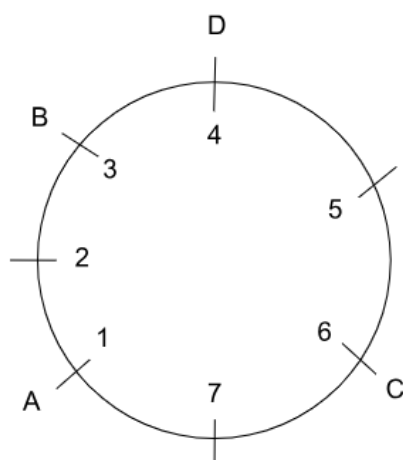
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

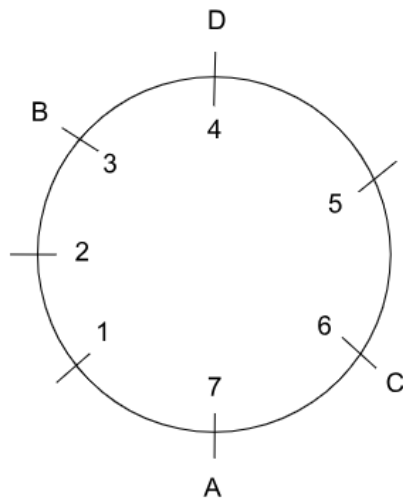
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

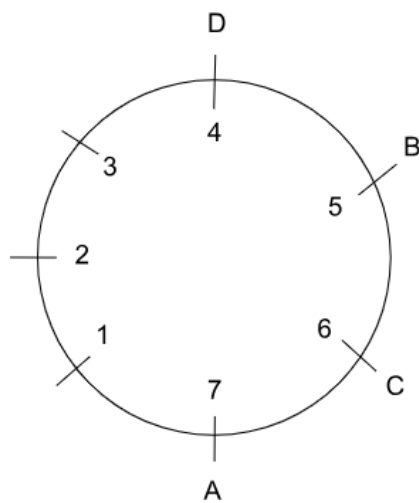
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



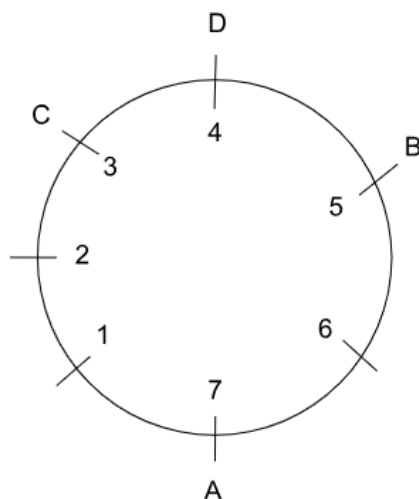
Turn 6: Bashir moves

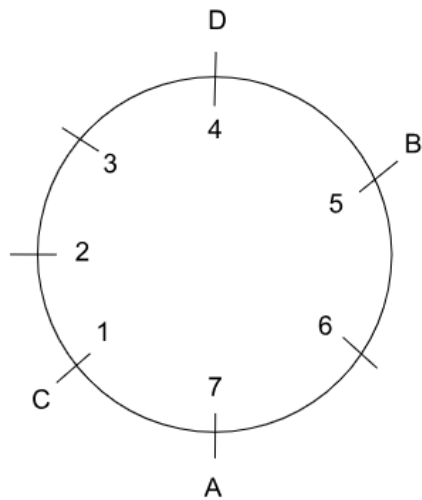
Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.





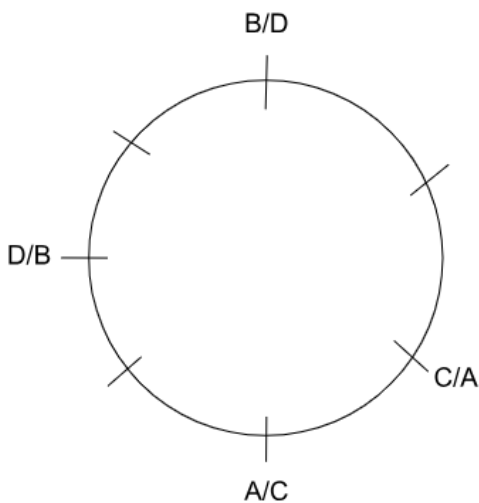
After turn 3, chair 4 is occupied by no one.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)

67. **B**

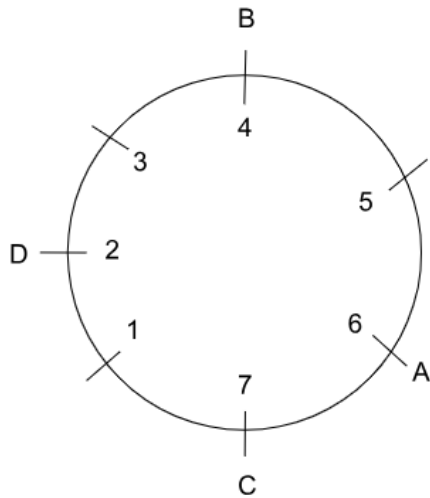
Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,



This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



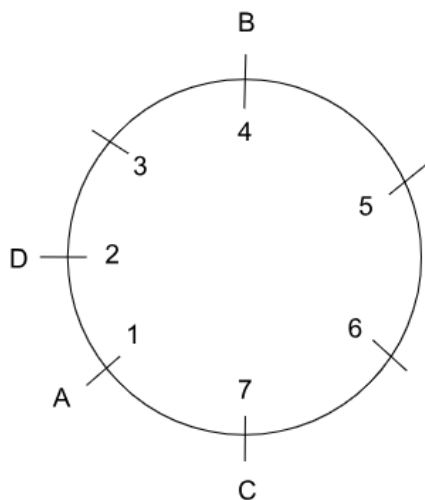
We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6. We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

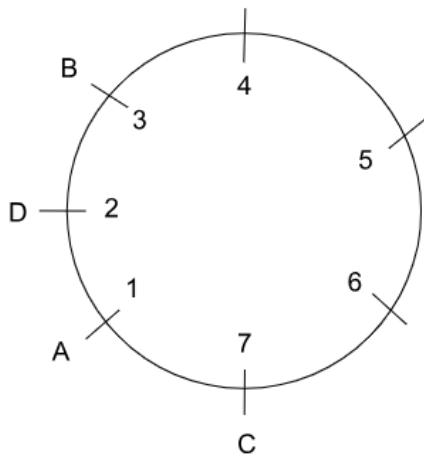
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



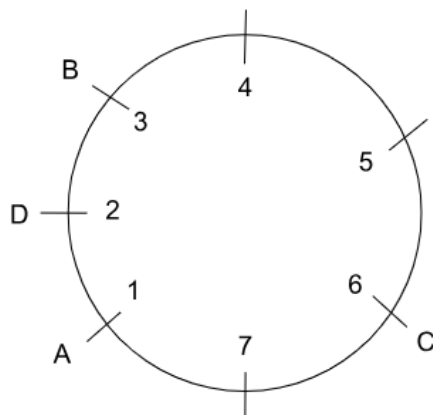
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



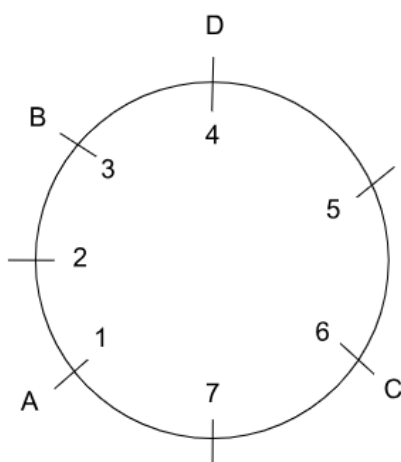
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

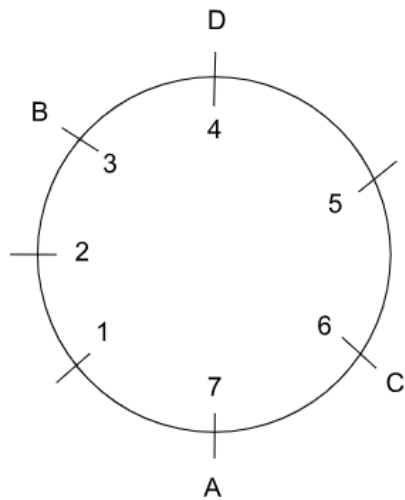
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

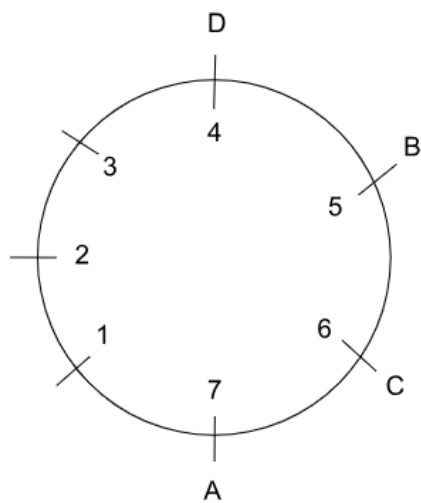
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



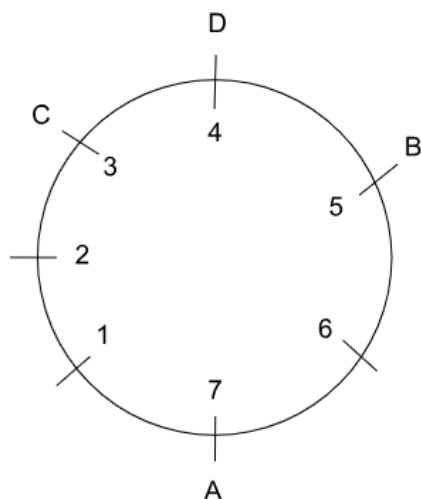
Turn 6: Bashir moves

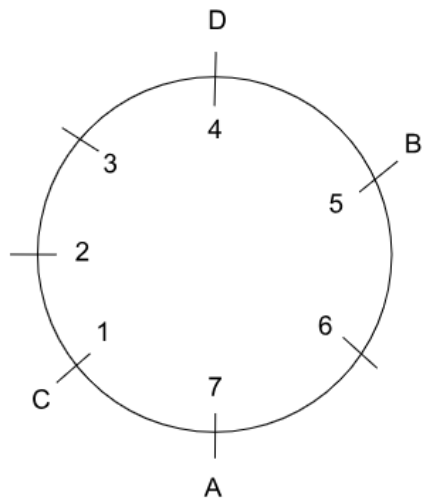
Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.





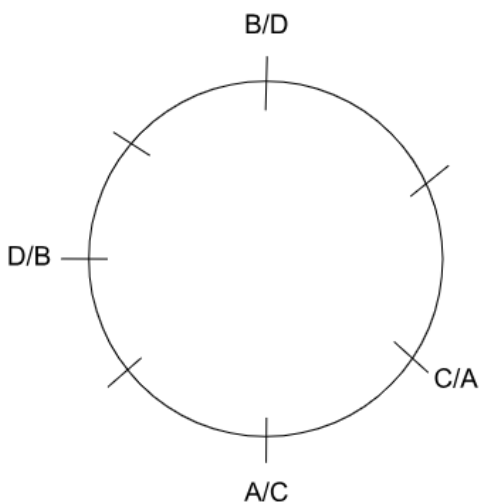
In either of the cases after turn 7, the seat adjacent to Bashir is occupied only by Davies, and the other one is empty.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

68.4

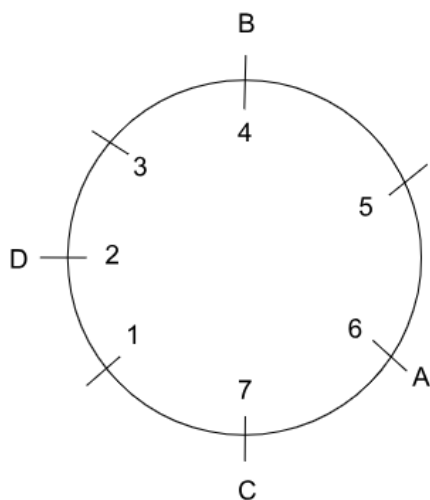
Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,



This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



So, Bashir's initial position is chair 4.

Hence, the correct answer is 4.

[VIEW SOLUTION](#)

69. A

	1st Performance				2nd Performance			
Finalists	Princess							Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items. The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.

From (ii) No composer who assigned item to Rani, assigned any item to Samragini.

Hence Dyu will compose for Samragini 1st Performance and Gagan will compose for Queen 1st Performance.

Also, Badal will compose for Samragini 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragini	Queen	Rani	Princess	Samragini	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

The sixth performance was composed by Badal. Hence C is the answer.

[VIDEO SOLUTION](#)

70. D

	1st Performance				2nd Performance			
Finalists	Princess							Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items. The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.

From (ii) No composer who assigned item to Rani, assigned any item to Samragni.

Hence Dyu will compose for Samragni 1st Performance and Gagan will compose for Queen 1st Performance.

Also, Badal will compose for Samragni 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragni	Queen	Rani	Princess	Samragni	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

The first and the sixth items were composed by Badal. Hence D is the answer.

VIDEO SOLUTION



71. A

	1st Performance				2nd Performance			
Finalists	Princess							Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items. The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.

From (ii) No composer who assigned item to Rani, assigned any item to Samragini.

Hence Dyu will compose for Samragini 1st Performance and Gagan will compose for Queen 1st Performance.

Also, Badal will compose for Samragini 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragini	Queen	Rani	Princess	Samragini	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

The second performance was composed by Dyu. Hence A is the answer.

 VIDEO SOLUTION

72. D

	1st Performance				2nd Performance			
Finalists	Princess			Rani				Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items.

The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.

From (ii) No composer who assigned item to Rani, assigned any item to Samragini.

Hence Dyu will compose for Samragini 1st Performance and Gagan will compose for Queen 1st Performance.

Also, Badal will compose for Samragini 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragini	Queen	Rani	Princess	Samragini	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

Option A: Samragini did not perform in any item composed by Ashman. This statement is true.

Option B: Princess did not perform in any item composed by Dyu. This is also true.

Option C: Rani did not perform in any item composed by Badal. This statement is true.

Option D: Queen did not perform in any item composed by Gagan. This statement is false.

Hence D is the answer.

73. D

Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

We can see that all options except option D are definitely true. Option D cannot be ascertained to be true. Dorm 10 can cost Rs. 1 crore or Rs. 6 crores to repair. Therefore, option D is the right answer.

74. 19

Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

Cost = 3 + 3 + 6 + 3 + 4 = Rs.19 crores. Therefore, 19 is the correct answer.

▶ VIDEO SOLUTION

75.3

Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to

repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

There are 3 dorms from 6 to 10 which are women's dorms.

It has been given that the cost of repairing the woman dorms add up to 20. Therefore, the distribution of the costs should be 6+6+5+3.

Dorm 4 is the dorm whose number is below 5 but is a woman's dorm. Therefore, dorm 9 should cost Rs.3 crores to repair. Dorm 8 cannot be a woman's dorm. Therefore, dorm 10 should be a woman's dorm and should cost Rs. 6 crore to repair.

Dorm 9 will cost Rs.3 crore to repair and hence, 3 is the correct answer.

 VIDEO SOLUTION

100+ CAT Quant Questions



76. D

Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to

repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

It has been given that the cost of repairing the woman dorms add up to 20. Therefore, the distribution of the costs should be 6+6+5+3.

Dorm 4 is the dorm whose number is below 5 but is a woman's dorm. Therefore, dorm 9 should cost Rs.3 crores to repair. Dorm 8 cannot be a woman's dorm. Therefore, dorm 10 should be a woman's dorm and should cost Rs. 6 crore to repair.

Hence, Option D is the right answer.

 VIDEO SOLUTION

77.7

Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.

Let's take cases:-

1. The lowest rating is 4

If the lowest rating is 4 then the other ratings will be in the range 5-10.

From this we need 4 odd and 1 even numbers.

This is not possible as there are only 3 odd numbers from 5-10.

Thus, the lowest rating is not 4.

2. The lowest rating is 2.

If the lowest rating is 2 then the other ratings will be in the range 5-10.

From this we need 4 odd and 1 even numbers.

This is possible when the odd ratings are 3,5,7 and 9.

We are given that the highest rating is not even. Thus, 10 rating is not possible.

We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.

Thus, the rating of the tea in cup 3 is an even number.

Thus, the rating of the tea in cup 5 must be an odd number.

Only 1 such pair is possible of 3 and 6.

Thus, the tea in cup 2 got the rating of 2.

The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.

We are given that:-

Tea in Cup 3 got a higher rating than that in Cup 1.

Thus, the tea in cup 1 got a rating of 5.

Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.

Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.

The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

Hence, 7 is the 2nd highest rating given.

 VIDEO SOLUTION

78.4

Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.

Let's take cases:-

1. The lowest rating is 4

If the lowest rating is 4 then the other ratings will be in the range 5-10.

From this we need 4 odd and 1 even numbers.

This is not possible as there are only 3 odd numbers from 5-10.

Thus, the lowest rating is not 4.

2. The lowest rating is 2.

If the lowest rating is 2 then the other ratings will be in the range 3-10.

From this we need 4 odd and 1 even numbers.

This is possible when the odd ratings are 3, 5, 7 and 9.

We are given that the highest rating is not even. Thus, 10 rating is not possible.

We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.

Thus, the rating of the tea in cup 3 is an even number.

Thus, the rating of the tea in cup 5 must be an odd number.

Only 1 such pair is possible of 3 and 6.

Thus, the tea in cup 2 got the rating of 2.

The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.

We are given that:-

Tea in Cup 3 got a higher rating than that in Cup 1.

Thus, the tea in cup 1 got a rating of 5.

Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.

Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.

The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

Thus, 4 was the number of the cup that contained tea from Ooty.

 VIDEO SOLUTION

79.B

Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.

Let's take cases:-

1. The lowest rating is 4

If the lowest rating is 4 then the other ratings will be in the range 5-10.

From this we need 4 odd and 1 even numbers.

This is not possible as there are only 3 odd numbers from 5-10.

Thus, the lowest rating is not 4.

2. The lowest rating is 2.

If the lowest rating is 2 then the other ratings will be in the range 3-10.

From this we need 4 odd and 1 even numbers.

This is possible when the odd ratings are 3, 5, 7 and 9.

We are given that the highest rating is not even. Thus, 10 rating is not possible.
 We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.
 Thus, the rating of the tea in cup 3 is an even number.
 Thus, the rating of the tea in cup 5 must be an odd number.
 Only 1 such pair is possible of 3 and 6.
 Thus, the tea in cup 2 got the rating of 2.
 The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.
 We are given that:-
 Tea in Cup 3 got a higher rating than that in Cup 1.
 Thus, the tea in cup 1 got a rating of 5.
 Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.
 Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.
 The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

If the tea from Munnar did not get the minimum rating then it must have got the 2nd lowest rating as we know, Assam>Wyanand>Munnar.
 Thus, Wyanand must have got a rating of 5.

 VIDEO SOLUTION

80. B

Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.
 Let's take cases:-

1. The lowest rating is 4

If the lowest rating in 4 then the other ratings will be in the range 5-10.
 From this we need 4 odd and 1 even numbers.
 This is not possible as there are only 3 odd numbers from 5-10.
 Thus, the lowest rating is not 4.

2. The lowest rating is 2.

If the lowest rating in 4 then the other ratings will be in the range 5-10.
 From this we need 4 odd and 1 even numbers.
 This is possible when the odd ratings are 3,5,7 and 9.
 We are given that the highest rating is not even. Thus, 10 rating is not possible.
 We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.
 Thus, the rating of the tea in cup 3 is an even number.
 Thus, the rating of the tea in cup 5 must be an odd number.
 Only 1 such pair is possible of 3 and 6.
 Thus, the tea in cup 2 got the rating of 2.
 The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.
 We are given that:-
 Tea in Cup 3 got a higher rating than that in Cup 1.
 Thus, the tea in cup 1 got a rating of 5.
 Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.
 Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.
 The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

It is given that the rating of Assam>Wayanad>Munnar

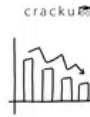
Hence, since Wayanad and Ooty are in consecutive cups, Wayanad can be either in cup number 3 or 5.

So Wayanad can only be in cup number 5, then Munnar will be in cup number 2. So Darjeeling and Assam can be in cup 1 and 3 in any order.

Hence B is a possibility.

 VIDEO SOLUTION

100+ CAT DILR Questions



81. A

We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.

Let us see how the friends are supposed to pay for the seats they choose:-

In row 1-1000

In row 2-10 - 300 for window and 500 for aisle

In row 11 - 200 for window and 400 for aisle

In row 12,13 - 1000

In row 14-20 - 200 for window and 400 for aisle

In row 21-30 - 0

Thus, As we can see 10, 11 and 12 are the only consecutive seats in which the amounts is different.

Thus, Jayanth, Ajit and Byomkesh sat in row 10, row 11 and row 12.

Manik sat beside Jayantha and thus Manik is also sitting in row 10.

Now we are given that 7 of the 8 friends paid a total of 4600 Rs.

Let's start with the cases:-

It is obvious that 5 friends cannot pay 1000 Rs for their seat because the amount will exceed 4600

Case 1:- 4 friends pay 1000 Rs each. Thus, the remaining friends will pay 600 Rs.

This is possible only when each of them pay 200 Rs.

So the case is- 1000×4 , 200×3

Case 2 :- 3 friends pay 1000 Rs each. Thus, the remaining friends will pay 1600 Rs.

There are 2 cases where this is possible:-

1000×3 , 500×2 , 400, 200

1000×3 , 400 $\times 4$

Case 3:- 2 friends pay 1000 Rs each. Thus, the remaining 5 friends will pay 2600 Rs.

This is not possible as each friend can pay a maximum of 500 Rs.

Thus, the possible cases are

1000×4 , 200×3

1000×3 , 500×2 , 400, 200

1000×3 , 400 $\times 4$

As there is no case in which a friend has to pay 300 Rs thus, Jayantha must be sitting in row 10 aisle seat.

Thus, Jayantha paid 500 Rs.

Thus, the case is:-

1000×3 , 500×2 , 400, 200

Thus, Manik must have also paid 500 sitting in row 10 aisle seat

Ajit must be sitting in row 11 aisle seat paying 400 Rs.

Byomyesh must be sitting row 12 aisle seat paying 1000 Rs.

Thus, among Gargi, Kikira, Pradosh and Tapes 2 must have paid 1000, 1 must have paid 200 and the remaining person must have paid nothing.

Now we know Gargi and Kikira are sitting adjacent to each other and thus, either both or none of them must have paid 1000 Rs.

Among Pradosh and Tapes a maximum of 1 person could have paid 1000 Rs.

Thus, the only possible case here is :-

Gargi and Kikira paid 1000 each.

Pradosh is sitting ahead of Tapes and one of them paid 200 Rs.

Since, both of them were sitting in seats marked by the same letter, in consecutive rows thus, the only possibility is Pradosh sitting in row 20 window seat and paying 200 and Tapesesh sitting in row 21 paying nothing.

Thus, the amount paid by each friend is as shown below:

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapesesh	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Manik is sitting in row 10.

VIDEO SOLUTION

82. D

We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.

Let us see how the friends are supposed to pay for the seats they choose:-

In row 1-1000

In row 2-10 - 300 for window and 500 for aisle

In row 11 - 200 for window and 400 for aisle

In row 12,13 - 1000

In row 14-20 - 200 for window and 400 for aisle

In row 21-30 - 0

Thus, As we can see 10, 11 and 12 are the only consecutive seats in which the amounts is different.

Thus, Jayanth, Ajit and Byomkesh sat in row 10, row 11 and row 12.

Manik sat beside Jayantha and thus Manik is also sitting in row 10.

Now we are given that 7 of the 8 friends paid a total of 4600 Rs.

Let's start with the cases:-

It is obvious that 5 friends cannot pay 1000 Rs for their seat because the amount will exceed 4600

Case 1:- 4 friends pay 1000 Rs each. Thus, the remaining friends will pay 600 Rs.

This is possible only when each of them pay 200 Rs.

So the case is- 1000×4 , 200×3

Case 2 :- 3 friends pay 1000 Rs each. Thus, the remaining friends will pay 1600 Rs.

There are 2 cases where this is possible:-

1000×3 , 500×2 , 400, 200

1000×3 , 400 $\times 4$

Case 3:- 2 friends pay 1000 Rs each. Thus, the remaining 5 friends will pay 2600 Rs.

This is not possible as each friend can pay a maximum of 500 Rs.

Thus, the possible cases are

1000×4 , 200×3

1000×3 , 500×2 , 400, 200

1000×3 , 400 $\times 4$

As there is no case in which a friend has to pay 300 Rs thus, Jayantha must be sitting in row 10 aisle seat.

Thus, Jayantha paid 500 Rs.

Thus, the case is:-

1000*3, 500*2, 400, 200

Thus, Manik must have also paid 500 sitting in row 10 aisle seat

Ajit must be sitting in row 11 aisle seat paying 400 Rs.

Byomyesh must be sitting row 12 aisle seat paying 1000 Rs.

Thus, among Gargi, Kikira, Pradosh and Tapes 2 must have paid 1000, 1 must have paid 200 and the remaining person must have paid nothing.

Now we know Gargi and Kikira are sitting adjacent to each other and thus, either both or none of them must have paid 1000 Rs.

Among Pradosh and Tapes a maximum of 1 person could have paid 1000 Rs.

Thus, the only possible case here is :-

Gargi and Kikira paid 1000 each.

Pradosh is sitting ahead of Tapes and one of them paid 200 Rs.

Since, both of them were sitting in seats marked by the same letter, in consecutive rows thus, the only possibility is Pradosh sitting in row 20 window seat and paying 200 and Tapes sitting in row 21 paying nothing.

Thus, the amount paid by each friend is as shown below:-

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapes	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Gargi paid 1000 rs for her choice of seat

 VIDEO SOLUTION

83. C

We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.

Let us see how the friends are supposed to pay for the seats they choose:-

In row 1-1000

In row 2-10 - 300 for window and 500 for aisle

In row 11 - 200 for window and 400 for aisle

In row 12,13 - 1000

In row 14-20 - 200 for window and 400 for aisle

In row 21-30 - 0

Thus, As we can see 10, 11 and 12 are the only consecutive seats in which the amounts is different.

Thus, Jayanth, Ajit and Byomkesh sat in row 10, row 11 and row 12.

Manik sat beside Jayantha and thus Manik is also sitting in row 10.

Now we are given that 7 of the 8 friends paid a total of 4600 Rs.

Let's start with the cases:-

It is obvious that 5 friends cannot pay 1000 Rs for their seat because the amount will exceed 4600

Case 1:- 4 friends pay 1000 Rs each. Thus, the remaining friends will pay 600 Rs.

This is possible only when each of them pay 200 Rs.

So the case is- 1000*4 , 200*3

Case 2 :- 3 friends pay 1000 Rs each. Thus, the remaining friends will pay 1600 Rs.

There are 2 cases where this is possible:-

1000*3, 500*2, 400, 200

1000*3, 400*4

Case 3:- 2 friends pay 1000 Rs each. Thus, the remaining 5 friends will pay 2600 Rs.

This is not possible as each friend can pay a maximum of 500 Rs.

Thus, the possible cases are

1000*4, 200*3

1000*3, 500*2, 400, 200

1000*3, 400*4

As there is no case in which a friend has to pay 300 Rs thus, Jayantha must be sitting in row 10 aisle seat.

Thus, Jayantha paid 500 Rs.

Thus, the case is:-

1000*3, 500*2, 400, 200

Thus, Manik must have also paid 500 sitting in row 10 aisle seat

Ajit must be sitting in row 11 aisle seat paying 400 Rs.

Byomyesh must be sitting row 12 aisle seat paying 1000 Rs.

Thus, among Gargi, Kikira, Pradosh and Tapes 2 must have paid 1000, 1 must have paid 200 and the remaining person must have paid nothing.

Now we know Gargi and Kikira are sitting adjacent to each other and thus, either both or none of them must have paid 1000 Rs.

Among Pradosh and Tapes a maximum of 1 person could have paid 1000 Rs.

Thus, the only possible case here is :-

Gargi and Kikira paid 1000 each.

Pradosh is sitting ahead of Tapes and one of them paid 200 Rs.

Since, both of them were sitting in seats marked by the same letter, in consecutive rows thus, the only possibility is Pradosh sitting in row 20 window seat and paying 200 and Tapes sitting in row 21 paying nothing.

Thus, the amount paid by each friend is as shown below:-

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapes	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Jayanta paid 500 for her choice of seat.

 VIDEO SOLUTION

84. D

We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.

Let us see how the friends are supposed to pay for the seats they choose:-

In row 1-1000

In row 2-10 - 300 for window and 500 for aisle

In row 11 - 200 for window and 400 for aisle

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapesh	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Tapesh did not paid any amount.

 VIDEO SOLUTION

85.C

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let ' $4x$ ' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	$3x$	$3x$	$4x$	$10x$
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that ' y ' is the number of educationalists in the research committee and ' d ' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	$3x$	$3x$	$4x$	$10x$
Educationalists	y	$y-d$	$y+d$	$3y$
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let ' $5z$ ' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

$$\Rightarrow 10x+3y+5z = 24$$

We can see that each of x, y and z has to be a natural number integer. If $x > 1$, then both y and z can't take any natural number.

Hence, we can say that $x = 1$.

At $x = 1$, $3y+5z = 14$. If $y = 1$ or 2 , Z is not an integer.

At $x = 1$ and $y = 3$, $z = 1$ which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

We can see that 'd' can assume two possible values. $d = 1$ or 2 .

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

Let us check the option one by one.

Option A: In the teaching committee the number of educationalists is equal to the number of politicians. We can see that in the teaching committee the number of educationalists can be equal to the number of politicians when both the numbers are '1'. Hence, this statement can be correct.

Option B: In the administration committee the number of bureaucrats is equal to the number of educationalists. We can see that in the administration committee the number of bureaucrats can be equal to the number of educationalists when both the numbers are '4'. Hence, this statement can be correct.

Option C: The size of the research committee is less than the size of the teaching committee. We can see the maximum size of teaching committee can be '6' which is less than the size of the research committee. Hence, the sentence is incorrect.

Option D: The size of the research committee is less than the size of the administration committee. We can see the minimum size of Administration committee can be '11' which is more than the size of the research committee. Hence, this statement is correct.

Therefore, we can say that option C is the correct answer.

 VIDEO SOLUTION

100+ CAT VARC Questions



86.4

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let '4x' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that 'y' is the number of educationalists in the research committee and 'd' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let '5z' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

$$\Rightarrow 10x + 3y + 5z = 24$$

We can see that each of x, y and z has to be a natural number integer. If $x > 1$, then both y and z can't take any natural number.

Hence, we can say that $x = 1$.

At $x = 1$, $3y + 5z = 14$. If $y = 1$ or 2 , Z is not an integer.

At $x = 1$ and $y = 3$, $z = 1$ which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

We can see that 'd' can assume two possible values. $d = 1$ or 2 .

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

From the table, we can see that the number of bureaucrats in the administration committee = 4.

 VIDEO SOLUTION

87.3

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let ' $4x$ ' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that ' y ' is the number of educationalists in the research committee and ' d ' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let ' $5z$ ' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

$$\Rightarrow 10x+3y+5z = 24$$

We can see that each of x, y and z has to be a natural number integer. If $x > 1$, then both y and z can't take any natural number.

Hence, we can say that $x = 1$.

At $x = 1$, $3y+5z = 14$. If $y = 1$ or 2 , Z is not an integer.

At $x = 1$ and $y = 3$, $z = 1$ which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

We can see that 'd' can assume two possible values. $d = 1$ or 2 .

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

From the table, we can see that the number of educationalists in the research committee = 3.

[▶ VIDEO SOLUTION](#)

88. A

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let '4x' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that 'y' is the number of educationalists in the research committee and 'd' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let '5z' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

$$\Rightarrow 10x + 3y + 5z = 24$$

We can see that each of x, y and z has to be a natural number integer. If $x > 1$, then both y and z can't take any natural number.

Hence, we can say that $x = 1$.

At $x = 1$, $3y + 5z = 14$. If $y = 1$ or 2 , Z is not an integer.

At $x = 1$ and $y = 3$, $z = 1$ which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

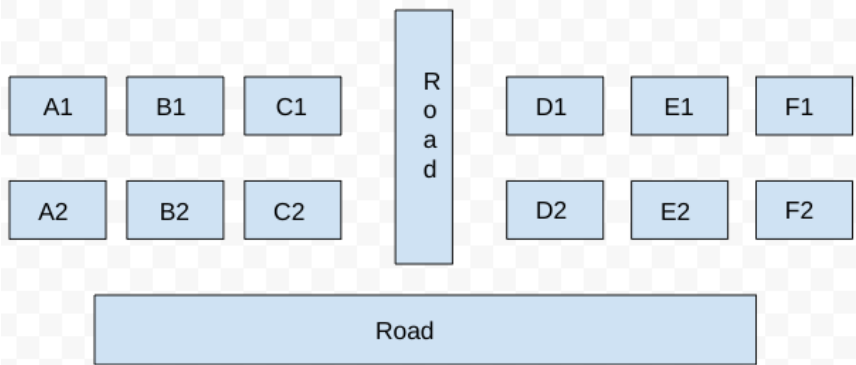
We can see that 'd' can assume two possible values. $d = 1$ or 2 .

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

From the table, we can not uniquely determine the size of the teaching committee. Hence, option A is the correct answer.

VIDEO SOLUTION

89. C



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

$$\Rightarrow 12 + 5a + 3b = 24 \Rightarrow 5a + 3b = 12 \quad (a = \text{road adjacency value}, b = \text{neighbor count})$$

The only value for which the equation satisfies is ($a = 0$, and $b = 4$). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

$$\Rightarrow 10 + 5a + 3b = 24 \Rightarrow 5a + 3b = 14$$

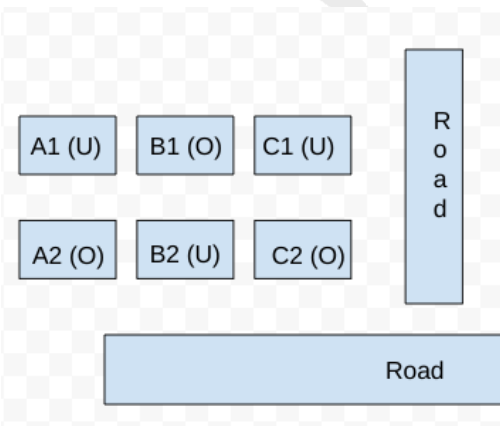
$$\Rightarrow (a, b) = (1, 3)$$

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2.

Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and O = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: $12 + 5 \cdot 1 + 3 \cdot b = 15 \Rightarrow 3b = -2$ (which is not possible)

Hence, the house has no parking space $\Rightarrow 10 + 5 \cdot 1 + 3b = 15 \Rightarrow b = 0$

$b = 0$ implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

i) If the house has no working space, then: $10 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 5/3$ (this is not possible since b has to be an integer value)

Hence, the house has parking space $\Rightarrow 12 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 1 \Rightarrow$ One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10 (there is no parking space) + $(5 \cdot 0) + (3 \cdot \text{the number of neighbours})$

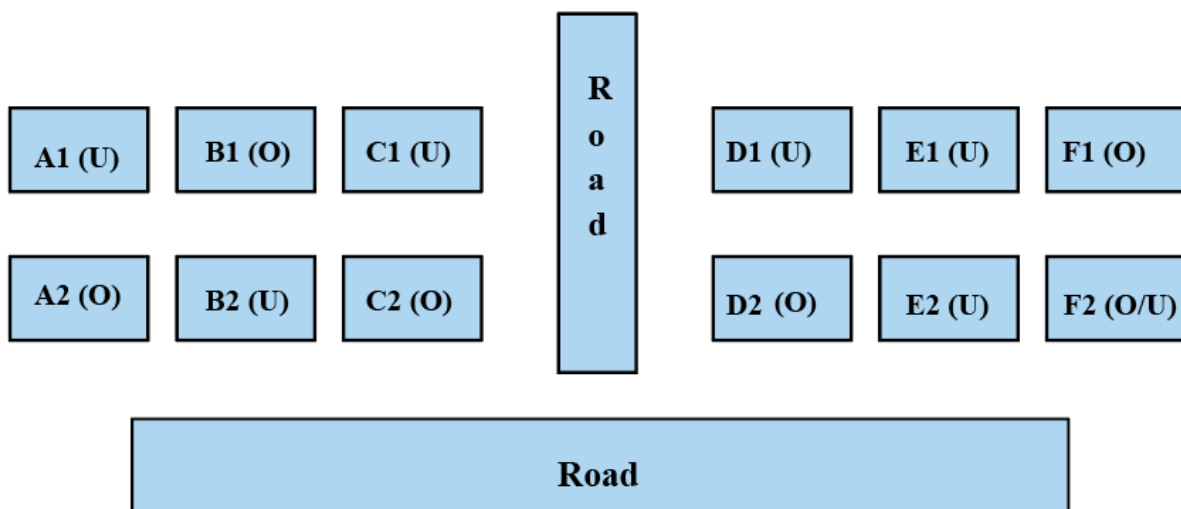
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.

Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.

Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:

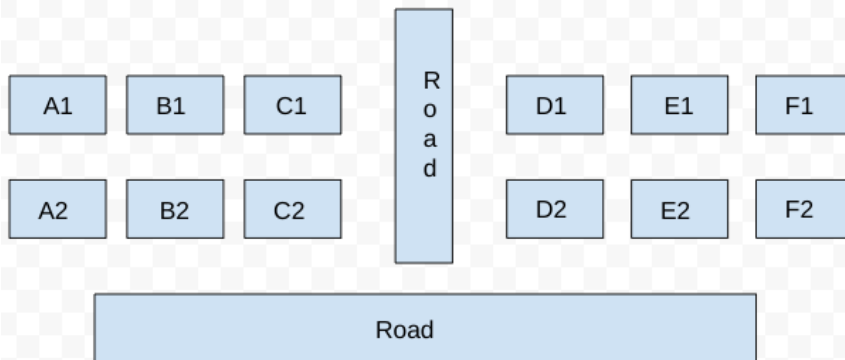


From the diagram, we can see that B1 and D2 are definitely occupied. The rest of the options are not definitely correct.

The correct options are both B and C.

VIDEO SOLUTION

90.3



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

$$\Rightarrow 12 + 5a + 3b = 24 \Rightarrow 5a + 3b = 12 \quad (a = \text{road adjacency value}, b = \text{neighbor count})$$

The only value for which the equation satisfies is ($a = 0$, and $b = 4$). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

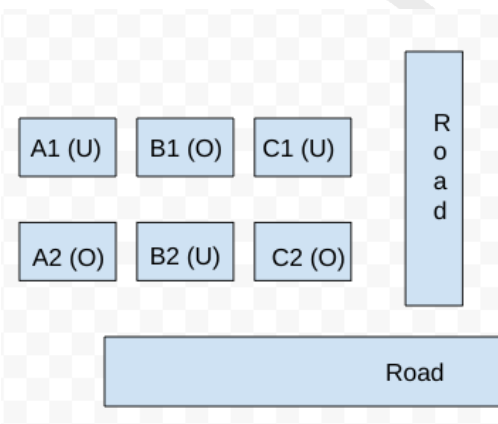
$$\Rightarrow 10 + 5a + 3b = 24 \Rightarrow 5a + 3b = 14$$

$$\Rightarrow (a, b) = (1, 3)$$

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and O = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: $12 + 5 \times 1 + 3 \times b = 15 \Rightarrow 3b = -2$ (which is not possible)

Hence, the house has no parking space $\Rightarrow 10 + 5 \times 1 + 3b = 15 \Rightarrow b = 0$

$b = 0$ implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

i) If the house has no working space, then: $10 + 5 \times 0 + 3b = 15 \Rightarrow b = 5/3$ (this is not possible since b has to be an integer value)

Hence, the house has parking space $\Rightarrow 12 + 5 \times 0 + 3b = 15 \Rightarrow b = 1 \Rightarrow$ One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10 (there is no parking space) + $(5 \times 0) + (3 \times \text{the number of neighbours})$

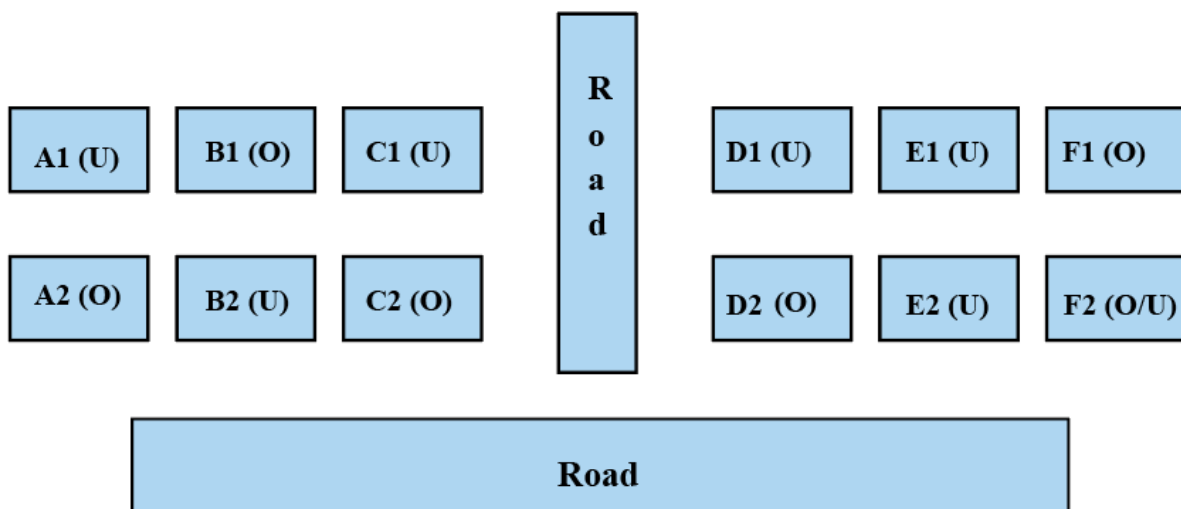
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.

Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.

Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:



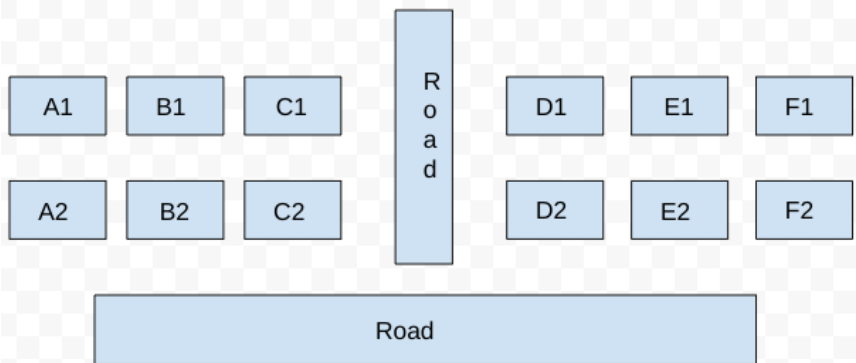
From the diagram, we can see that 3 houses are vacant in block YY.

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91.D



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

$$\Rightarrow 12 + 5a + 3b = 24 \Rightarrow 5a + 3b = 12 \quad (a = \text{road adjacency value}, b = \text{neighbor count})$$

The only value for which the equation satisfies is ($a = 0$, and $b = 4$). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

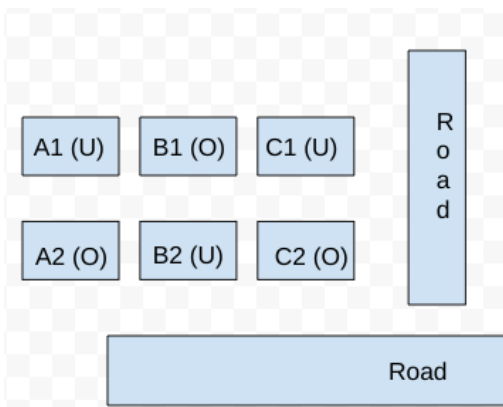
$$\Rightarrow 10 + 5a + 3b = 24 \Rightarrow 5a + 3b = 14$$

$$\Rightarrow (a, b) = (1, 3)$$

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and O = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: $12 + 5 \cdot 1 + 3 \cdot b = 15 \Rightarrow 3b = -2$ (which is not possible)

Hence, the house has no parking space $\Rightarrow 10 + 5 \cdot 1 + 3b = 15 \Rightarrow b = 0$

$b = 0$ implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

i) If the house has no working space, then: $10 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 5/3$ (this is not possible since b has to be an integer value)

Hence, the house has parking space $\Rightarrow 12 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 1 \Rightarrow$ One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10 (there is no parking space) + $(5 \cdot 0) + (3 \cdot \text{the number of neighbours})$

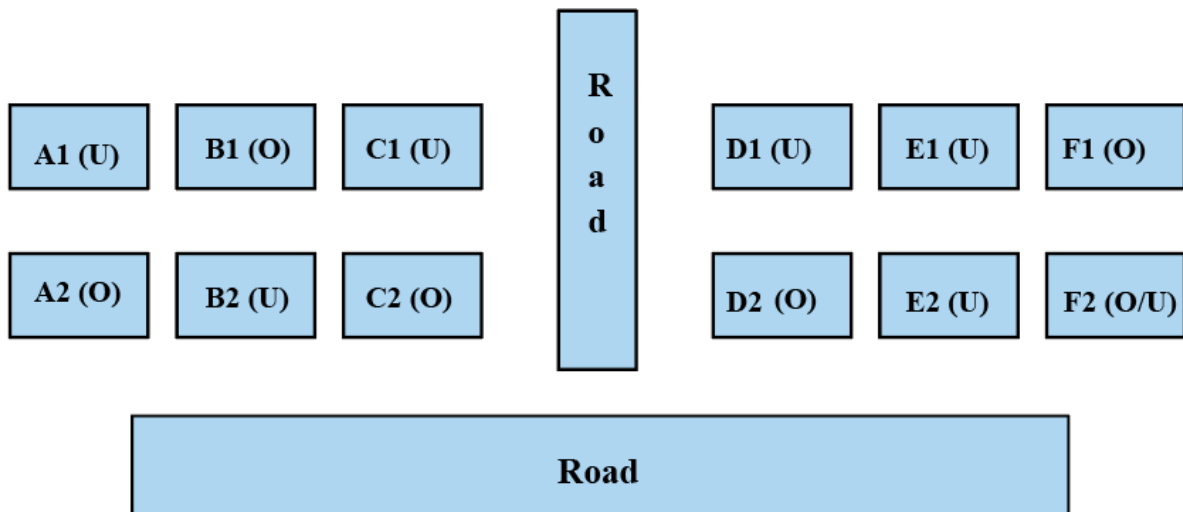
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.

Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.

Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:



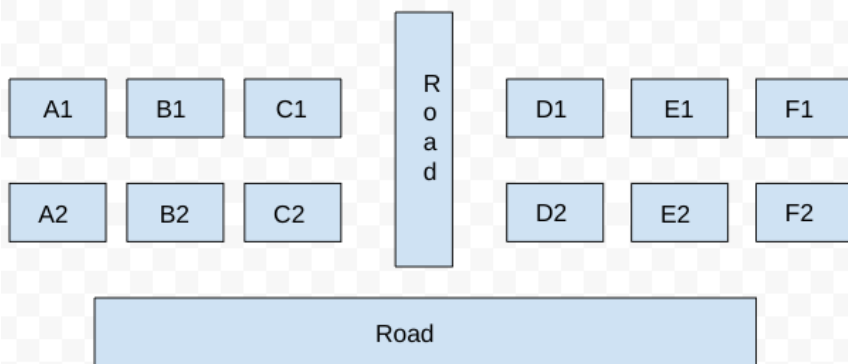
From the diagram, we can say that the number of vacant houses in Row 2 in Block XX is 1, and the number of vacant houses in Row 2 in Block YY is either 1 or 2.

Hence, the total number of vacant houses is either 2 or 3

The correct option is D

[VIDEO SOLUTION](#)

92.21



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

$$\Rightarrow 12 + 5a + 3b = 24 \Rightarrow 5a + 3b = 12 \quad (a = \text{road adjacency value}, b = \text{neighbor count})$$

The only value for which the equation satisfies is ($a = 0$, and $b = 4$). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

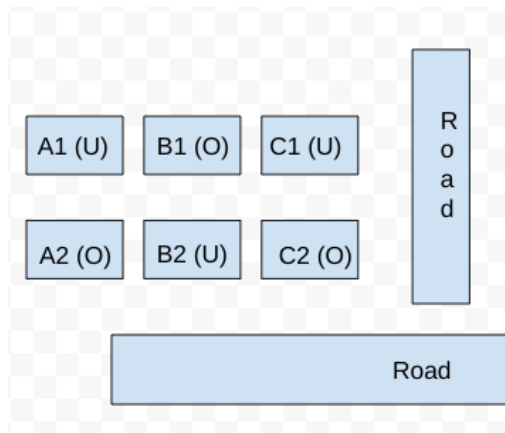
$$\Rightarrow 10 + 5a + 3b = 24 \Rightarrow 5a + 3b = 14$$

$$\Rightarrow (a, b) = (1, 3)$$

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and O = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: $12 + 5 \cdot 1 + 3 \cdot b = 15 \Rightarrow 3b = -2$ (which is not possible)

Hence, the house has no parking space $\Rightarrow 10 + 5 \cdot 1 + 3b = 15 \Rightarrow b = 0$

$b = 0$ implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

i) If the house has no working space, then: $10 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 5/3$ (this is not possible since b has to be an integer value)

Hence, the house has parking space $\Rightarrow 12 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 1 \Rightarrow$ One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10 (there is no parking space) $+ (5 \cdot 0) + (3 \cdot \text{the number of neighbours})$

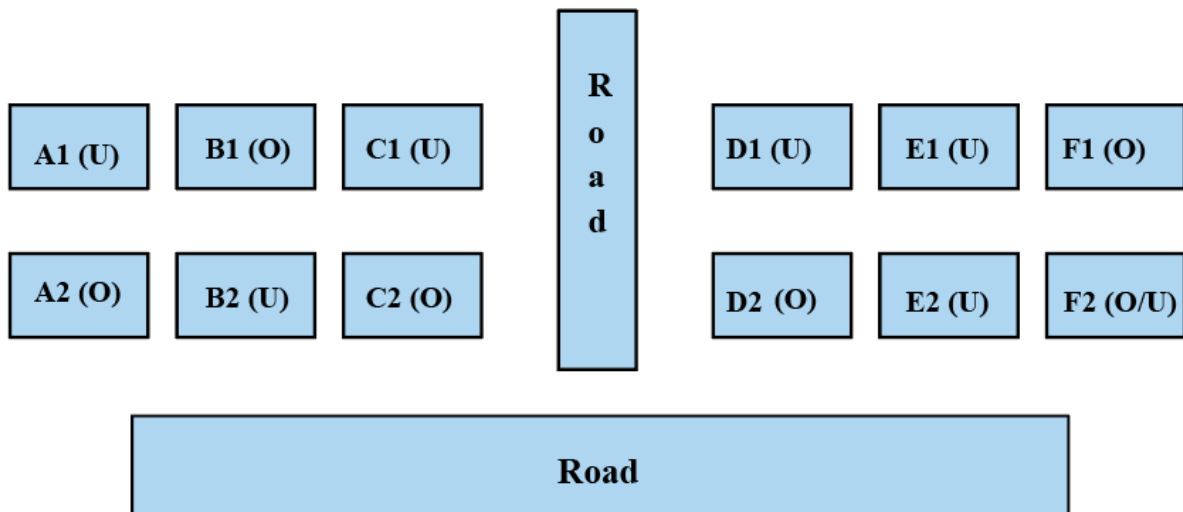
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.

Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.

Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:

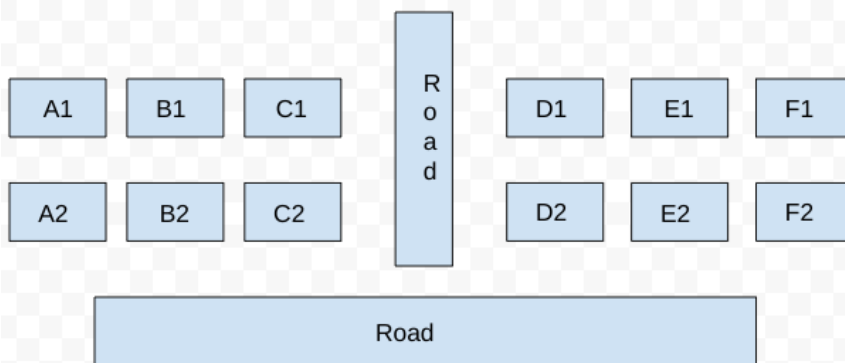


From the diagram, the vacant house with the maximum possible quoted price in column E is E2 when F2 is occupied.

The maximum possible quoted price of E2 is $10 + 5 \times 1 + 3 \times 2 = 21$ Lacs. (E2 has no parking space because E1 has the parking space, and it is given that there is only one house with parking space in Block YY.)

[VIDEO SOLUTION](#)

93. A



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + $5 \times$ (road adjacency value) + $3 \times$ (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

$$\Rightarrow 12 + 5a + 3b = 24 \Rightarrow 5a + 3b = 12 \quad (a = \text{road adjacency value}, b = \text{neighbor count})$$

The only value for which the equation satisfies is ($a = 0$, and $b = 4$). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

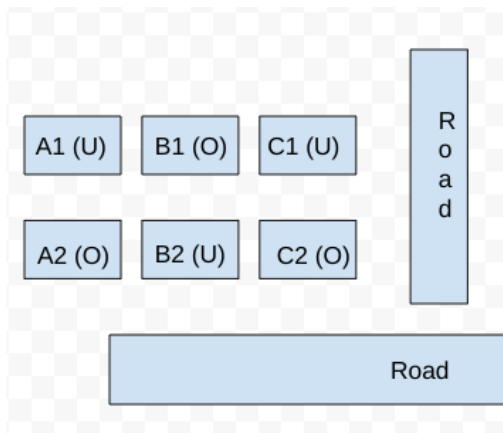
$$\Rightarrow 10 + 5a + 3b = 24 \Rightarrow 5a + 3b = 14$$

$$\Rightarrow (a, b) = (1, 3)$$

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and O = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: $12 + 5 \cdot 1 + 3 \cdot b = 15 \Rightarrow 3b = -2$ (which is not possible)

Hence, the house has no parking space $\Rightarrow 10 + 5 \cdot 1 + 3b = 15 \Rightarrow b = 0$

$b = 0$ implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

i) If the house has no working space, then: $10 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 5/3$ (this is not possible since b has to be an integer value)

Hence, the house has parking space $\Rightarrow 12 + 5 \cdot 0 + 3b = 15 \Rightarrow b = 1 \Rightarrow$ One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10 (there is no parking space) $+ (5 \cdot 0) + (3 \cdot \text{the number of neighbours})$

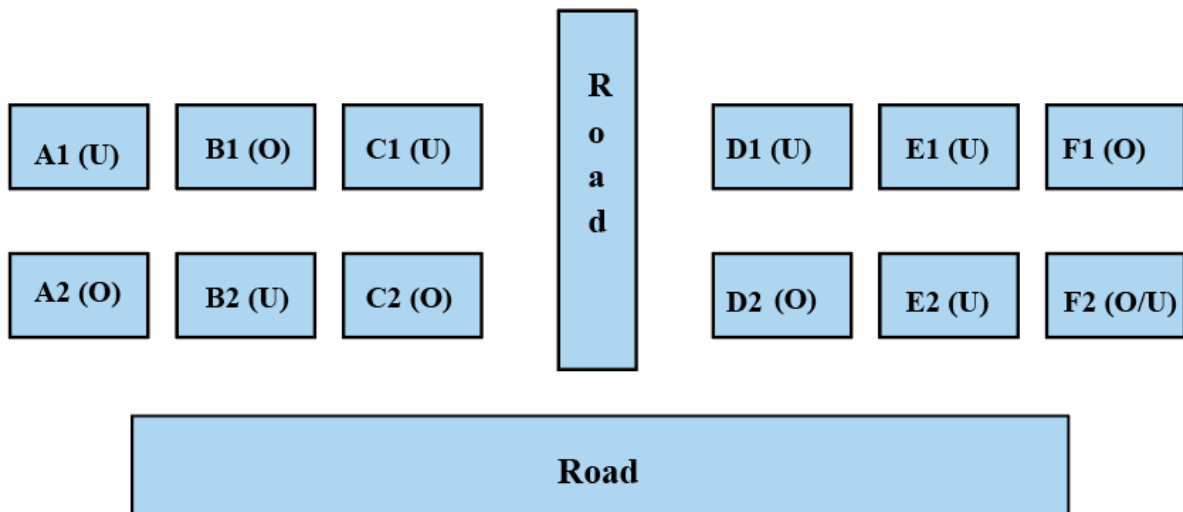
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.

Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.

Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:



From the diagram, we can see that E1 has the parking space (case 2).

The correct option is A

[VIDEO SOLUTION](#)

94. C

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

In statement 1, it is given that contamination levels at three pumps among P1 - P5 were recorded as high. This is only possible when pumps 1, 3 and 5 have high level of contamination. Also, P6 was the only pump among P1 - P10 where the contamination level was recorded as low. Therefore, we can say that pumps 2 and 4 have medium level of contamination.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10. (Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = $8/2 = 4$

Also, the number of pumps that have medium contamination level = $20 - 8 - 4 = 12$

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Low
P12	Medium
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	Low
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Low
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	Low
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Low
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

We know that medium contamination level was recorded at exactly 8 pumps. Hence, option C is the correct answer.

 VIDEO SOLUTION

95. B

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

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Pumps	Contamination level
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P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10. (Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = $8/2 = 4$

Also, the number of pumps that have medium contamination level = $20 - 8 - 4 = 12$

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
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P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Low
P12	Medium
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	Low
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Low
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	Low
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Low
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

We can see that in case 1 the contamination level at P15 was recorded as medium. Let us check all the option one by one.

(Option :A) Contamination levels at P13 and P17 were recorded as the same. From the table, we can see that the contamination levels at P13 and P17 were recorded as medium. Hence, we can say that this statement is correct.

(Option :B) Contamination levels at P11 and P16 were recorded as the same. From the table, we can see that the contamination levels at P11 was recorded as Medium whereas at P16 it was recorded as Low. Hence, we can say that this statement is incorrect. Thus, option B is the correct answer.

 VIDEO SOLUTION



CAT Important Topics




96. D

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

In statement 1, it is given that contamination levels at three pumps among P1 - P5 were recorded as high. This is only possible when pumps 1, 3 and 5 have high level of contamination. Also, P6 was the only pump among P1 - P10 where the contamination level was recorded as low. Therefore, we can say that pumps 2 and 4 have medium level of contamination.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10. (Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = $8/2 = 4$

Also, the number of pumps that have medium contamination level = $20 - 8 - 4 = 12$

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Low
P12	Medium
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	Low
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Low
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	Low
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Low
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Let us check the options one by one.

(Option:A) The contamination level at P20 was recorded as medium. This need not be true as we can see that in Case 1 at P20 low contamination level is recorded.

(Option:B) The contamination level at P13 was recorded as low. This need not be true as we can see that in Case 2(a), at P13 high contamination level is recorded.

(Option:C) The contamination level at P12 was recorded as high. This need not be true as we can see that in Case 2(a), at P12 medium contamination level is recorded.

(Option:D) The contamination level at P10 was recorded as high.. This is true for all cases. Hence, we can say that option D is the correct answer.

 VIDEO SOLUTION

97. D

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

In statement 1, it is given that contamination levels at three pumps among P1 - P5 were recorded as high. This is only possible when pumps 1, 3 and 5 have high level of contamination. Also, P6 was the only pump among P1 - P10 where the contamination level was recorded as low. Therefore, we can say that pumps 2 and 4 have

medium level of contamination.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10. (Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = $8/2 = 4$

Also, the number of pumps that have medium contamination level = $20 - 8 - 4 = 12$

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level	Pumps	Contamination level	Pumps	Contamination level	Pumps	Contamination level	Pumps	Contamination level
P1	High	P1	High	P1	High	P1	High	P1	High
P2	Medium	P2	Medium	P2	Medium	P2	Medium	P2	Medium
P3	High	P3	High	P3	High	P3	High	P3	High
P4	Medium	P4	Medium	P4	Medium	P4	Medium	P4	Medium
P5	High	P5	High	P5	High	P5	High	P5	High
P6	Low	P6	Low	P6	Low	P6	Low	P6	Low
P7	High	P7	High	P7	High	P7	High	P7	High
P8	High	P8	High	P8	High	P8	High	P8	High
P9	Medium	P9	Medium	P9	Medium	P9	Medium	P9	Medium
P10	High	P10	High	P10	High	P10	High	P10	High
P11	Low	P11	Medium	P11	Medium	P11	Medium	P11	Medium
P12	Medium	P12	Low	P12	High	P12	High	P12	High
P13	High	P13	High	P13	Low	P13	Medium	P13	Medium
P14	Medium	P14	Medium	P14	Medium	P14	Low	P14	High
P15	High	P15	High	P15	High	P15	High	P15	Low
P16	Medium	P16	Medium	P16	Medium	P16	Medium	P16	Medium
P17	Low	P17	Low	P17	Low	P17	Low	P17	Low
P18	Medium	P18	Medium	P18	Medium	P18	Medium	P18	Medium
P19	Low	P19	Low	P19	Low	P19	Low	P19	Low
P20	Medium	P20	Medium	P20	Medium	P20	Medium	P20	Medium

We can see that in case 2(a) the contamination level at P11 was recorded as low. Let us check all the option one by one.

(Option : A) The contamination level at P12 was recorded as high. This statement is incorrect as we can see in the table, the contamination level at P12 was recorded as medium.

(Option : B) The contamination level at P15 was recorded as medium. This statement is incorrect as we can see in the table, the contamination level at P15 was recorded as High.

(Option : C) The contamination level at P18 was recorded as low. This statement is incorrect as we can see in the table, the contamination level at P18 was recorded as Medium.

(Option : D) The contamination level at P14 was recorded as medium. This statement is correct as we can see in the table, the contamination level at P14 was recorded as Medium. Hence, we can say that option D is the correct answer.

 VIDEO SOLUTION

98. D

All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

$$5 \times 4 = 20, 10 \times 3 = 30, 15 \times 2 = 30$$

The total possible with considering the minimum number of questions of each type = $20 + 30 + 30 = 80$ marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = $9 + 4 = 13$ questions

2) {5,5,10} = $9 + 3 = 12$ questions

3) {10,10} = $9 + 2 = 11$ questions

4) {5,15} = $9 + 2 = 11$ questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

$$\{5,5,5,5\}\{5,15\}$$

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT. we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is D

 VIDEO SOLUTION

99. A

All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

$$5 \times 4 = 20, 10 \times 3 = 30, 15 \times 2 = 30$$

The total possible with considering the minimum number of questions of each type = $20 + 30 + 30 = 80$ marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = $9 + 4 = 13$ questions

2) {5,5,10} = $9 + 3 = 12$ questions

3) {10,10} = $9 + 2 = 11$ questions

4) {5,15} = $9 + 2 = 11$ questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

{5,5,5,5}{5,15}

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT.
we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is A

[VIDEO SOLUTION](#)

100. D

All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

$$5 \times 4 = 20, 10 \times 3 = 30, 15 \times 2 = 30$$

The total possible with considering the minimum number of questions of each type = $20 + 30 + 30 = 80$ marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = $9 + 4 = 13$ questions

2) {5,5,10} = $9 + 3 = 12$ questions

3) {10,10} = $9 + 2 = 11$ questions

4) {5,15} = $9 + 2 = 11$ questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

{5,5,5,5}{5,15}

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT.
we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

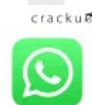
	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is D

 VIDEO SOLUTION



CAT WhatsApp Group



101. D

All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

$$5 \times 4 = 20, 10 \times 3 = 30, 15 \times 2 = 30$$

The total possible with considering the minimum number of questions of each type = $20 + 30 + 30 = 80$ marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = $9 + 4 = 13$ questions

2) {5,5,10} = $9 + 3 = 12$ questions

3) {10,10} = $9 + 2 = 11$ questions

4) {5,15} = $9 + 2 = 11$ questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

{5,5,5,5}{5,15}

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT.
we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is D

 VIDEO SOLUTION

102. C

Considering (i), Ignesh has to eat 6 vadas, since 6 is the only multiple of 3.

Also, using the same information, we can say that a person consumes 2 vadas and 4 idlis.

Using (vii), Bimal eats 2 more idlis than Ignesh.

Possibilities:

Bimal - 8, Ignesh - 6

Bimal - 6, Ignesh - 4

But Ignesh cannot have 4 idlis because the person who eats 4 idlis eats 2 vadas.

Hence we take Bimal - 8 and Ignesh - 6.

Also, we get that Bimal eats $6 - 2 = 4$ vadas.

So far, we get the following information.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4

Using (vi), there is a person who eats twice as many idlis as Mukesh. The only pair satisfying is 8, 4.

So, Mukesh eats 4 idlis. Plus the person who eats 4 idlis eats 2 vadas. Hence, we get

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2

Daljit also eats Vada as per info (v), so we get the following

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit		1
Sandip		0

(iv) gives us the information that the one who eats 1 idli does not have vada.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit	5	1
Sandip	1	0

Considering the persons who had chutney and those who didn't, 3 persons do not have chutney. Bimal is one of them(the one eating 4 vadas).

Mukesh is the second one to not take chutney(last hint). Also, Sandip does not take chutney. Hence, we get this information as well.

	Idlis	Vadas	Chutney
Ignesh	6	6	✓
Bimal	8	4	~
Mukesh	4	2	~
Daljit	5	1	✓
Sandip	1	0	~

Hence, Ignesh eating 6 vadas and 6 idlis eat Chutney.

 VIDEO SOLUTION

103. **A**

Considering (i), Ignesh has to eat 6 vadas, since 6 is the only multiple of 3.

Also, using the same information, we can say that a person consumes 2 vadas and 4 idlis.

Using (vii), Bimal eats 2 more idlis than Ignesh.

Possibilities:

Bimal - 8, Ignesh - 6

Bimal - 6, Ignesh - 4

But Ignesh cannot have 4 idlis because the person who eats 4 idlis eats 2 vadas.

Hence we take Bimal - 8 and Ignesh - 6.

Also, we get that Bimal eats $6 - 2 = 4$ vadas.

So far, we get the following information.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4

Using (vi), there is a person who eats twice as many idlis as Mukesh. The only pair satisfying is 8, 4.

So, Mukesh eats 4 idlis. Plus the person who eats 4 idlis eats 2 vadas. Hence, we get

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2

Daljit also eats Vada as per info (v), so we get the following

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit		1
Sandip		0

(iv) gives us the information that the one who eats 1 idli does not have vada.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit	5	1
Sandip	1	0

Considering the persons who had chutney and those who didn't, 3 persons do not have chutney. Bimal is one of them(the one eating 4 vadas).

Mukesh is the second one to not take chutney(last hint). Also, Sandip does not take chutney. Hence, we get this information as well.

	Idlis	Vadas	Chutney
Ignesh	6	6	✓
Bimal	8	4	~
Mukesh	4	2	~
Daljit	5	1	✓
Sandip	1	0	~

Option A, stating that Daljit eats 5 idlis, is right.

▶ VIDEO SOLUTION

104. C

Considering (i), Ignesh has to eat 6 vadas, since 6 is the only multiple of 3.

Also, using the same information, we can say that a person consumes 2 vadas and 4 idlis.

Using (vii), Bimal eats 2 more idlis than Ignesh.

Possibilities:

Bimal - 8, Ignesh - 6

Bimal - 6, Ignesh - 4

But Ignesh cannot have 4 idlis because the person who eats 4 idlis eats 2 vadas.

Hence we take Bimal - 8 and Ignesh - 6.

Also, we get that Bimal eats $6 - 2 = 4$ vadas.

So far, we get the following information.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4

Using (vi), there is a person who eats twice as many idlis as Mukesh. The only pair satisfying is 8, 4.

So, Mukesh eats 4 idlis. Plus the person who eats 4 idlis eats 2 vadas. Hence, we get

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2

Daljit also eats Vada as per info (v), so we get the following

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit		1
Sandip		0

(iv) gives us the information that the one who eats 1 idli does not have vada.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit	5	1
Sandip	1	0

Considering the persons who had chutney and those who didn't, 3 persons do not have chutney. Bimal is one of them(the one eating 4 vadas).

Mukesh is the second one to not take chutney(last hint). Also, Sandip does not take chutney. Hence, we get this information as well.

	Idlis	Vadas	Chutney
Ignesh	6	6	✓
Bimal	8	4	~
Mukesh	4	2	~
Daljit	5	1	✓
Sandip	1	0	~

Hence, Ignesh eats 6 Vadas.

[VIDEO SOLUTION](#)

105. A

Let a,d,j,g be the shots put by Ashish, Dhanraj, Ganesh and Jugraj respectively.

According to given conditions we have ,

$$g = a - 8; d + r = 37; j = d + 8; a = 5 + d; a + g = 40$$

Solving these equations, we have $a = 24$, $d = 19$ and $j = 27$ and $r = 18$. Hence option A is the correct answer.

[VIEW SOLUTION](#)



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106. A

Let a,d,j,g be the shots put by ashish,dhanraj,ganesh and jugraj respectively. According to given conditions we have ,

$$g = a - 8;$$

$$d + r = 37;$$

$$j = d + 8;$$

$$a = 5 + d;$$

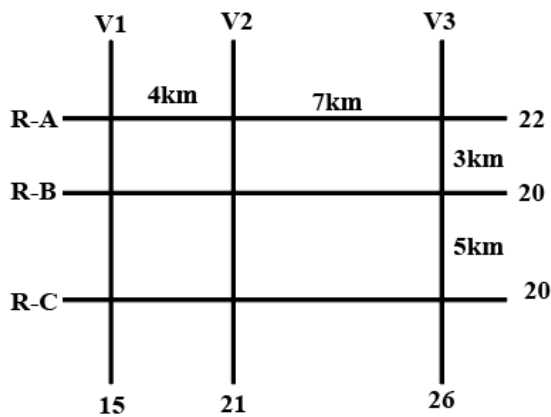
$$a + g = 40 .$$

Solving we have $a = 24$, $d = 19$ and $j = 27$.so $d + j = 45$

[VIEW SOLUTION](#)

107. 3

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

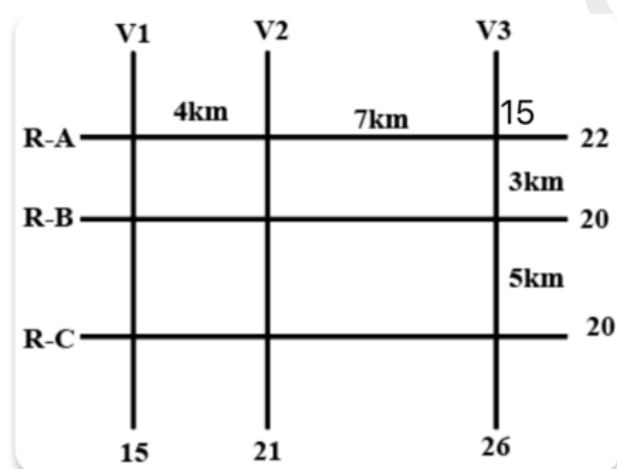
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

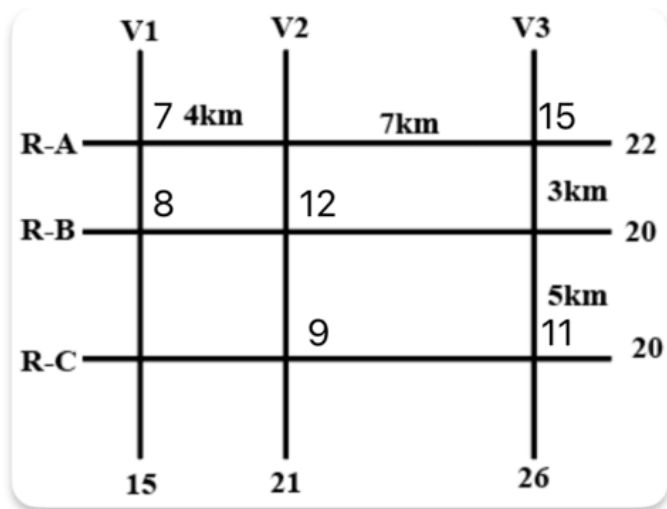
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)



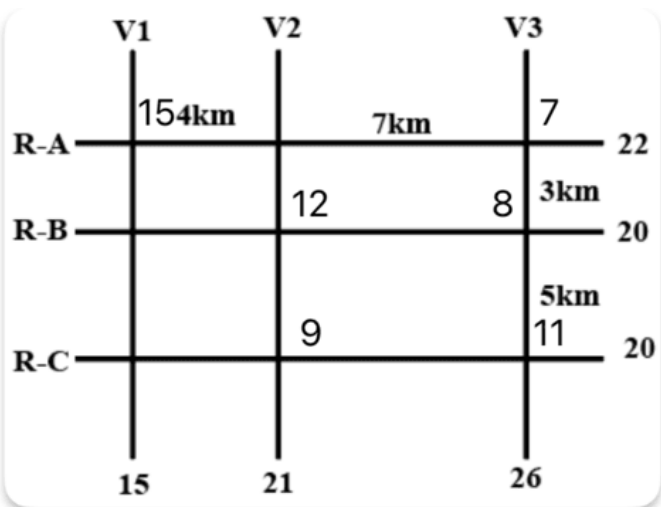
We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



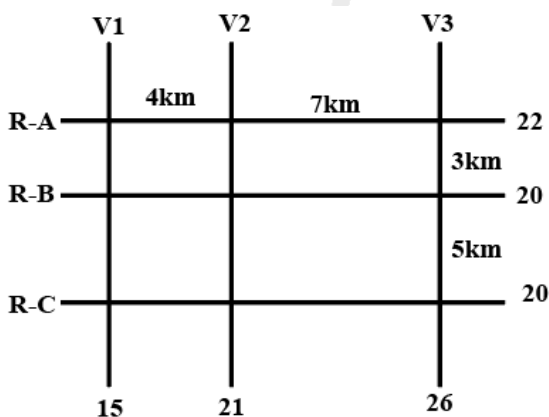
Using the two cases, we can answer the given questions.

Three ATM's with 10L or more: 15L, 12L and 11L. Answer is 3.

[VIDEO SOLUTION](#)

108. B

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity

being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

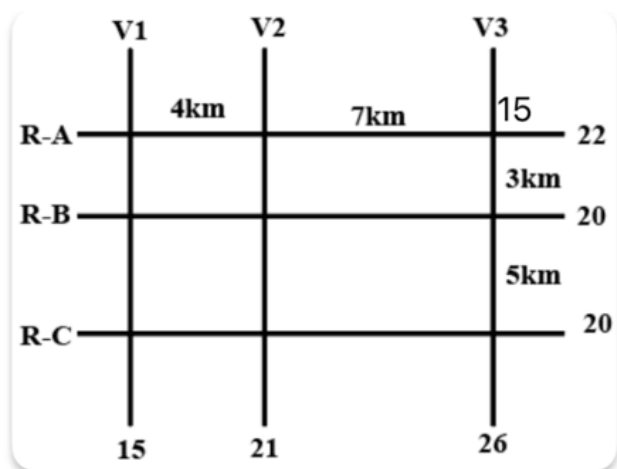
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

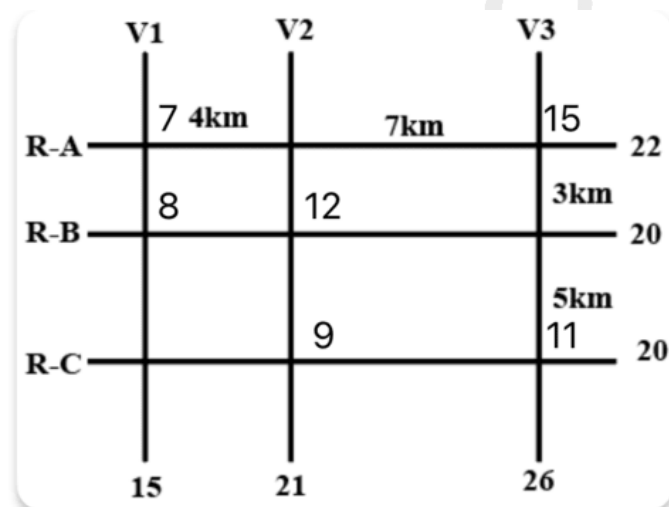
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)



We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

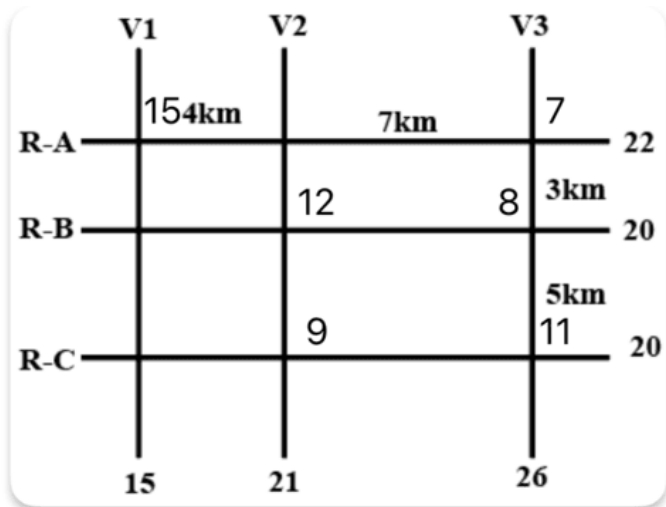
We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and

RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



Using the two cases, we can answer the given questions.

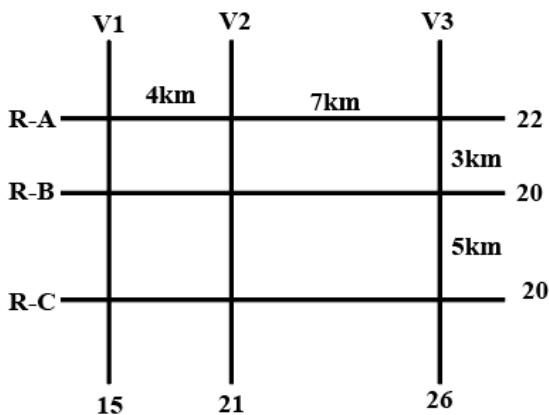
The only statement that is correct considering both the cases is Option B

"The ATM placed at the (R-C, V2) intersection has a cash requirement of Rs. 9 Lakhs."

[VIDEO SOLUTION](#)

109. A

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not

possible since there cannot be an ATM with 5L capacity.

The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)

	V1	V2	V3	
R-A		4km	7km	15
R-B				3km
R-C				5km
	15	21	26	

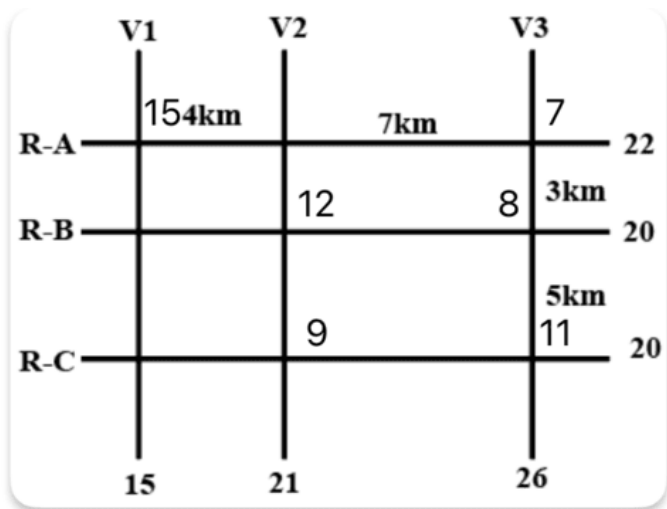
We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.

	V1	V2	V3	
R-A	7	4km	7km	15
R-B	8	12		3km
R-C		9		5km
	15	21	26	

Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



Using the two cases, we can answer the given questions.

We see that in both cases, RA RB and RC have two ATMs

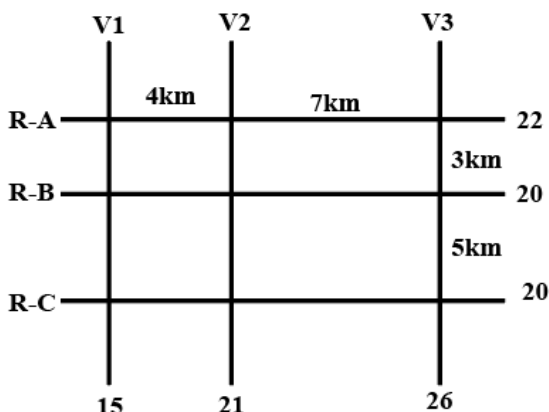
Only in the first case V1 V2 and V3 have two ATM's, in the second case V3 has 3 ATMs and V1 has 1.

Hence only Statement A is correct.

[VIDEO SOLUTION](#)

110. D

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)

	V1	V2	V3	
R-A		4km	7km	15
R-B				3km
R-C				5km
	15	21	26	

We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.

	V1	V2	V3	
R-A	7	4km	7km	15
R-B	8	12		3km
R-C		9		5km
	15	21	26	

Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.

	V1	V2	V3	
R-A	15	4km	7km	7
R-B		12	8	3km
R-C		9		5km
	15	21	26	

Using the two cases, we can answer the given questions.

Second highest is 12L and second lowest is 8L

In Case 1 the distance between 12L and 8L ATM is 4km and in Case 2 the distance between 12L and 8L ATM is 7km.

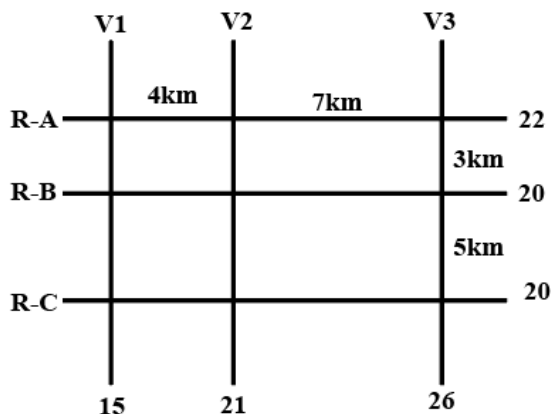
Hence the answer is either 4km or 7km.

 VIDEO SOLUTION



111.3

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cash requirement and the ATM located at the intersection of R-C and V3 is 12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

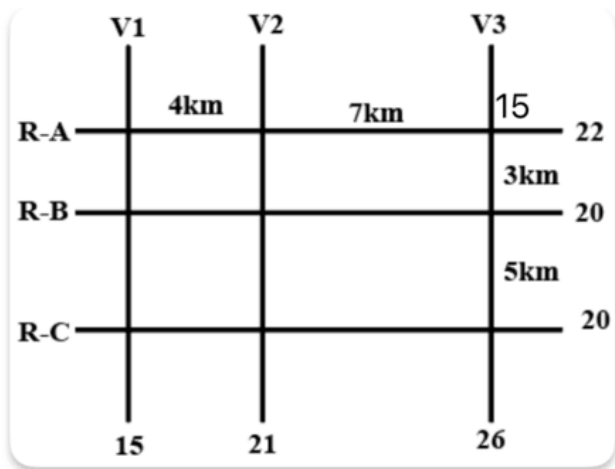
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

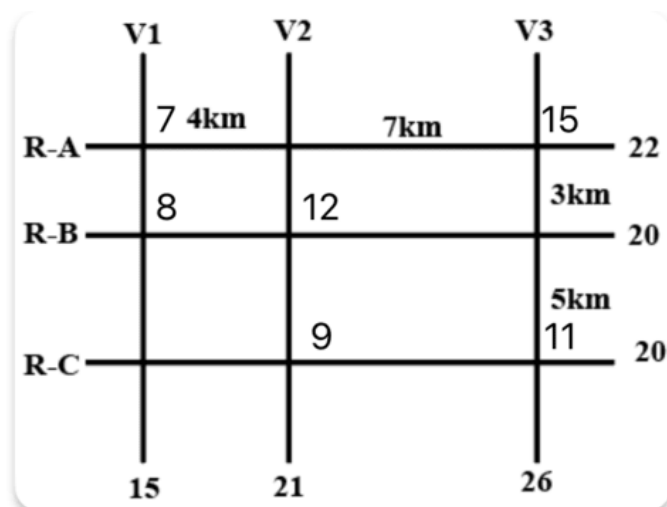
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)



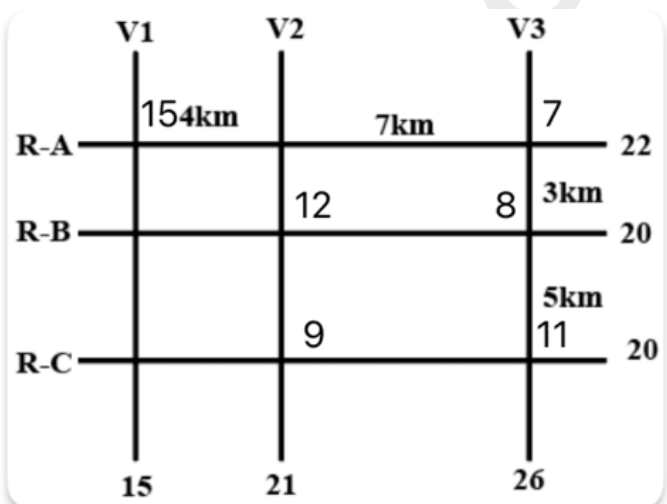
We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



Using the two cases, we can answer the given questions.

ATMs that can be uniquely determined are the ATMs with cash 9L, 11L and 12L. Hence the answer is 3.

 VIDEO SOLUTION

112. **D**

Let's look at option 4.

Order of cars is 8,2,3,V,5,6,7. This sequence is easily possible.

Let's say cars 1,2,3,4,5,6,7 arrive one after the another.

Now Car 1 leaves and Car 8 takes that place.

Finally Car 4 leaves. Hence we can see that this combination of cars is possible

 VIDEO SOLUTION

113. **C**

The original sequence as given in the question is 4,5,6,V,3

This is possible when cars 1,2,3 arrived and then cars 1 and 2 leave. After that cars 4,5 and 6 arrive.

Now there are 4 slots to the left of car 3. This is only possible when cars 1 and 2 were SUVs. Now out of these 4 slots,

3 slots are occupied by cars 4,5 and 6. As a result these are compact cars. Car 3 can be a SUV or a Compact car and it won't impact the final solution.

 VIDEO SOLUTION

114. **D**

Here we can see that cars 3 and 5 are still in their position. Thus car 4 was not the first car to leave, either 1 or 2 left before 4. Let's say only car 2 left before car 4. Now supposingly if car 2 is an SUV, car 6 was parked in that lot. Thus car 2 and car 7 are compact cars. Option 4 is correct.

 VIDEO SOLUTION

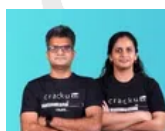
115. **D**

following is the order of arrival and departure of cars

	1 (SUV)		2	3	4 (SUV)	
Car 1 leaves	V	V	2	3	4 (SUV)	
Car 5 Arrives	5	V	2	3	4 (SUV)	
Car 6 (SUV) Arrives	5	V	2	3	4 (SUV)	6 (SUV)
Car 4 leaves	5	V	2	3	V	V
Car 7 (SUV) Arrives	5	V	2	3	7 (SUV)	6 (SUV)
Car 8 Arrives	5	8	2	3	7 (SUV)	6 (SUV)

As we can see that car 2 and car 7 are parked next to car 3

 VIDEO SOLUTION



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116. **6**

To solve this question we can use the answer of the previous question, since maximum 9 red beads are possible, filling the remaining space with green and blue beads, in such a way that number of blue beads is minimised

R	G	B	R	G
G	R	G	B	R
B	G	R	G	B
R	B	G	R	G
G	R	B	G	R

Hence number of blue beads is 6

[VIDEO SOLUTION](#)

117.6

6 more beads can be placed as shown

R			R	
		R		
	R			R
R			R	
		R		

[VIDEO SOLUTION](#)

118.9

R			R	
	R			R
		R		
R			R	
	R			R

Maximum 9 red beads are possible as shown here

[VIDEO SOLUTION](#)

119.2

Since we are required to use only two colours, these can be either

1. Green + Blue
2. Blue + Red
3. Green + Red

But we know that Between any two Red coloured beads in a row or a column, there have to be both a Green and a Blue coloured bead. Hence without both Green and Blue, Red beads cannot be used to fill the grid. Thus if we use only Green and Blue beads, the two configurations that are possible are:

G	B	G	B	G
B	G	B	G	B
G	B	G	B	G
B	G	B	G	B
G	B	G	B	G

There are only 2 configurations possible

B	G	B	G	B
G	B	G	B	G
B	G	B	G	B
G	B	G	B	G
B	G	B	G	B

[VIDEO SOLUTION](#)

120.B

The different possible cases are:

	Amar	Akbar	Antony
1	Green	Red	Red
2	Blue	Red	Red
3	Green	Red	Green
4	Green	Blue	Green
5	Blue	Blue	Green
6	Blue	Blue	Red

From the table we can see among all of the statements, only fourth statement is false, as if amar wears green, anthony has to wear green.

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121. **A**

Akbar can wear either red or blue.

Anthony can wear either red or green.

So when both of them wear same colour that has to be red.

And Amar can wear either blue or green.

Hence our answer will be A.

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